

Mzuzu University

Centre for Open and Distance Learning

Module 1

End User Computing

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Mzuzu University

Centre for Open and Distance Learning

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Introduction

Welcome to "End User Computing (BICT 1101) offered by the Department of Information and Communication Technology at Mzuzu University. You are required to study this course on the programme you are pursuing.

The aim of this course is to introduce basic computer skills to students and bring awareness to them of the roles that information and communication technology plays in the information sector.

Upon successful completion of this course, you will be able to: Operate a personal computer running MS Windows; create documents using a word processing application; produce documents using a spreadsheet application; create presentation using a presentation application; build databases using a database application; find authentic information on the web using search engines and send and receive electronic mail.

This course is expected to be covered in one semester. For each topic, where necessary, specific activities are provided for self-testing. You are advised to attempt all the activities for effective learning. A sample examination paper is provided at the very end. We hope you will find it useful when preparing for your examination.

Visual icons

The following is a list of icons you may find in this module and an explanation of the kind of information they represent.



Key Words

This is a list of important new terms and phrases used in the unit. Ensure that you understand their meaning in the context that they are used.



Learning outcomes

These are unit objectives. They identify what you are expected to know and understand by the end of the unit. Use them to test your understanding at the end of each unit.



Activity

This is a practice activity within/and at the end of each reading text. You have to work out this activity in a separate notebook.

 **Self-assessment test**

This is a test at the end of each unit. Work out the answers and compare them with the suggested answers at the end of each unit. Do not turn it in for marking.

 **Unit Summary**

This is a brief account of the main ideas you have read and learnt in the unit. Make sure you understand the unit very well before you move to the next.

 **References**

This is a list of books and websites pertaining to information in the unit. This is a useful guide on choosing appropriate reading materials. Make all possible efforts to supplement information in the unit by additional reading from this list. This will enrich your content of the course.

 **Module test**

This is a sample of the end of module examination. Make sure you are familiar with the format of the examination.

UNIT 1

USING THE COMPUTER AND MANAGING FILES

Introduction

As a University student you will be given tasks and assignments that you need to do as part of your academic requirements. You will be required to hand over on or before the due date specified. The work is to be typed before handing it over to your lecturers. In order to do this, you need to understand how to use a computer that you will use to type on your work. Before that, you need to understand what a computer is and what exactly makes it work.

This unit will introduce to you to the computer basics and its parts. The unit will also elaborate to you on how you can use menus, taskbars, manage files and folders in windows environment. Furthermore you will also be able to know how you can also make a choice of a good printer to print your work.

Areas of emphasis:

- What makes a Computer work
- Menus, commands and dialogue boxes
- The start Menu and Taskbar
- Working with folders
- Working with files
- Printing.



Key Words

- Memory, Input device, Output device, Software, Hardware, Windows, Linux, CPU, Viruses, RAM, ROM, Click, Desktop, Dialog box, Double-click, Drag, Mouse, Shortcut, Start button, Taskbar, Windows et al.

Prerequisite knowledge

Ability to read and write. This knowledge is very important and it will assist you to easily understand this unit.



Learning outcomes

By the end of this unit, you should be able to:

- define the term computer and its characteristics
- demonstrate an understanding of the components of a computer

- distinguish between a hardware and software parts of a computer
- describe the differences between RAM and ROM
- demonstrate an understanding of the role of a CPU and its parts
- manage menus, taskbar, commands, files and folders on the computer

Resources needed

- A working desktop computer or a laptop, preferably running windows to practice and refer to on the issues being discussed.

1.1.0 The Computer

Let us begin by defining what a computer is and how it works as well as what is it that makes it work the way it does.

A computer is an electronic device, operating under the control of instructions stored in its own **memory** that can accept data (**input**), process the data according to specified rules, produce information (**output**), and **store** the information for future use.

Any digital computer carries out **five** functions in gross terms as follows:

a. Takes data as input;
b. Stores the data/instructions in its memory and use when required;
c. Processes the data and converts it into useful information;
d. Generates output;
e. Controls all the above steps.

1.1.1 Characteristics of Computer

Speed, accuracy, diligence, storage capability and versatility are some of the key characteristics of a computer. A brief overview of these characteristics are:

• *Speed*: The computer can process data very fast, at the rate of millions of instructions per second. Some calculations that would have taken hours and days to complete otherwise, can be completed in a few seconds using the computer. For example, calculation and generation of salary slips of thousands of employees of an organization, weather forecasting that requires analysis of a large amount of data related to temperature, pressure and humidity of various places, etc.

• *Accuracy*: Computer provides a high degree of accuracy. For example, the computer can accurately give the result of division of any two numbers up to 10 decimal places.

• *Diligence*: When used for a longer period of time, the computer does not get tired or fatigued. It can perform long and complex calculations with the same speed and accuracy from the start till the end.

• *Storage Capability*: Large volumes of data and information can be stored in the computer and also retrieved whenever required. A limited amount of data can be stored, temporarily, in the primary memory. Secondary storage devices like floppy disk and compact disk can store a large amount of data permanently.

• *Versatility*: Computer is versatile in nature. It can perform different types of tasks with the same ease. At one moment you can use the computer to prepare a letter document and in the next moment you may play music or print a document.

Computers have several limitations too. Computer can only perform tasks that it has been programmed to do. Computer cannot do any work without instructions from the user. It executes instructions as specified by the user and does not take its own decisions.

1.1.2 Computer Components

Any kind of computers consists of ***Hardware*** and ***Software***.

Hardware:

Computer hardware is the collection of physical elements that constitutes a computer system. Computer hardware refers to the physical parts or components of a computer such as the monitor, mouse, keyboard, computer data storage, hard drive disk (HDD), system unit (graphic cards, sound cards, memory, motherboard and chips), etc. all of which are physical objects that can be touched

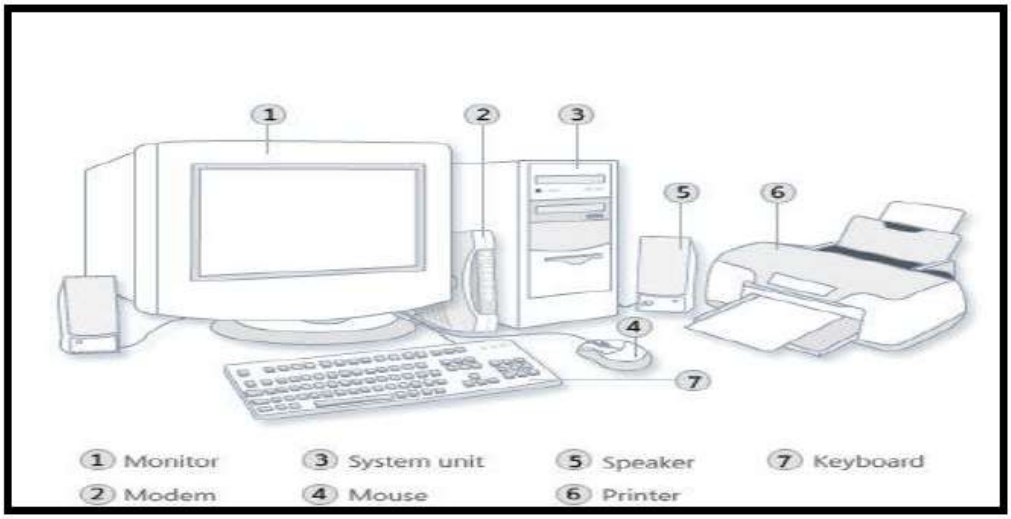


Figure 1.1: Hardware components of the Computer

1.Input Devices

Input device is any peripheral (piece of computer hardware equipment to provide data and control signals to an information processing system such as a computer or other information appliance.

Input device Translate data from form that humans understand to one that the computer can work with. Most common are keyboard and mouse

1. Keyboard	2. Mouse (pointing device)	3. Microphone
4. Touch screen	5. Scanner	6. Webcam
7. Touchpads	8. MIDI keyboard	9. Light pens
10.Graphics Tablets	11.Cameras	12.Pen Input
13.Video Capture Hardware	14.Microphone	15.Trackballs
16.Barcode reader	17.Digital camera	18.Joystick
19.Gamepad	20.Electronic Whiteboard	21. Mouse

Table 1: Example of input devices

2. Central Processing Unit (CPU)

A CPU is brain of a computer. It is responsible for all functions and processes. Regarding computing power, the CPU is the most important element of a computer system. The CPU is comprised of three main parts:

* *Arithmetic Logic Unit (ALU):* Executes all arithmetic and logical operations. Arithmetic calculations like as addition, subtraction, multiplication and division. Logical operation like compare numbers, letters, or special characters

* *Control Unit (CU):* controls and co-ordinates computer components. Read the code for the next instruction to be executed. Increment the program counter so it points to the next instruction. Read whatever data the instruction requires from cells in memory. Provide the necessary data to an ALU or register. If the instruction requires an ALU or specialized hardware to complete, instruct the hardware to perform the requested operation.

* *Registers* : Stores the data that is to be executed next, "very fast storage area".

3. Primary Memory:-

1. RAM (Random Access Memory): RAM is a memory scheme within the computer system responsible for storing data on a temporary basis, so that it can be promptly accessed by the processor as and when needed. It is volatile in nature, which means that data will be erased once supply to the storage device is turned off. RAM stores data randomly and the processor accesses these data randomly from the RAM storage. RAM is considered "random access" because you can access any memory cell directly if you know the row and column that intersect at that cell.

2. ROM (Read Only Memory): ROM is a permanent form of storage. ROM stays active regardless of whether power supply to it is turned on or off. ROM devices do not allow data stored on them to be modified.

4. Secondary Memory:-

Stores data and programs permanently: its retained after the power is turned off

a. Hard drive (HD): A hard disk is part of a unit, often called a "disk drive," "hard drive," or "hard disk drive," that store and provides relatively quick access to large amounts of data on an electromagnetically charged surface or set of surfaces.

b. Optical Disk: an optical disc drive (ODD) is a disk drive that uses laser light as part of the process of reading or writing data to or from optical discs. Some drives can only read from discs, but recent drives are commonly both readers and recorders, also called burners or writers. Compact discs, DVDs, and Blu-ray discs are common types of optical media which can be read and recorded by such drives. Optical drive is the generic name; drives are usually described as "CD" "DVD", or "Blu-ray", followed by "drive", "writer", etc. There are three main types of optical media: CD, DVD, and Blu-ray disc. CDs can store up to 700 megabytes (MB) of data and DVDs can store up to 8.4 GB of data. Blu-ray discs, which are the newest type of optical media, can store up to 50 GB of data. This storage capacity is a clear advantage over the floppy disk storage media (a magnetic media), which only has a capacity of 1.44 MB.
c. Flash Disk: A storage module made of flash memory chips. A Flash disk have no mechanical platters or access arms, but the term "disk" is used because the data are accessed as if they were on a hard drive. The disk storage structure is emulated.

5. Output devices

An output device is any piece of computer hardware equipment used to communicate the results of data processing carried out by an information processing system (such as a computer) which converts the electronically generated information into human readable form.

1. Monitor	2. LCD Projection Panels
3. Printers (all types)	4. Computer Output Microfilm (COM)
5. Plotters	6. Speaker(s)
7. Projector	

Table 2: Examples of out devices

1.1.3 Software

Software is a generic term for organized collections of computer data and instructions, often broken into two major categories: system software that provides the basic non-task-specific functions of the computer, and application software which is used by users to accomplish specific tasks.

Software Types

a. ***System software*** is responsible for controlling, integrating, and managing the individual hardware components of a computer system so that other software and the users of the system see it as a functional unit without having to be concerned with the low-level details such as transferring data from memory to disk, or rendering text onto a display. Generally, system software consists of an operating system and some fundamental utilities such as disk formatters, file managers, display managers, text editors, user authentication (login) and management tools, and networking and device control software.

b. ***Application software*** is used to accomplish specific tasks other than just running the computer system. Application software may consist of a single program, such as an image viewer; a small collection of programs (often called a software package) that work closely together to accomplish a task, such as a spreadsheet or text processing system; a larger collection (often called a software suite) of related but independent programs and packages that have a common user interface or shared data format, such as Microsoft Office, which consists of closely integrated word processor, spreadsheet, database, etc.; or a software system, such as a database management system, which is a collection of fundamental programs that may provide some service to a variety of other independent applications.

Comparison Application Software and System Software

	System Software	Application Software
	Computer software, or just software is a general term primarily used for digitally stored data such as computer programs and other kinds of information read and written by computers. App comes under computer software though it has a wide scope now.	Application software, also known as an application or an "app", is computer software designed to help the user to perform specific tasks.
Example:	1) Microsoft Windows 2) Linux 3) Unix 4) Mac OSX 5) DOS	1) Opera (Web Browser) 2) Microsoft Word (Word Processing) 3) Microsoft Excel (Spreadsheet software) 4) MySQL (Database Software) 5) Microsoft PowerPoint (Presentation Software) 6) Adobe Photoshop (Graphics Software)
Interaction:	Generally, users do not interact with system software as it works in the background.	Users always interact with application software while doing different activities.
Dependency:	System software can run independently of the application software.	Application software cannot run without the presence of the system software.

Table 3: Application software vs System software

1.1.4 Computer Viruses

A virus is a small piece of software that piggybacks on real programs. For example, a virus might attach itself to a program such as a spreadsheet program. Each time the spreadsheet program runs, the virus runs, too, and it has the chance to reproduce (by attaching to other programs) or wreak havoc.

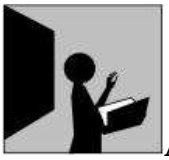
•E-mail viruses: An e-mail virus travels as an attachment to e-mail messages, and usually replicates itself by automatically mailing itself to dozens of people in the victim's e-mail address book. Some e-mail viruses don't even require a double-click -- they launch when you view the infected message in the preview pane of your e-mail software [source: Johnson].

•Trojan horses: A Trojan horse is simply a computer program. The program claims to do one thing (it may claim to be a game) but instead does damage when you run it (it may erase your hard disk). Trojan horses have no way to replicate automatically.

•Worms: A worm is a small piece of software that uses computer networks and security holes to replicate itself. A copy of the worm scans the network for another machine that has a specific security hole. It copies itself to the new machine using the security hole, and then starts replicating from there, as well.

How to avoid viruses and lessen their impact?

a. Install anti-virus software from a reputable vendor. Update it and use it regularly.
b. In addition to scanning for viruses on a regular basis, install an "on access" scanner (included in most anti-virus software packages) and configure it to start each time you start up your computer. This will protect your system by checking for viruses each time you run an executable file.
c. Use a virus scan before you open any new programs or files that may contain executable code. This includes packaged software that you buy from the store as well as any program you might download from the Internet.
d. If you are a member of an online community or chat room, be very careful about accepting files or clicking links that you find or that people send you within the community.
e. Make sure you back up your data (documents, important email messages, etc.) on disc so that in the event of a virus infection, you do not lose valuable work.

 **Activity 1**

1. What is a computer?
2. Give examples of an Input device?
3. Give examples of an output device?
4. Give one use of the parts of the CPU?
5. What is the difference between Application software and System software?

1.1.5 Getting started with Windows and Linux

As we have already looked at the software part of a computer and learned that Windows and Linux are examples of the System software. This means that they play a greater role in providing support for the application software to run. It has to be noted that both Windows and Linux being operating systems do not interact with the users directly. Their role in the system is in the background.

Windows is a licensed operating system designed for the individuals with the perspective of having no computer programming knowledge and for business and other commercial users. It is very simple and straightforward to use. As Windows OS is licensed it mean you will be required to buy in order to use it.

Linux is a free and open source operating system based on UNIX standards. It provides programming interface as well as user interface compatible with UNIX based systems and

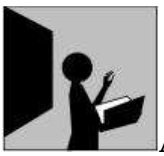
provides large variety applications. A Linux operating system also contains many separately developed elements, resulting in UNIX system which is fully compatible and free from proprietary code.

Key Differences between Linux and Windows Operating System

- Linux is free and open source operating system whereas Windows is a commercial operating system.
- Windows is not customizable while Linux is customizable and a user can modify the code and can change its the look and feel.
- Linux provides high security than windows because Linux is open source.
- The separation of the directories is done using a backslash in windows. On the contrary, in Linux, these are separated by using forward slash.
- In Linux, file names are case sensitive while windows file name are case-insensitive.
- Linux uses the monolithic kernel which consumes more running space whereas Windows uses the micro kernel which takes less space but system running efficiency is lower than Linux.

Linux and Windows both Operating systems have their advantages and disadvantages. Windows is simple to use but is not a free and open source OS, whereas Linux is free, open source, customizable and secure but kind of complex for the users having no programming background. Linux is more reliable than windows.

Therefore this module will focus on the Windows platform which is the most used and popular Operating System among University students.

 **Activity 2**

1. What is the difference between Windows operating System and Linux operating system?

1.1.6 The Basics – Beginning to Use a Computer

Using menus

Most programs contain dozens or even hundreds of commands (actions) that you use to work the program. Many of these commands are organized under menus. Like a restaurant menu, a program menu shows you a list of choices. To keep the screen uncluttered, menus are hidden until you click their titles in the menu bar, located just underneath the title bar.

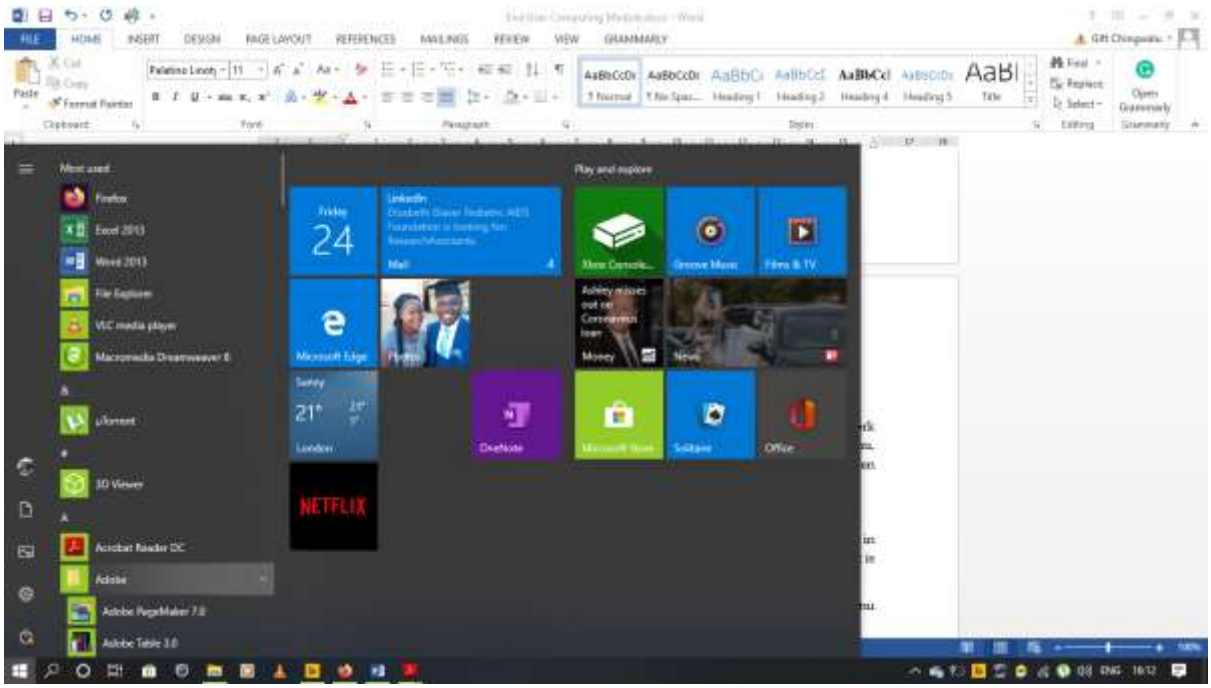


Figure 2 : Task bar with menus

To choose one of the commands listed in a menu, click it. Sometimes a dialog box appears, in which you can select further options. Some menu items are not commands at all. Instead, they open other menus. In the submenu would open yet another submenu.

If you don't see the command you want, try looking at another menu. Move your mouse pointer along the menu bar and its menus open automatically; you don't need to click the menu bar again. To close a menu without selecting any commands, click the menu bar or any other part of the window.

Recognizing menus isn't always easy, because not all menu controls look alike or even appear on a menu bar.

Menu Conventions

Several Office applications maintain Window menus that have nearly identical commands. In addition, all Office applications have *File*, *Edit*, and *Help* menus that have many of the same commands—so once you learn how to use these menus in one Office application, you'll be able to use them in all Office applications. What's more, each program follows a series of conventions for the presentation of commands, buttons, and information on menus and in dialog boxes.

Using command buttons

A command button performs a command (makes something happen) when you click it. You'll most often see them in dialog boxes, which are small windows that contain options for completing a task.

To close an open page, you must first click either the Yes or No button. Clicking Yes saves the page and any changes you've made, and clicking No deletes the page and discards any

changes you've made. Clicking Cancel dismisses the dialog box and returns you to the program.

Dialog Box

This dialog box in figure above appears when you choose Print from 2013 Word's File menu.) Dialog boxes present you with one or more command options by using list boxes, buttons, and other components. To complete a dialog box, indicate your preferences, and then click the OK button. To move from one item to the next in a dialog box, click the item, or use the Tab key until the item is highlighted. Many dialog boxes don't require you to fill every blank.

If you have second thoughts about using the command after you see the dialog box, you can cancel the command by clicking the Cancel button or by clicking the Close button on the dialog box's title bar. You can also get help by clicking the Help button on the title bar and then clicking the item you want to learn more about. Finally, if you need to see document information displayed beneath a dialog box, move the dialog box out of the way by dragging its title bar.

The start Menu and Taskbar

The Start menu is invoked by clicking the Windows key (previously the Start button). This brings up a list of installed programs as well as shortcuts to documents, pictures and your login session.

1.1.7 Working with files and folders

What is a file?

There are many different types of files you can use. For example, Microsoft Word documents, digital photos, digital music, and digital videos are all types of files. You might even think of a file as a digital version of a real-world thing you can interact with on your computer. When you use different applications, you'll often be viewing, creating, or editing files.

Files are usually represented by an icon. In the image below, you can see a few different types of files below the Recycle Bin on the desktop.



Figure 3 : files on the desktop

To open a file:

There are two main ways to open a file:

- Find the file on your computer and double-click it**. This will open the file in its** default application**.** In our example, we'll open a Microsoft Word document (**BdayInvites.docx**), which will open in Microsoft Word**.**

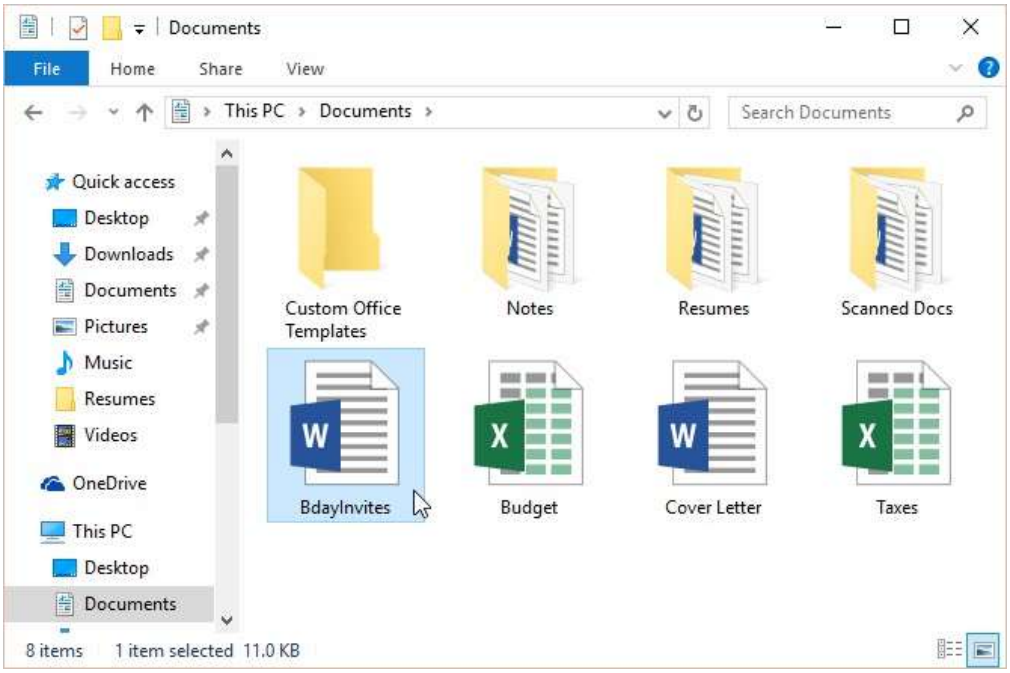


Figure 4: files in the Documents folder

- Open the application, then use the application to open the file**.** Once the application is open, you can go to the **File** menu at the top of the window and select Open**.**

Moving and deleting files

As you begin using your computer, you will start to collect more and more files, which can make it more difficult to find the files you need. Fortunately, Windows allows you to **move files** to different folders and **delete files** you no longer use.

To move a file:

It's easy to move a file from one location to another. For example, you might have a file on the **desktop** that you want to move to your **Documents** folder.

1. Click and drag the file to the desired location.

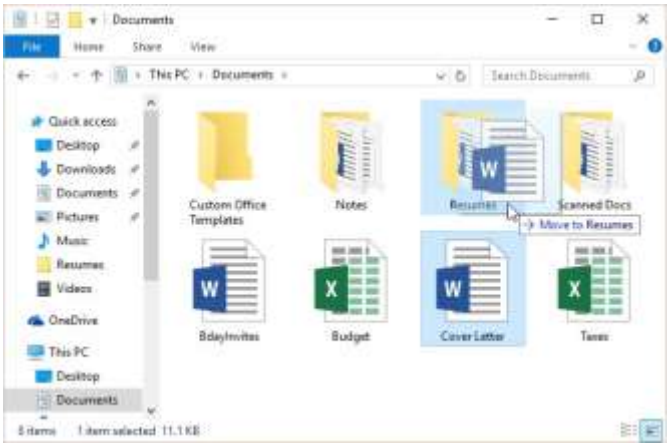


Figure 5: dragging a file in the Documents folder

2. Release the mouse. The file will appear in the new location. In this example, we have opened the folder to see the file in its new location.

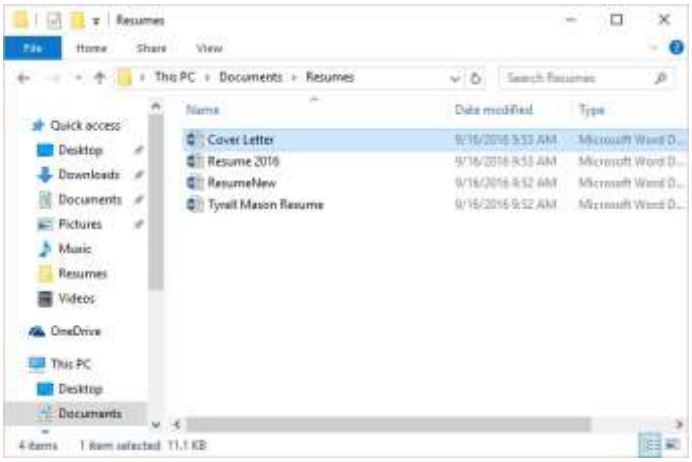


Figure 6: dragged file in the new location

You can use this same technique to **move an entire folder**. Note that moving a folder will also move all of the files within that folder.

What is a folder?

Windows uses **folders** to help you organize files. You can put files **inside a folder**, just like you would put documents inside a real folder. In the image below, you can see some folders on the desktop.

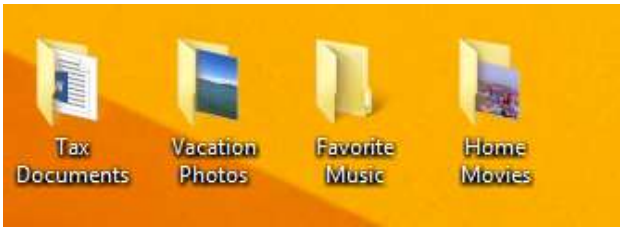


Figure 7 : Folders on the desktop

To create a new folder:

1. Within File Explorer, locate and select the **New folder** button. You can also **right-click** where you want the folder to appear, then select **New > Folder**.

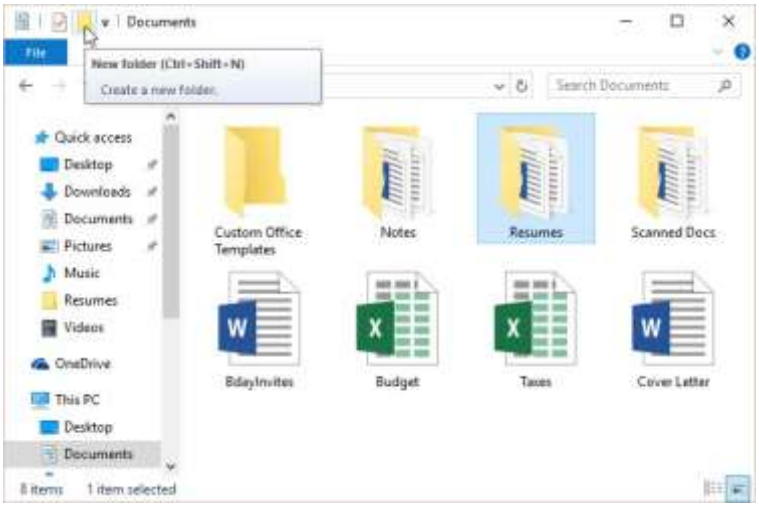


Figure 8: creating a new folder

2. The new folder will appear. Type the desired **name** for the folder and press **Enter**. In our example, we'll call it **School Documents**.

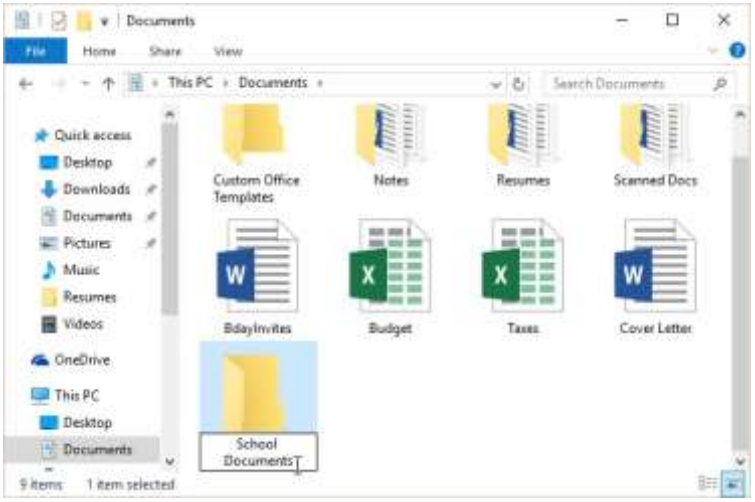


Figure 9: Naming a new folder

3. The new folder will be created. You can now move files into this folder.

To rename a file or folder:

You can change the name of any file or folder. A unique name will make it easier to remember what type of information is saved in the file or folder.

1. Click the file or folder, **wait about one second**, and click again. An editable text field will appear.
2. Type the desired name on your keyboard and press **Enter**. The name will be changed.

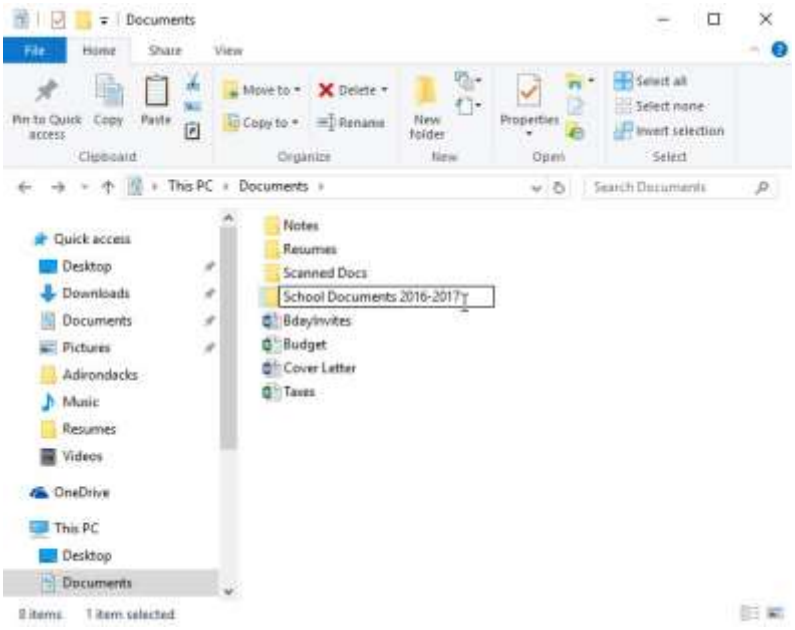


Figure 8: renaming a folder

You can also **right-click** the folder and select **Rename** from the menu that appears.

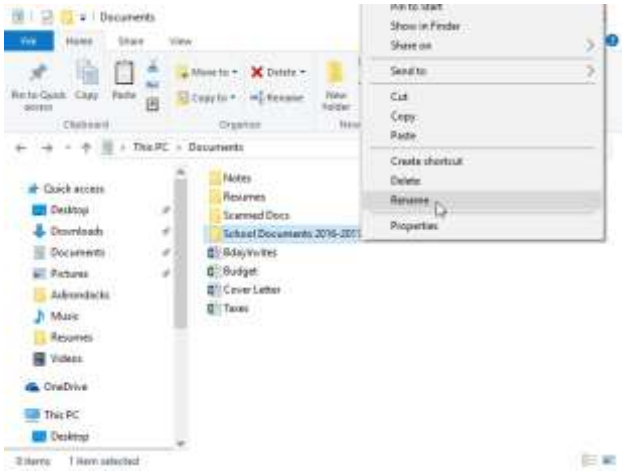


Figure 9: creating a folder

To delete a file or folder:

If you no longer need to use a file, you can delete it. When you delete a file, it is moved to the **Recycle Bin**. If you change your mind, you can move the file from the Recycle Bin back to its original location. If you're sure you want to permanently delete the file, you will need to **empty the Recycle Bin**.

1. Click and drag the file to the **Recycle Bin** icon on the **desktop**. You can also click the file to select it and press the **Delete** key on your keyboard.

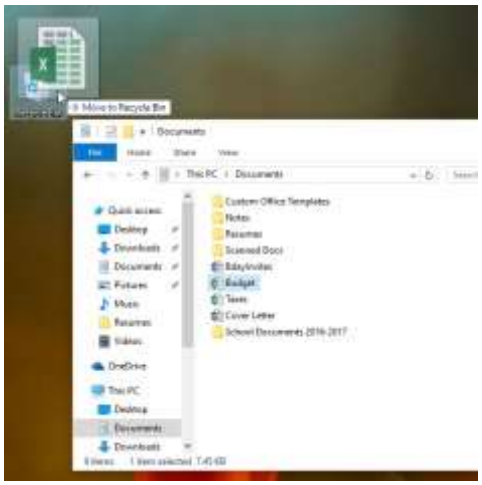


Figure 10: deleting a folder

2. To permanently delete the file, right-click the **Recycle Bin** icon and select **Empty Recycle Bin**. All files in the Recycle Bin will be permanently deleted.



Figure 11: emptying a recycle bin

Note that deleting a folder will also delete all of the files within that folder.

Selecting multiple files

Now that you know the basics, here are a few tips to help you move your files even faster. There are a few ways to select **more than one file at a time**:

- If you're viewing your files as icons, you can **click and drag the mouse to draw a box** around the files you want to select. When you're done, release the mouse; the files will be selected. You can now move, copy, or delete all of these files at the same time.

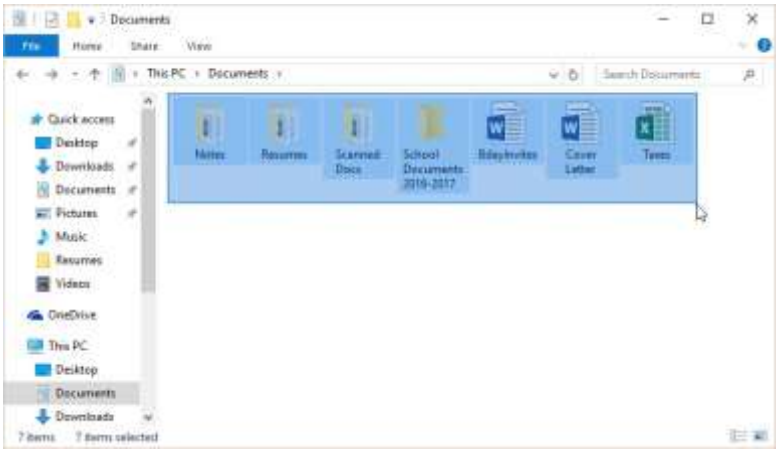


Figure 11: Selecting multiple files

- To select **specific files** from a folder, press and hold the **Control** key on your keyboard, then click the files you want to select.

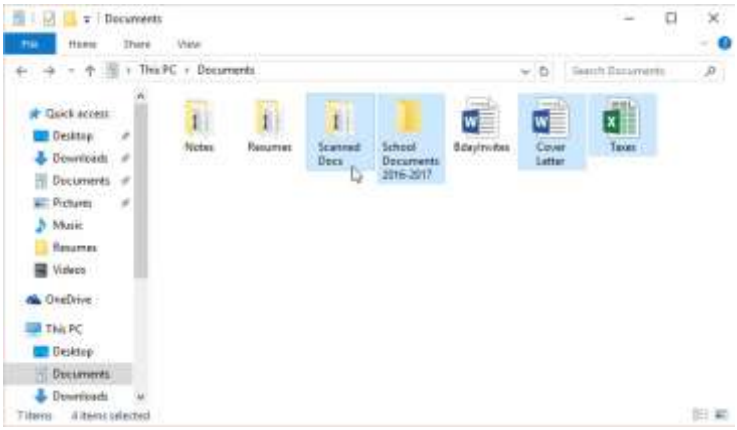


Figure 12: specific file selection

- To select a **group of files** from a folder, click the first file, press and hold the **Shift** key on your keyboard, then click the last file. All of the files between the first and last ones will be selected.

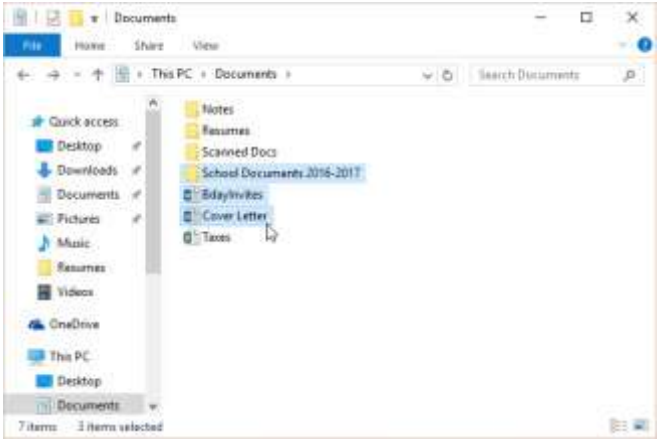


Figure 13: group file selection

Selecting all files

If you want to select **all files in a folder** at the same time, open the folder in File Explorer and press **Ctrl+A** (press and hold the **Control** key on your keyboard, then press **A**). All of the files in the folder will be selected.

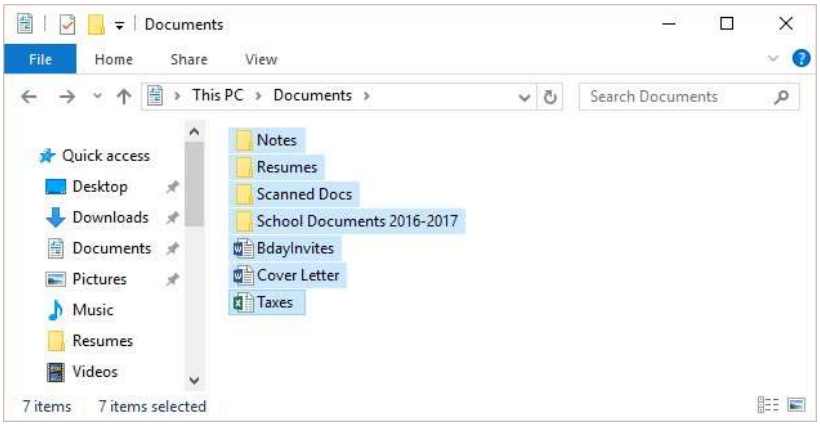


Figure 14: all file selection

Shortcuts

If you have a file or folder you use frequently, you can save time by creating a **shortcut** on the desktop. Instead of navigating to the file or folder each time you want to use it, you can simply double-click the shortcut to open it. A shortcut will have a small arrow in the lower-left corner of the icon.

Note that creating a shortcut does **not** create a duplicate copy of the folder; it's simply a way to access the folder more quickly. If you delete a shortcut, it will not delete the actual folder or the files it contains. Also note that copying a shortcut onto a flash drive will not work; if you want to bring a file with you, you'll need to navigate to the actual location of the file and copy it to the flash drive.

1. Locate and right-click the desired folder, then select **Send to Desktop (create shortcut)**.

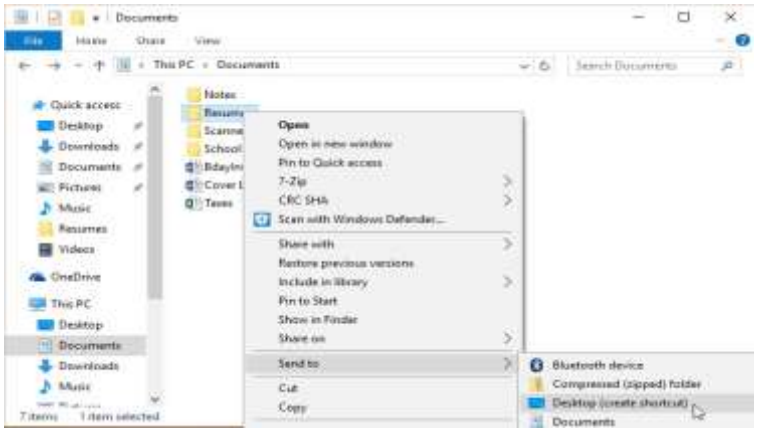
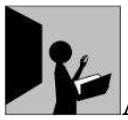


Figure 15: creating a folder shortcut

2. A shortcut to the folder will appear on the desktop. Notice the arrow in the lower-left corner of the icon. You can now double-click the shortcut to open the folder at any time.

Understanding how to work with files and folders is an important part of using your computer. Once you understand how files and folder work, you'll use them all the time.

Your work on the computer is to be stored in a folder. You can have as many folder as you can to stare your academic work. Each folder can be named either by course code or course titles for example *BICT1101* to store all assignments for this module.

 **Activity 3**

If working with files and folders feels a little tricky right now, don't worry! Like anything else, working with files and folders is largely a matter of practice. You'll start to feel more comfortable as you continue using your computer. Keep practicing.

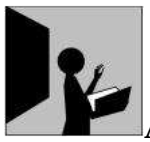
1.1.8 Windows and Linux applications

Windows operating system comes with several useful applications already installed. These applications can help you with many common tasks, including browsing the Internet, managing your calendar, and shopping for music. These applications also allow you to do basic things such as write and edit text files, do simple editing of image files etc.

In Windows, these applications are found in the Accessories sub-menu of the All programs option in the Start menu.

Most of these apps can be opened by clicking the Start button, then clicking the icon of the app you want to open. You may need to navigate the Start menu to find the app you're looking for.

Linux too has its own specific applications that it supports. And they include Firefox or chrome for browsing the internet, Libre office for editing documents and gscan2pdf for opening protected document files.

 **Activity 4**

1. List ANY five applications that are available in Windows operating system and their use?

1.1.9 Customizing the Desktop

Customizing your desktop is a handy way to optimize workflow while adding a personal touch to the appearance of your computer. Windows mostly enables you to customize several elements of the desktop on your computers, including the wallpaper, desktop gadgets and taskbar. The wallpaper is the background image situated behind the application icons and files on your desktop.

1.1.10 Printing.

A computer printer is a piece of hardware for a computer. It is an output device. It allows a user to print items on paper, such as letters and pictures. Mostly a printer prints under the control of a computer. Many can also work as a copying machine or with a digital camera to print directly without using a computer.

A printer is a device that accepts text and graphic output from a computer and transfers the information to paper, usually to standard size sheets of paper. Printers vary in size, speed, sophistication, and cost. In general, more expensive printers are used for higher-resolution color printing.

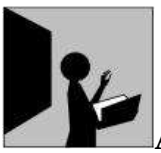
The four printer qualities of most interest to most users are:

Color: Color is important for users who need to print pages for presentations or maps and other pages where color is part of the information. Color printers can also be set to print only in black-and-white. Color printers are more expensive to operate since they use two ink cartridges (one color and one black ink) that need to be replaced after a certain number of pages. Users who don't have a specific need for color and who print a lot of pages will find a black-and-white printer cheaper to operate.

Resolution: Printer resolution (the sharpness of text and images on paper) is usually measured in dots per inch (dpi). Most inexpensive printers provide sufficient resolution for most purposes at 600 dpi.

Speed: If you do much printing, the speed of the printer becomes important. Inexpensive printers print only about 3 to 6 sheets per minute. Color printing is slower. More expensive printers are much faster.

Memory: Most printers come with a small amount of memory (for example, one megabyte) that can be expanded by the user. Having more than the minimum amount of memory is helpful and faster when printing out pages with large images or tables with lines around them (which the printer treats as a large image).

 **Activity 5**

1. List four qualities of a printer?



Mzuzu University

Self-Assessment Test

1. What are the **Five** functions that each digital computer carriers?

2. Differentiate RAM and ROM?

3. Explain the role of Application software in a computer.

4. List **Five** ways of avoiding computer viruses?

5. Linux and Windows Operating systems are examples of System software. Explain their role in a computer system.



Unit Summary

In this unit we have learnt more about the computer. As a machine that input data, processes and output information. It has both hardware and software components that make it work. There also viruses that poses a threat to the operations of a computer hence a need to avoid introducing them to your computer. The unit also introduced us to the windows environment with its taskbar, menus, dialog boxes and commands. We have also learnt on how we can work with files and folders, from creating, deleting up to creating shortcuts on the desktop for easy access.



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UNIT 2

WORD PROCESSING

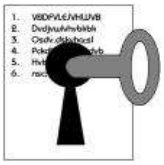
Introduction

MS Word is the most popular word processing software used today. A word processor is essentially a computerized version of the standard typewriter. However, the computer adds features typewriters never dreamed of having like spell check, the ability to save and store documents, copy and paste functions, and the ability to add images and shapes to documents, and many more. When attached to an email, electronic documents, created by MS Word can be delivered in seconds. Another benefit is that it helps the user to type faster and more accurate.

This unit will help to know and understand how to use the software to create, edit, and format written documents in the workplace, at school, and at home. Examples include personal and formal business letters, Cvs, coversheets, and assignments. Intermediate and advanced level knowledge of this software could lead to job opportunities since MS Word is used a lot in the workplace.

Areas of emphasis:

- how to use Ms word
- Editing and Format documents
- Using Proofing tools
- Document layout and Mail merge
- Inserting graphics in a document
- Creating Tables



Key Words

- Mail merge, Formatting, Tables, Graphics, Save, Copy, Paste, Ribbon, et al.

Prerequisite knowledge

This unit assumes you are familiar with the basic operation of a personal computer. Specific skills required for this tutorial are:

- Use of the mouse and keyboard
- Opening and saving files on the hard disk and on USB memory stick (thumb drive)
- Managing files in general i.e. the difference between drive letters, hard disk, etc.
- Running programs from the Windows Start menu
- Minimizing, maximizing and resizing windows

You should have a USB memory stick (thumb drive) or other portable memory device available to store the files that will be created in this unit.



Learning outcomes

By the end of this unit, you should be able to:

- Create and edit documents
- Use proofing tools
- Use the document layout feature
- Use tables and the mail merge tool

Resources needed

- a working desktop computer, with latest Microsoft office Word installed and running Windows OS to practice and refer to it on the issues being discussed in the unit.

2.1.0 Getting started with Microsoft Word

Before you begin creating documents in Word, you may want to **set up your Word environment** and become familiar with a few **key tasks** such as how to minimize and maximize the Ribbon, configure the Quick Access Toolbar, display the ruler, and use the Word Count and Zoom tools.

To minimize and maximize the Ribbon:

- Right-click anywhere in the main menu.
- Select **Minimize the Ribbon** in the menu that appears. This will **toggle** the Ribbon **on** and **off**.

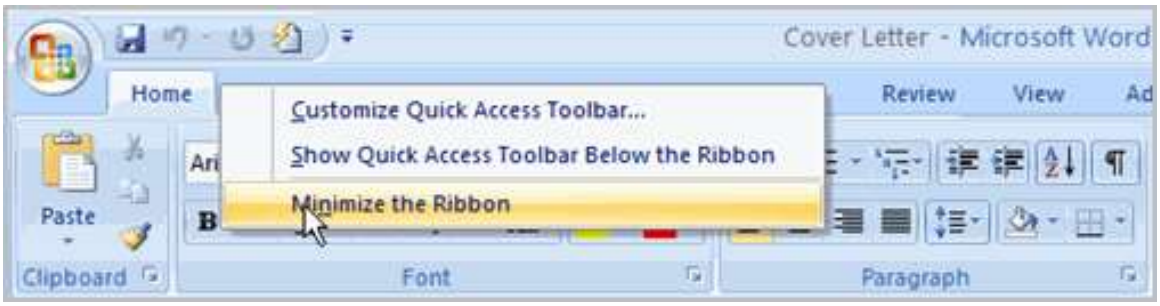


Figure 1: Microsoft word ribbon

The **check mark** beside **Minimize the Ribbon** indicates that the feature is active.

The new **tabbed Ribbon system** replaces traditional menus in Word 2007. It is designed to be responsive to your current task and easy to use; however, you can choose to **minimize the Ribbon** if you would rather use different menus or keyboard shortcuts.

To add commands to the Quick Access Toolbar:

- Click the **arrow** to the right of the Quick Access Toolbar.
- Select the **command** you want to add from the drop-down list. It will appear in the Quick Access Toolbar.

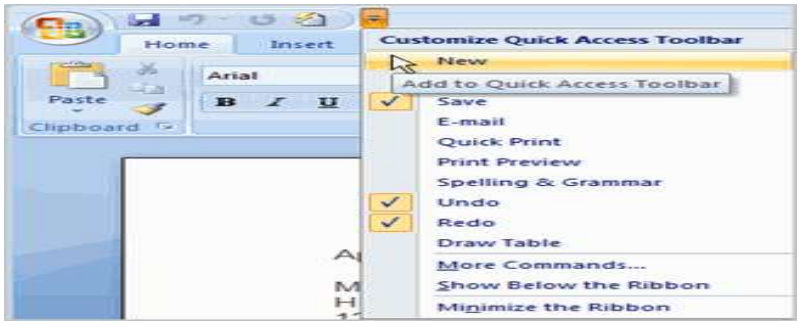


Figure 2: Quick Access Toolbar

The Save, Undo, and Redo commands appear by default in the Quick Access Toolbar. You may want to add other commands to make using specific Word features more convenient for you.

To display or hide the Ruler:

- Click the **View Ruler** icon over the scrollbar.

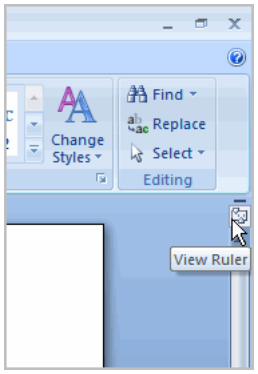
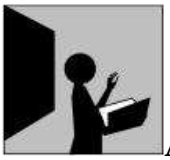


Figure 3: View ruler

The **View Ruler** icon works as a toggle button to turn the Ruler on and off.

 **Activity 1**

1. Open **Word** on your computer. A **new blank document** will appear on the screen.
2. Make sure your **Ribbon** is maximized.
3. Display the **Ruler**.
4. Add any commands you want to the **Quick Access** Toolbar.
5. Close **Word** without saving the document.

2.1.1 Working with Text

It's important to know how to perform **basic tasks with text** when working in a word processing application. You need to learn the basics of working with text, including how to insert, delete, select, copy, paste, and drag and drop text.

To insert text:

- Move your mouse to the location where you want text to appear in the document.
- Left-click the mouse. The **insertion point** appears.
- Type the text you want to appear.

To delete text:

- Place your cursor next to the text you want to delete.
- Press the **Backspace** key on your keyboard to delete text to the left of the cursor.
- Press the **Delete** key on your keyboard to delete text to the right of the cursor.

To select text:

- Place the **insertion point** next to the text you want to select.
- Left-click your mouse. While holding it down, drag your mouse over the text to select it.
- Release the mouse button. You have selected the text. A **highlighted box** will appear over the selected text.

When you select text or images in Word, a **hover toolbar** with formatting options appears. This makes formatting commands easily accessible, which can save you time.

To copy and paste text:

- Select the text you want to copy.
- Click the **Copy** command on the Home tab.
- Place the insertion point where you want text to appear.
- Click the **Paste** command on the Home tab. The text will appear.

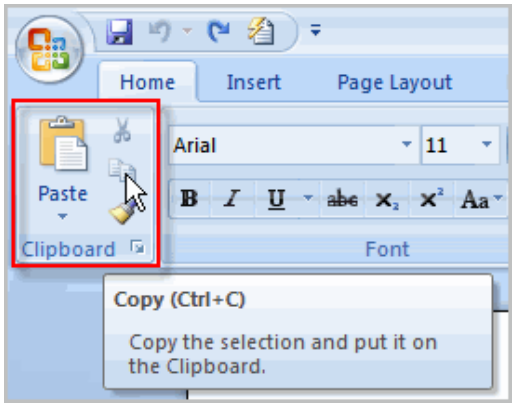


Figure 4: the clipboard

To drag and drop text:

- Select the text you want to copy.
- Left-click your mouse, and **drag the text** to the location where you want it to appear. The cursor will have a text box underneath it to indicate that you are moving text.

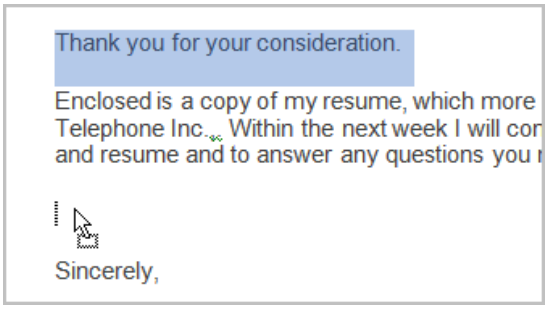
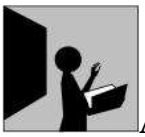


Figure 5: copying text

- Release the mouse button, and the text will appear.

If text **does not appear** in the exact location you want, you can click the **Enter** key on your keyboard to move the text to a new line.

 **Activity 2**

Use the cover letter or any other Word document you choose to complete this challenge.

- Open an **existing Word document.**
- Select a **sentence.**
- **Copy and paste** the sentence from one location in the document to another.
- Select **another sentence**.
- **Drag and drop** it to another location in the document.
- Select the **entire document**.
- Change the **font style**, **size,** and **color**.

2.1.2 Saving your work

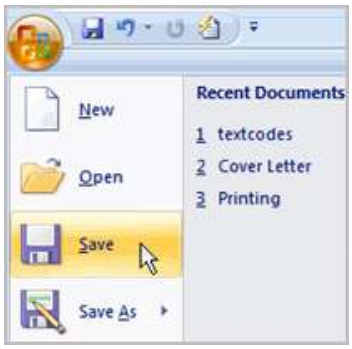


Figure 6: copying text

It's important to know how to save the documents you are working with. There are many ways you **share and receive documents**, which will affect how you need to save the file.

Are you downloading the document? Saving it for the first time? Saving it as another name? Sharing it with someone who doesn't have Word 2007? All of these factors will affect how you **save Word documents**. Let us see how to use the Save and Save As commands, how to save as a Word 97-2003 compatible document, and how to save as a PDF.

To use the Save As command:

- Click the **Microsoft Office button**.
- Select **Save As** → **Word Document**. The **Save As** dialog box appears.



Figure 7: saving the file

- Select the **location** where you want to save the document using the drop-down menu.
- Enter a **name** for the document.

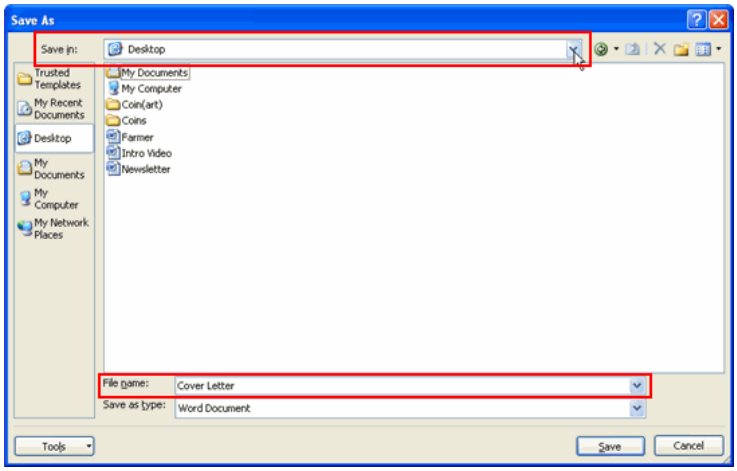


Figure 8: Saving location for a file

- Click the **Save** button.

To use the Save command:

- Click the **Microsoft Office button**.
- Select **Save** from the menu.

Using the Save command saves the document in its current location using the same file name. If you are saving for the first time and select **Save**, the **Save As** dialog box will appear.

To Save As a Word 97 - 2003 document:

- Click the Microsoft Office button.
- Select **Save As** → **Word 97-2003 Document**.



Figure 9: Save as dialog box

- Select the location where you want to save the document using the drop-down menu.
- Enter a name for the document.
- Click the Save button.

To download the PDF extension:

- Click the Microsoft Office button.
- Select **Save As** → **Find add-ins** for other file formats. This will open your web browser to the Microsoft site.
- Follow the instructions on the Microsoft site for downloading the extension.

To save as a PDF:

- Click the Microsoft Office button.
- Select **Save As** → **PDF**. The Save As dialog box will appear.

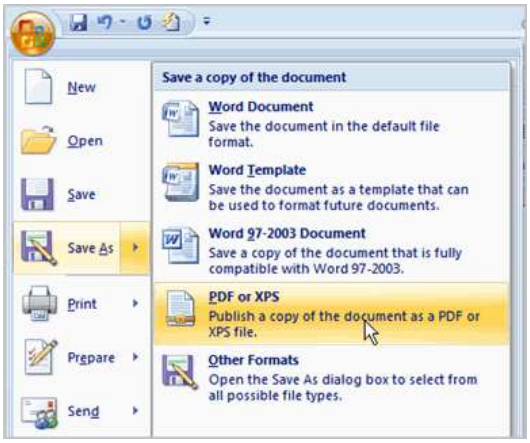


Figure 10: Save as pdf

- Select the location where you want to save the document using the drop-down menu.
- Enter a name for the document.
- Click the Publish button.

2.1.3 Compatibility mode

Sometimes you may need to work with documents that were created in earlier versions of Microsoft Word, such as Word 2003 or Word 2000. When you open these types of documents, they will appear in **Compatibility mode**.

Compatibility mode **disables** certain features, so you'll only be able to access commands found in the program that was used to create the document. For example, if you open a document created in Word 2003, you can only use tabs and commands found in Word 2003.

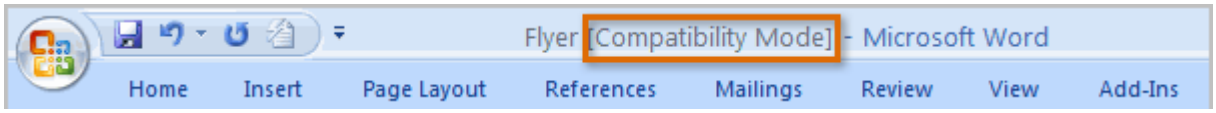


Figure 11: the compatibility menu bar

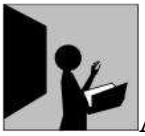
If you want access to all of the Word 2007 features, you can **save** the document in the Word 2007 file format.

To exit Compatibility mode:

- Click the **Microsoft Office button**.
- Select **Save As** → **Word Document**.



Figure 12: Saving mode

 **Activity 3**

Use the cover letter or any other Word document you choose to complete this challenge.

- **Open** an existing Word document.
- Save the document with the file name **trial**.
- Save the **same document** as a PDF file.
- **Close** the document.
- Open **another** existing Word document.
- Save the document so it is **compatible** with Word 2003.
- **Close** the document.

2.1.4 Editing documents & Formatting documents

To create and design effective documents, you need to know how to **format text**. In addition to making your document more appealing, **formatted text** can draw the reader's attention to specific parts of the document and help communicate your message.

Let us learn to format the font size, style, and color of text, as well as how to use the Bold, Italic, Underline, and Change Case commands.

To format font size:

- Select the text you want to modify.
- Left-click the **drop-down arrow** next to the **font size box** on the Home tab. The font size drop-down menu appears.
- Move your cursor over the various font sizes. A **live preview** of the font size will appear in the document.

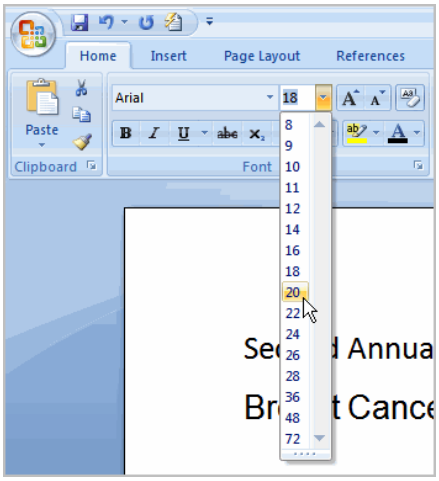


Figure 13: Font sizes

- Left-click the font size you want to use. The font size will change in the document.

To format font style:

- Select the text you want to modify.
- Left-click the **drop-down arrow** next to the **font style box** on the Home tab. The font style drop-down menu appears.
- Move your cursor over the various font styles. A **live preview** of the font will appear in the document.

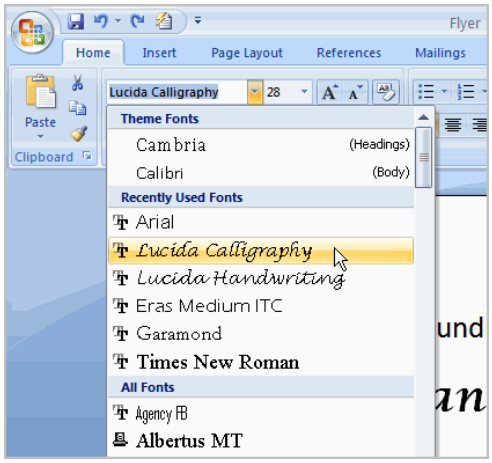


Figure 14: Font types

- Left-click the font style you want to use. The font style will change in the document.

To format font color:

- Select the text you want to modify.
- Left-click the drop-down arrow next to the **font color box** on the Home tab. The font color menu appears.
- Move your cursor over the various font colors. A live preview of the color will appear in the document.

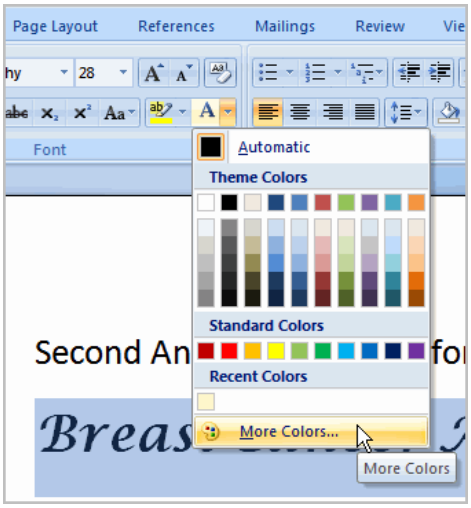


Figure 15: Font color

- Left-click the font color you want to use. The font color will change in the document.

Your color choices aren't limited to the drop-down menu that appears. Select **More Colors** at the bottom of the list to access the Colors dialog box. Choose the color you want, then click OK.

To use the Bold, Italic, and Underline commands:

- Select the text you want to modify.
- Click the Bold, Italic, or Underline command in the **Font group** on the Home tab.

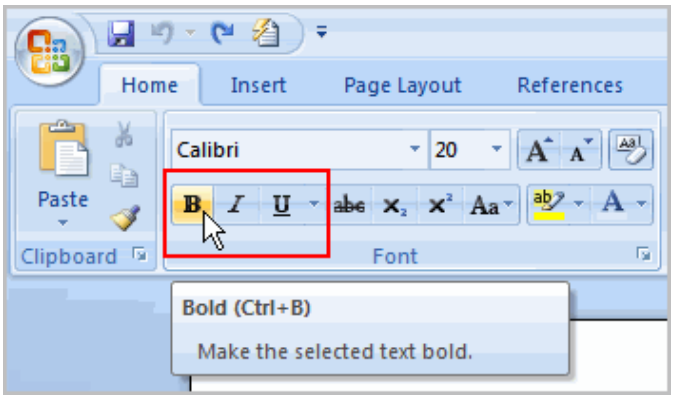


Figure 16: Font group commands

To change the text case:

- Select the text you want to modify.
- Click the **Change Case** command in the **Font** group on the Home tab.
- Select one of the case options from the list.

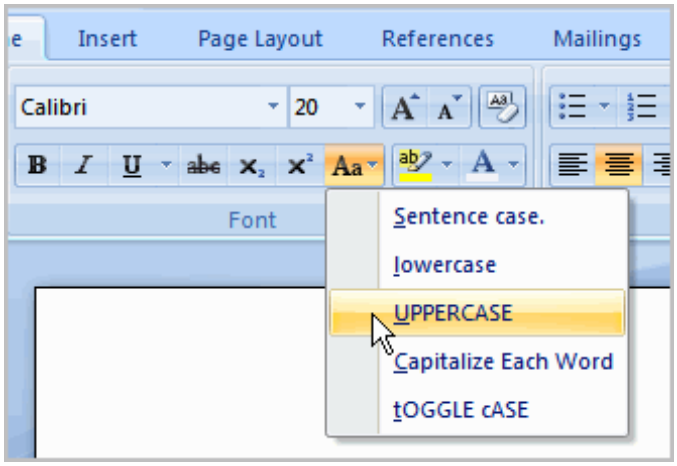


Figure 17: Font case commands

To change text alignment:

- Select the text you want to modify.
- Select one of the four **alignment options** from the Paragraph group on the Home tab.
  - **Align Text Left:** Aligns all of the selected text to the left margin
  - **Center:** Aligns text an equal distance from the left and right margins
  - **Align Text Right:** Aligns all of the selected text to the right margin
  - **Justify:** Aligns text equally to the right and left margins; used in many books, newsletters, and newspapers

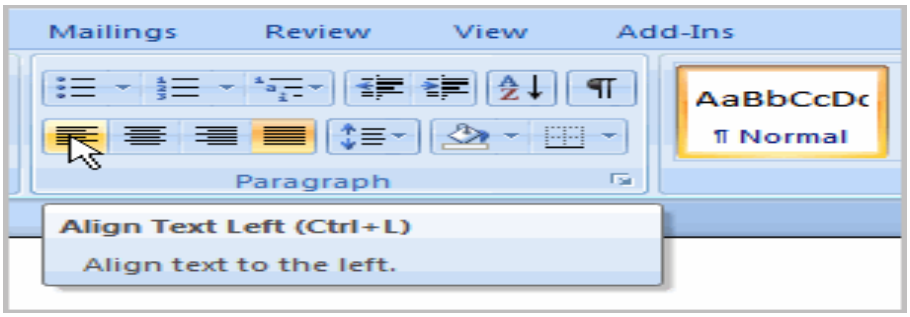
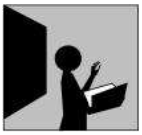


Figure 18: Alignment options



Activity 4

- Create a **new blank** Word document.
- **Insert text** into the document.
- Change the **font size** of some text.
- Change the **font style** of some text.
- Change the **font color** of some text.
- Try various cases using the **Change Case** command.
- Try the four alignment commands.

Using indents and Tabs

A great way to draw attention to specific text is to indent it. There are several ways you can indent text in Word; however, it's important to use these tools appropriately and indent correctly each time. This can save time and make the editing process go smoothly.

To indent using the Tab key:

The most common way to indent is to use the Tab key. This method is best for indenting one line of text rather than multiple lines.

- Place the insertion point to the left of the text you want to indent.
- Press the Tab key. This indents the line 1/2 inch by default.

To use the Indent commands:

Using the Tab key to **indent multiple lines** can make formatting difficult if you add or remove text later. Indenting multiple lines is best done using the **Indent commands**.

- Select the text you want to indent.
- Click the **Increase Indent** command to **increase** the indent. The default is 1/2 inch. You can press the command multiple times.

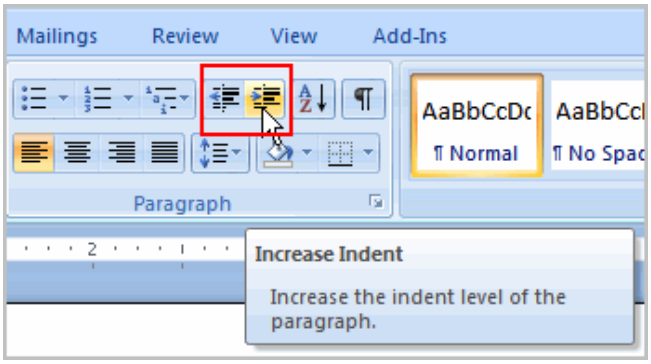


Figure 19: Indent commands

- Click the **Decrease Indent** command to **decrease** the indent.

To modify the default indent settings:

- Select the **Page Layout** tab.
- Select the text you want to modify.
- Use the **arrows** or enter text in the fields to modify the **Left** and **Right Indents**.



Figure 20: Indent settings

The tab selector

The **tab selector** is located above the **vertical ruler** on the left. Hover over the tab selector to see the name of the type of tab that's active.

Tab options include:

- **First Line Indent**: Inserts the indent marker on the ruler, and indents the first line of text in a paragraph
- **Hanging Indent**: Inserts the hanging indent marker, and indents all lines other than the first line
- **Left Tab**: Moves text to the right as you type
- **Center Tab**: Centers text according to the tab
- **Right Tab**: Moves text to the left as you type
- **Decimal Tab**: Aligns decimal numbers using the decimal point
- **Bar Tab**: Draws a vertical line on the document

To set a tab stop to indent the first line of text:

- Click the **tab selector** until the **First Line Indent** icon is visible.
- Left-click at any point on the **horizontal ruler**. The First Line Indent icon will appear.

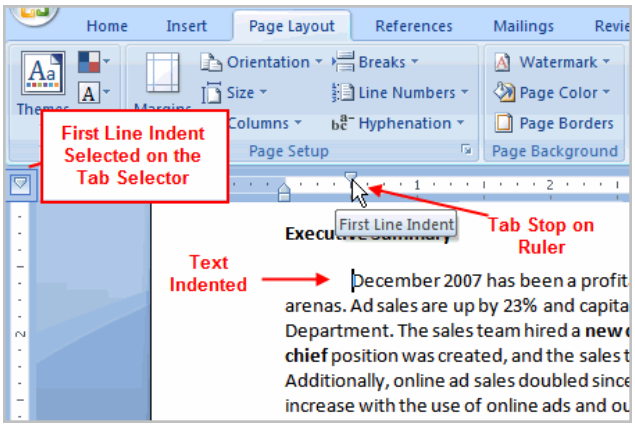


Figure 21: tab selector

- Press the **Enter** key to start a new paragraph, and your insertion point will automatically indent to that point.

To move a tab stop once you have inserted it, left-click and drag the tab stop back and forth on the ruler.

You can set a **hanging indent** the same way; however, this tab stop changes all **other lines** in a paragraph.

To set the left, center, right, and decimal tab stops:

- Click the **tab selector** until the tab stop you want to use appears.

- Left-click the **location on the horizontal ruler** where you want your text to appear.

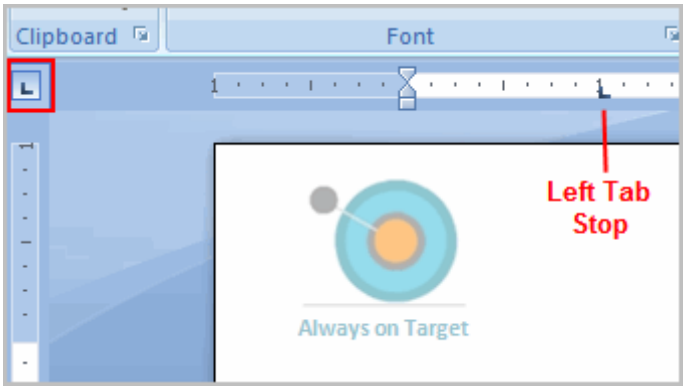


Figure 22: tab stop

- Press the **Tab** key to reach the tab stop.

To use Find and Replace to replace existing text:

- Click the **Replace** command on the Home tab. The Find and Replace dialog box appears.

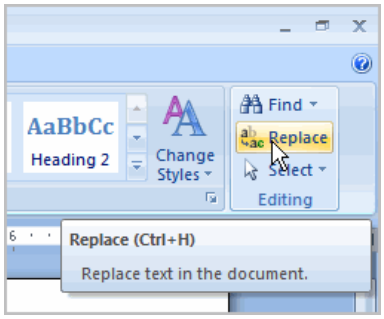


Figure 23: Find and Replace

- Enter text in the **Find** field that you want to locate in your document.
- Enter text in the **Replace** field that will replace the text in the Find box.

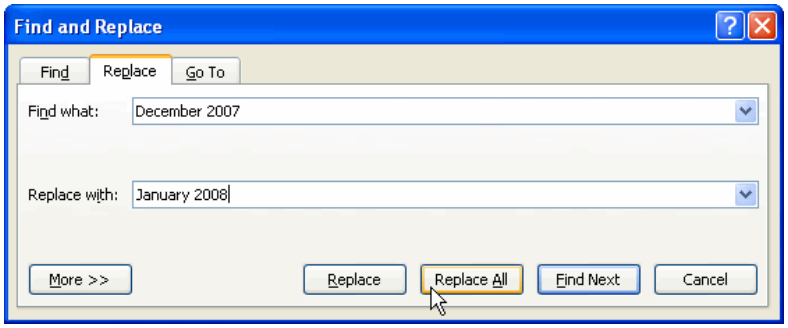
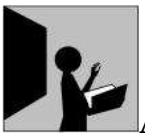


Figure 24: Find and Replace dialog box

- Click OK. The change is made in the document.

You can also use the **Find command** to locate **specific information** in a document. For example, if you are working with a 20-page report, it would be time consuming to search the

document for a specific topic. You can use the Find command to locate **all instances** of a word or phrase in the document. This is a great way to **save time** when working with longer documents.

 **Activity 5**

Use the report or any Word document you choose to complete this activity.

- Practice using the **Tab** key to indent the first line of a paragraph.
- Select some text, and use the **Increase** and **Decrease Indent commands** to see how they change the text.
- Explore the **tab selector** and all of the tab stops. Practice using each one.

2.1.5 Proofing tools



Figure 25: Proofing tools

Worried about making mistakes when you type? Don't be. Word provides you with several **proofing features** that will help you produce professional, error-free documents.

Line colors

By default, Word automatically checks your document for **spelling and grammar** errors. These errors are indicated by **colored wavy lines**.

- The **blue line** indicates a contextual spelling error.
    - A **contextual spelling** error is when an incorrect spelling of a word is chosen. For example, if you write **Deer Mr. Theodore** at the beginning of a letter, **deer** is a contextual spelling error because **dear** should have been used. **Deer** is spelled correctly, but it is used incorrectly in this letter.
- The **red line** indicates a misspelled word.
- The **green line** indicates a grammatical error.

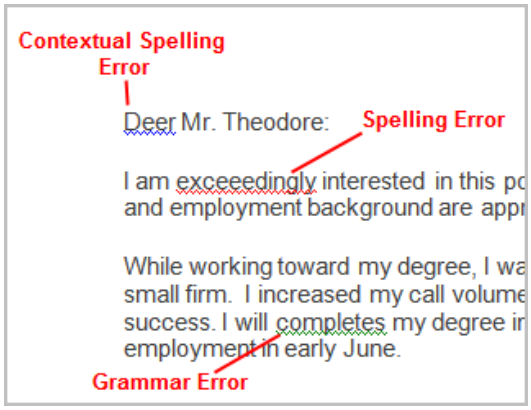


Figure 26: Line colors

To use the spelling check feature:

- Right-click the **underlined** word. A menu will appear.
- Select the **correct spelling** of the word from the **listed suggestions**.
- Left-click your mouse on the word. It will appear in the document.

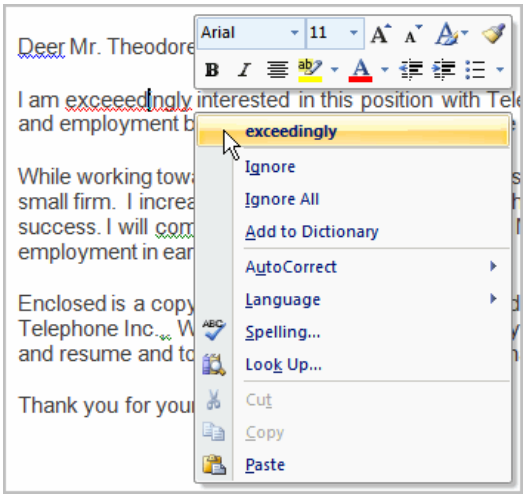


Figure 27: Spelling check feauture

You can choose to **ignore** an underlined word, add it to the **dictionary**, or go to the **Spelling** dialog box.

To use the grammar check feature:

- Right-click the **underlined** word. A menu will appear.
- Select the **correct word** from the **listed suggestions**.
- Left-click your mouse on the word. It will appear in the document.

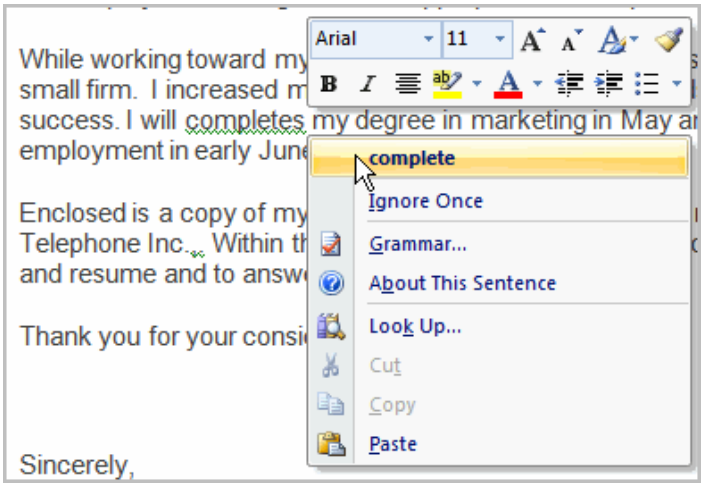
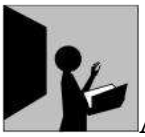


Figure 28: Grammar Check feature

You can also choose to **ignore** an underlined word, go to the **Grammar** dialog box, or find out **more information** about the word and its usage.

You can also wait and run the spelling and grammar check **after completing the document**. Click the **Spelling & Grammar command** on the Review tab.

 **Activity 6**

Use the cover letter or any other Word document you choose to complete this activity.

- Open an **existing** Word document.
- Type the following **sentences** at the beginning of the document:
  - I really enjoy learning an computers and about new skillz.
  - I like to take tutorials were I can learn independently.
- Correct the **spelling errors** in the sentences.
- Correct the **grammar mistake** that appears in one of the sentences.
- Use the **Spelling & Grammar command** to check the remainder of the document.

2.1.6 Printing

Once you complete your document, you may want to **print** it for various reasons. Let us cover the three basic features of printing in Word, including **Print Preview**, **Quick Print**, and **traditional Print**.

To preview the document before printing:

- Click the Microsoft Office button.
- Select **Print** → **Print Preview**. The document opens in Print Preview format.



Figure 29: Printing preview selection window

- Click **Print** to print the document or **Close Print Preview** to exit the preview format and make changes to the document.

In **Print Preview format**, you can perform many tasks, including:

- Modifying margins
- Changing page orientation
- Changing page size
- Zooming in and out to view various parts of the document
- Viewing multiple pages
- Accessing Word Options to change several Word settings

To print:

- Click the Microsoft Office button.
- Select **Print** → **Print**. The Print dialog box appears.
- Select the pages you want to print—either **all pages** or a **range** of pages.
- Select the **number of copies**.
- Check the **Collate** box if you are printing multiple copies of a multi-page document.
- Select a printer from the drop-down list.

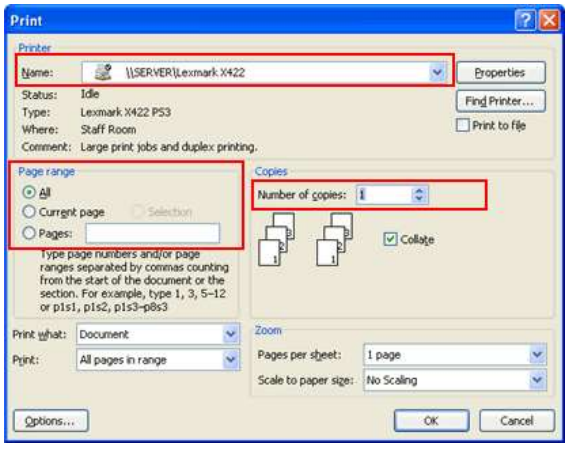


Figure 30: Printing dialog box

Click OK.

To print via Quick Print:

- Click the Microsoft Office button.
- Select **Print** → **Quick Print**.
- The document automatically prints to the default printer.

2.1.7 Document Layout

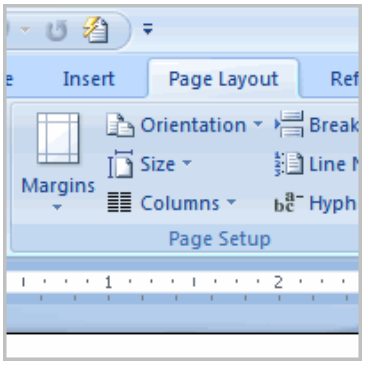


Figure 31: Page setup part of the ribbon

You may find that the **default page layout settings** in Word are not sufficient for the document you want to create, in which case you will want to modify these settings. In addition, you may want to change the **page formatting** depending on the document you're creating.

To change page orientation:

- Select the **Page Layout** tab.
- Click the **Orientation** command in the Page Setup group.

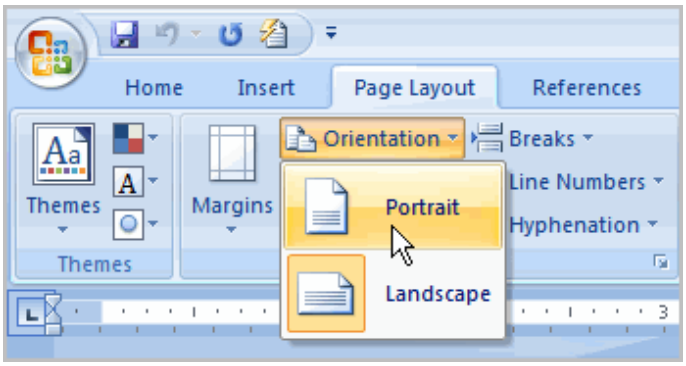


Figure 32: Page orientation options on the ribbon

- Left-click either **Portrait** or **Landscape** to change the page orientation.

Landscape format means everything on the page is oriented **horizontally**, while **portrait** format means everything is oriented **vertically**.

To change the paper size:

- Select the Page Layout tab.
- Left-click the **Size** command, and a drop-down menu will appear. The current paper size is highlighted.

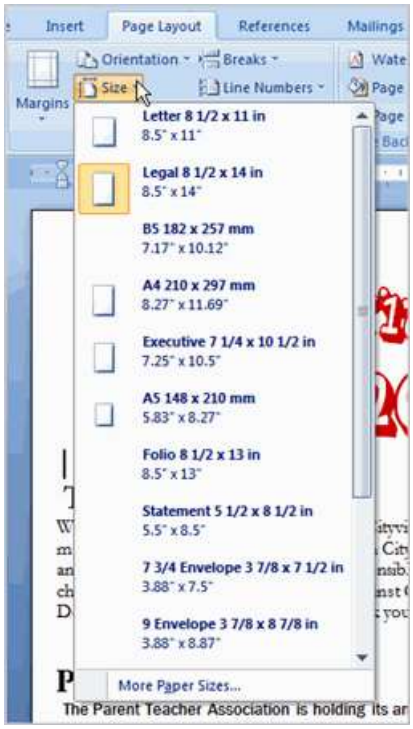


Figure 33: Paper size options on the ribbon

- Left-click a **size option** to select it. The page size of the document changes.

To format page margins:

- Select the **Page Layout** tab.
- Click the **Margins** command. A menu of options appears. **Normal** is selected by default.
- Left-click the predefined margin size you want.

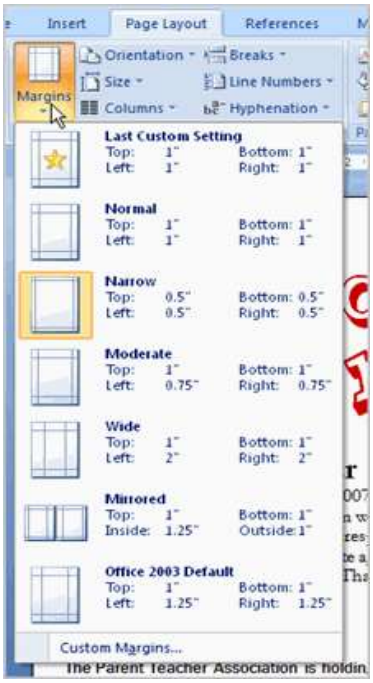


Figure 34: Paper size options using margins

OR

- Select **Custom Margins** from the menu. The Page Setup dialog box appears.

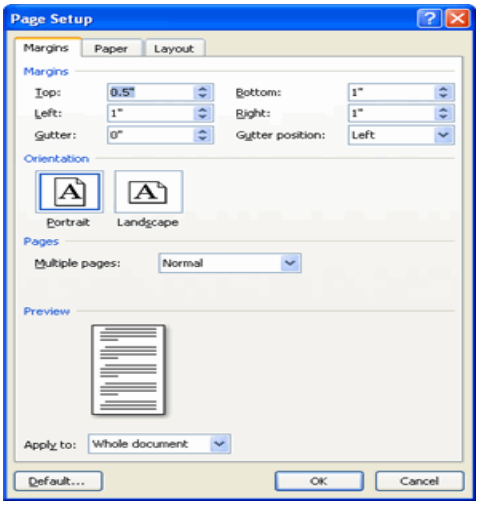


Figure 35: Page set up dialog box

- Enter the desired margin size in the appropriate fields.

You can always access the **Page Setup** dialog box by clicking the small arrow in the bottom-right corner of the Page Setup group. The dialog box should look familiar to people who have used previous versions of Word.

To insert a break:

- Place your **insertion point** where you want the **break** to appear.
- Select the **Page Layout** tab.

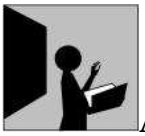
- Click the **Breaks** command. A menu appears.



Figure 36: Page layout tab

- Left-click a **break option** to select it. The break will appear in the document.

Why would you need to use a break? Each **type of break** serves a different purpose and will affect the document in different ways. **Page breaks** move text to a new page before reaching the end of a page, while **section breaks** create a barrier between parts of the document for formatting purposes. **Column breaks** split text in columns at a specific point. Practice using the various break styles to see how they affect the document.

 **Activity 7**

Use the newsletter or any Word document you choose to complete this activity.

- Change the **page orientation**.
- Change the **paper size**.
- Modify **at least** one margin.
- Insert a **section break**.

2.1.8 Tables

A **table** is a grid of cells arranged in **rows** and **columns**. Tables can be customized and are useful for various tasks such as presenting text information and numerical data.

Inserting and modifying tables:

To convert existing text to a table:

- Select the text you want to convert.
- Select the **Insert** tab.
- Click the **Table** command.

- Select **Convert Text to Table** from the menu. A dialog box appears.



Figure 37: Table layout

- Choose one of the options in the **Separate text at:** section. This is how Word knows what text to put in each column.

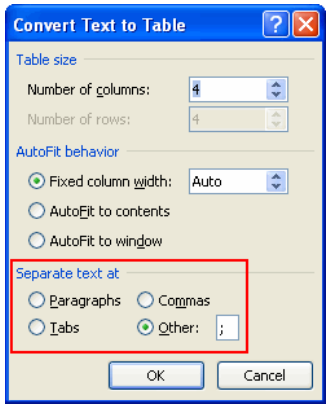


Figure 37: Text to Table dialog box

- Click OK. The text appears in a table.

To add a row above an existing row:

- Place the insertion point in a row below the location where you want to add a row.

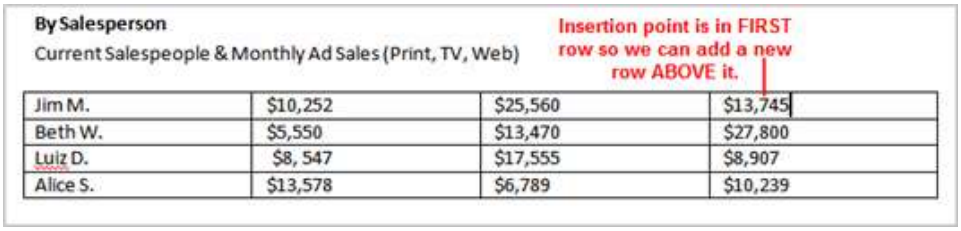


Figure 38: Converting Text to table

- Right-click the mouse. A menu appears.
- Select **Insert → Insert Rows Above**.

A new row appears **above** the insertion point.

You can also add rows below the insertion point. Follow the same steps, but select Insert Rows Below from the menu.

To add a column:

- Place the **insertion point** in a **column adjacent** to the location where you want the new column to appear.
- Right-click the mouse. A menu appears.
- Select **Insert → Insert Columns to the Left** or **Insert Columns to the Right**. A new column appears.

To delete a row or column:

- Select the row or column.
- Right-click your mouse, and a menu appears.
- Select **Delete Columns** or **Delete Rows**.

To apply a table style:

- Select the table. A **Table Tools Design** tab now appears on the Ribbon.
- Select the **Design** tab to access all of the **Table Styles** and **Options**.

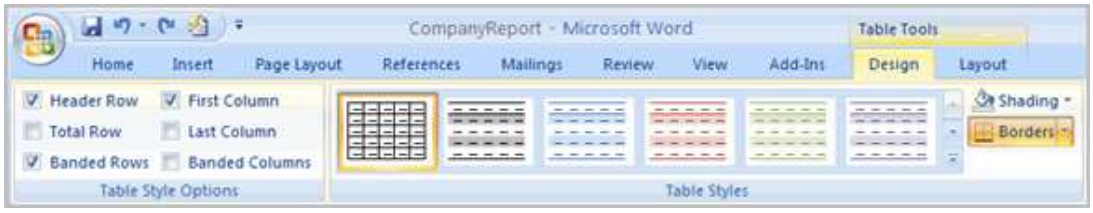


Figure 39: Table tools design tab

- Click through the various styles in the **Table Styles** section.
- Left-click a style to select it. The table style will appear in the document.

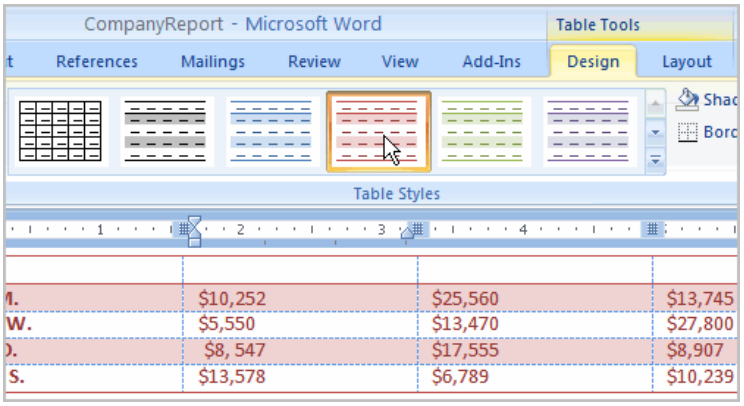


Figure 40: Table styles

You can modify which table styles are displayed. In the **Table Styles Options**, you can select and deselect various table options. For example, you can select banded rows, and only tables with banded rows will appear in the Tables Styles section.

Want to have a little more creative freedom when it comes to formatting your tables? You can manually change the table border or shading, change line weight, or erase part of the table.

To insert a blank table:

- Place your insertion point in the document where you want the table to appear.
- Select the **Insert** tab.
- Click the **Table** command.
- Drag your mouse over the diagram squares to select the number of columns and rows in the table.

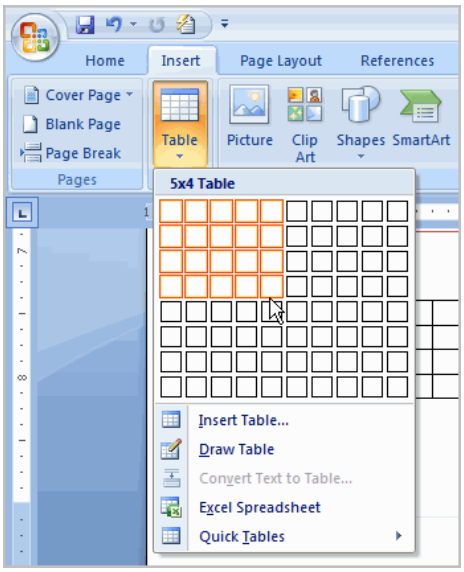


Figure 41: Inserting a blank table

- Left-click your mouse, and the table appears in the document.
- Enter text into the table.

Modifying a table using the layout tab

When you select a table in Word 2007, **Design** and **Layout** tabs appear under **Table Tools** on the Ribbon. Using commands on the Layout tab, you can make a variety of modifications to the table, including:

- Adding and deleting columns
- Adding and deleting rows
- Changing the cell size
- Aligning cell text
- Changing text direction
- Merging and splitting cells

2.1.9 Mail merge

Mail Merge is a useful tool that will allow you to easily produce multiple letters, labels, envelopes, and more using information stored in a list, database, or spreadsheet. In this lesson, you will learn how to use the **Mail Merge Wizard** to create a **data source** and a form **letter**, and explore other wizard features. Additionally, you will learn how to use the Ribbon commands to access Mail Merge tools outside of the wizard.

To use Mail Merge:

- Select the **Mailings** on the Ribbon.
- Select the **Start Mail Merge** command.

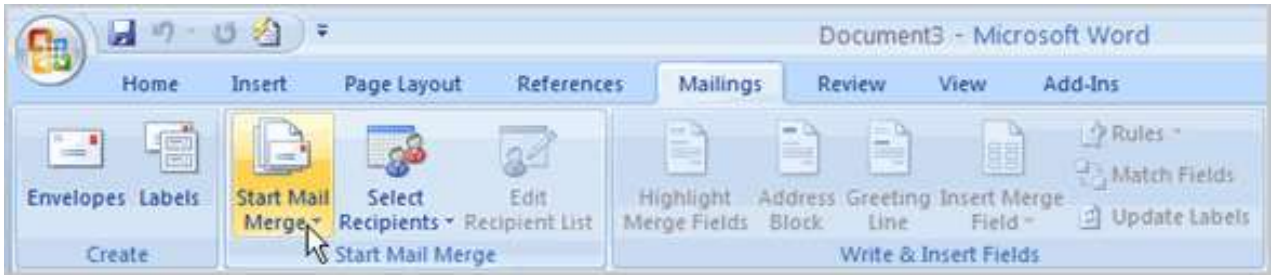


Figure 42: Mail merge tab on the ribbon

Select **Step by Step Mail Merge Wizard**.

The Mail Merge task pane appears and will guide you through the **six main steps** to complete a merge. You will have several decisions to make during the process. The following is an example of how to create a form letter and merge the letter with a data list.

Steps 1-3

- Choose the type of document you want to create. In this example, select **Letters**.
- Click **Next:Starting document** to move to Step 2.
- Select **Use the current document**.
- Click **Next:Select recipients** to move to Step 3.
- Select the **Type a new list** button.
- Click **Create** to create a data source. The **New Address List** dialog box appears.
    - Click **Customize** in the dialog box. The Customize Address List dialog box appears.
    - Select any field you do not need, and click **Delete**.
    - Click **Yes** to confirm that you want to delete the field.
    - Continue to delete any unnecessary fields.
    - Click **Add**. The Add Field dialog box appears.
    - Enter the new field name.
    - Click OK.
    - Continue to add any fields necessary.
    - Click **OK** to close the Customize Address List dialog box.

To customize the new address list:

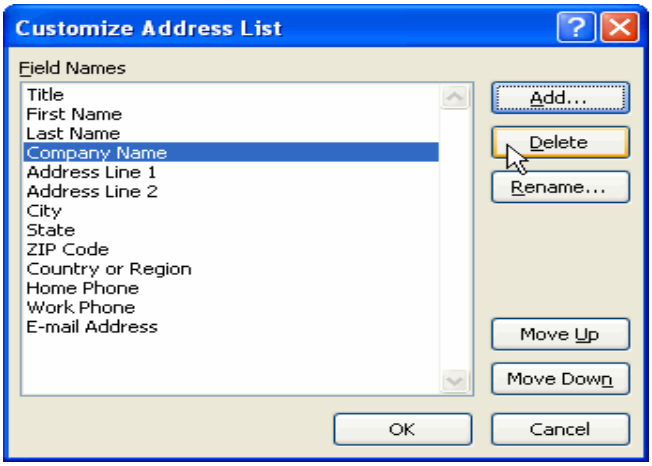


Figure 43: Mail merge address list customisation

- Enter the necessary data in the New Address List dialog box.
- Click **New Entry** to enter another record.
- Click **Close** when you have entered all of your data records.
- Enter the file name you want to save the data list as.
- Choose the location where you want to save the file.
- Click **Save**. The Mail Merge Recipients dialog box appears and displays all of the data records in the list.
- Confirm that the data list is correct, and click **OK**.
- Click **Next:Write your letter** to move to Step 4.

Steps 4-6

- Write a letter in the current Word document, or use an open existing document.

To insert recipient data from the list:

- Place the insertion point in the document where you want the information to appear.
- Select Address block, Greeting line, or Electronic postage from the task pane. A dialog box with options will appear based on your selection.

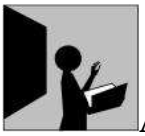
OR

- Select More Items. The Insert Merge Field dialog box will appear.
- Select the field you want to insert in the document.
- Click **Insert**. Notice that a placeholder appears where information from the data record will eventually appear.
- Repeat these steps each time you need to enter information from your data record.

- Click **Next: Preview your letters** in the task pane once you have completed your letter.

- Preview the letters to make sure the information from the data record appears correctly in the letter.
- Click **Next: Complete the merge**.
- Click **Print** to print the letters.
- Click **All**.
- Click **OK** in the Merge to Printer dialog box.
- Click **OK** to send the letters to the printer.

The Mail Merge Wizard allows you to complete the merge process in a variety of ways. The **best** way to learn how to use the different functions in Mail Merge is to try to develop several of the different documents—letters, labels, and envelopes—using the different types of data sources.

 **Activity 8**

Use the report or any Word document you choose to complete this activity.

- Open a **new blank Word document**.
- Open the **Mail Merge** task pane.
- Create a **data list**, and practice adding and removing fields.
- Explore the different **Mail Merge features** until you are familiar with them.

2.1.10 Graphics

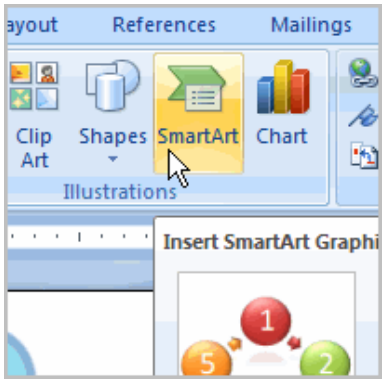


Figure 44: SmartArt

SmartArt allows you to **visually communicate information** rather than simply using text. Illustrations can make an impact in your document, and SmartArt makes using graphics especially easy.

SmartArt graphics

To insert a SmartArt illustration:

- Place the insertion point in the document where you want the graphic to appear.
- Select the **Insert** tab.
- Select the **SmartArt** command in the Illustrations group. A dialog box appears.

- Select a **category** on the left of the dialog box, and review the SmartArt graphics that appear in the **center**.
- Left-click a graphic to select it.

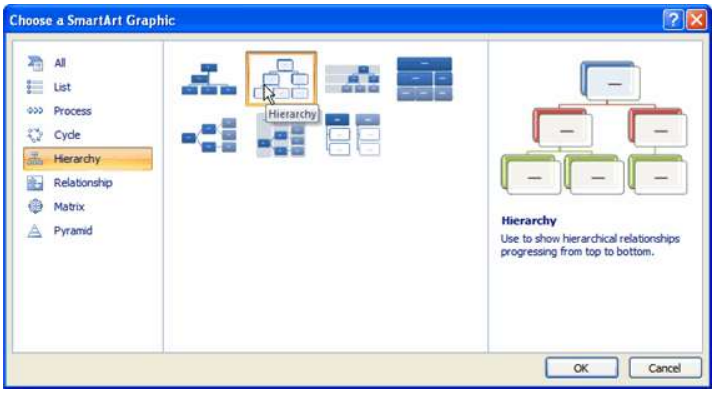


Figure 45: SmartArt illustration

- Click OK.

To see more details about a graphic, left-click any image, and a larger version of the graphic and **additional text details** will appear on the **right side** of the dialog box.

To add text to a SmartArt graphic:

- Select the graphic. The **first text box** is selected. If the task pane on the left of the graphic is visible, the **insertion point** appears in it. If the task pane is not visible, click the arrow to open the task pane.
- Enter text into the task pane fields. The information will appear in the graphic.

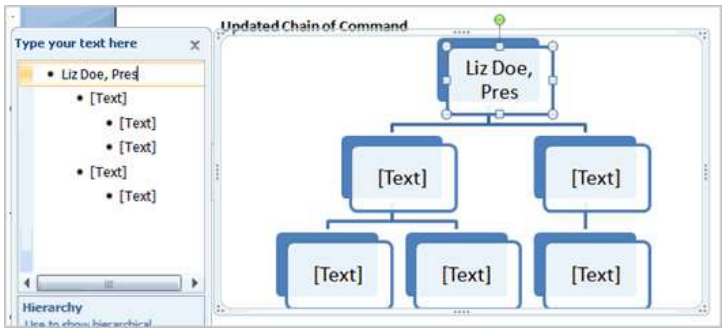


Figure 46: Adding text to a SmartArt illustration

OR

- Click **X** to close the task pane.
- Enter text into the **first text box** in the graphic.

- Continue to enter text in the text box graphics.

Notice that the text you enter **automatically** resizes to fit inside the box.

To add a shape to a graphic:

- Select the graphic. The **SmartArt Tools Design and Format tabs** appear on the Ribbon.
- Select the **Design** tab.
- Click the **Add Shape** command in the Graphics group.
- Decide where you want the **new shape** to appear, and select one of the shapes nearby the desired location.
- Select **Add Shape Before** or **Add Shape After**. If you wanted to add a superior or a subordinate, you could select the **Above** or **Below** options.

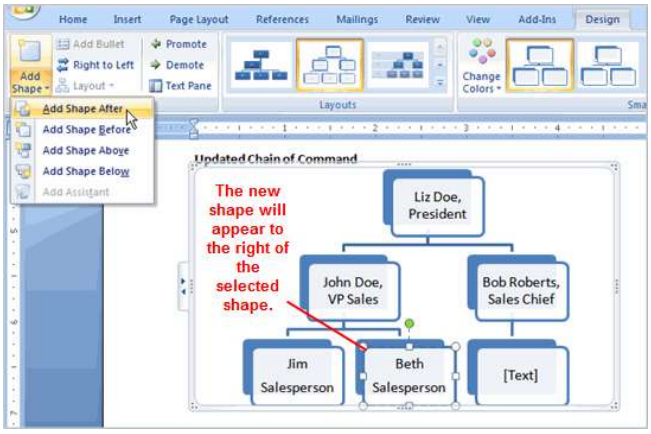


Figure 47: Adding shape to a SmartArt illustration

To move shapes using the Promote or Demote commands:

- Select the graphic. The SmartArt Tools Design and Format tabs appear on the Ribbon.
- Select the **Design** tab.
- Select the **shape** you want to move.
- Click the **Promote** or **Demote** command in the Create Graphic group.

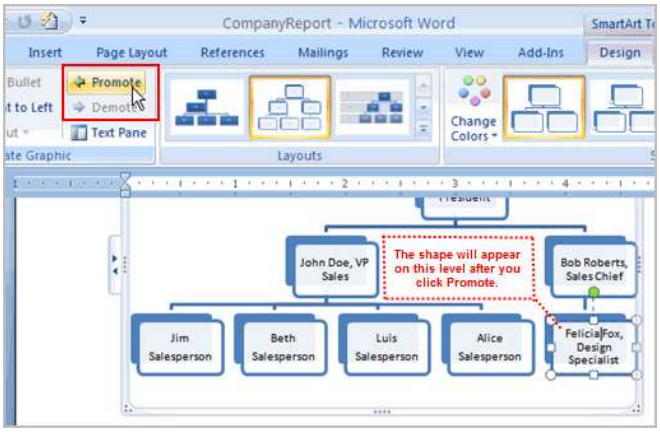


Figure 48: Promote and Demote SmartArt commands

The **Promote** and **Demote** commands allow you to move shapes and customize the graphic instead of having to use the predefined default illustration.

To change the graphic style:

- Select the graphic. The SmartArt Tools Design and Format tabs appear on the Ribbon.
- Select the Design tab.
- **Left-click a style** to select it.

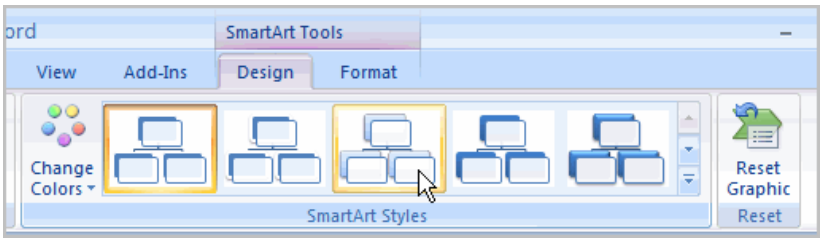


Figure 49: changing SmartArt Style

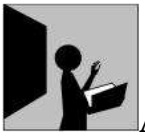
Click the **More** arrow to see all of the style options. Hover over each to display a live preview of the style in your document.

To change the color scheme:

- Select the graphic. The SmartArt Tools Design and Format tabs appear on the Ribbon.
- Select the Design tab.
- Click the **Change Color** command. A menu of color schemes appears.
- Left-click an option to select it.

Explore the options in the **Layout** group if you want to switch to another graphic but keep the existing text. You also can select the **Format** tab to access additional formatting options, including fill, text, and line color.

Some of the options will differ from graphic to graphic. Look carefully at the SmartArt Tools tabs, and explore the active commands.

 **Activity 9**

Use the report or any Word document you choose to complete this activity.

- Insert a **SmartArt illustration** into a document.
- Enter **text** into the graphic.
- Modify the **style.**

- Modify the **graphic** as needed.

Pictures

Pictures can be added to Word documents and then formatted in various ways. The **picture tools** in Word 2007 make it easy to incorporate images into your documents and **modify these images** in innovative ways.

To insert a picture:

- Place your **insertion point** where you want the image to appear.
- Select the **Insert** tab.
- Click the **Picture** command in the **Illustrations** group. The Insert Picture dialog box appears.

- Select the image file on your computer.



Figure 50: insert picture dialog box

- Click **Insert**, and it will appear in your document.

Left-click a **corner sizing handle**, and while holding down the mouse button resize the image. The image retains its proportions.

To wrap text around an image:

- Select the image.
- Select the **Picture Tools Format** tab.
- Click the **Text Wrapping** command in the Arrange group.

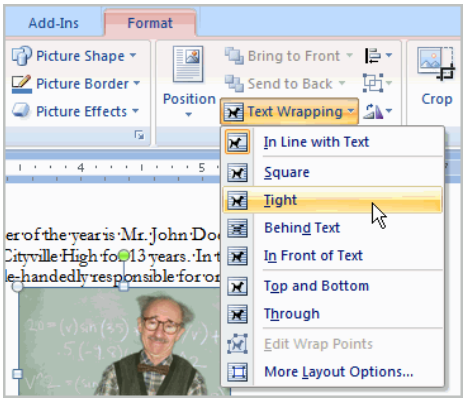


Figure 51: Wrap text on n image

- Left-click a menu option to select it. In this example, we selected **Tight**.
- Move the image around to see how the text wraps for each setting.
- The **Position** button has predefined text wrapping settings. The Position button is located to the left of the Text Wrapping button.
- Click the Position button, and a drop-down list of text wrapping options will appear. Hover over an option to see what it does.

If you can't get your text to wrap the way you want, click the **Text Wrapping** command and select **More Layout Options** from the menu. You can make more precise changes in the Advanced Layout dialog box that appears.

To crop an image:

- Select the image.
- Select the **Format** tab.
- Click the **Crop** command. The black cropping handles appear.

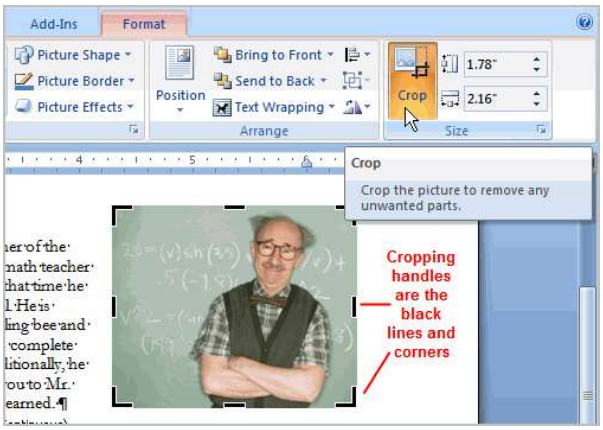


Figure 52: cropping an image

- Left-click and move a **handle** to crop an image.
- Click the **Crop** command to deselect the crop tool.

Corner handles will crop the image proportionally.

To compress a picture:

- Select the picture.
- Select the **Format** tab.
- Click the **Compress Pictures** command in the Adjust group. A dialog box appears.
- Click the **Options** button to access the **Compression Setting** dialog box.

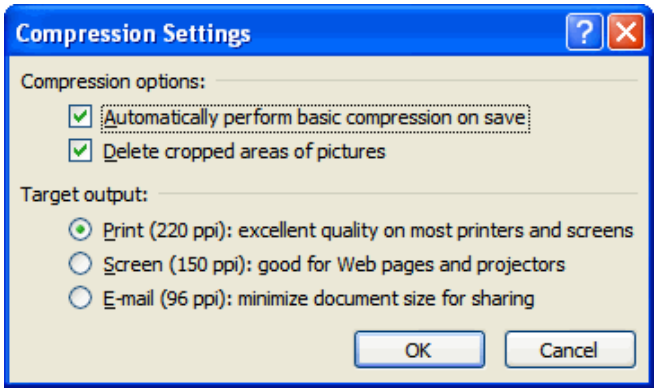


Figure 53: compressing picture options

- Choose the target output.
- Change any of the default picture settings you want.
- Click **OK** in the Compression Settings dialog box.
- Click **OK** in the Compress Pictures dialog box.

You need to monitor the **file size** of your images and documents that include images, especially if you send them via email. Cropping and resizing an image doesn't decrease the image file size, but compression does.

Other picture tools

There are many other things you can do to modify a picture. From the Format tab, some of the other useful commands include:

- **Change Picture**: Selects a new picture from your computer
- **Reset Picture**: Reverts to original image
- **Brightness**: Adjusts the brightness of an image
- **Color**: Adjusts the contrast of an image from light to dark
- **Recolor**: Modifies the color in a variety of ways, including black and white, sepia, pink, and purple

To change the shape of a picture:

- Select the picture.
- Select the **Format** tab.
- Click the **Picture Shape** icon. A menu appears.

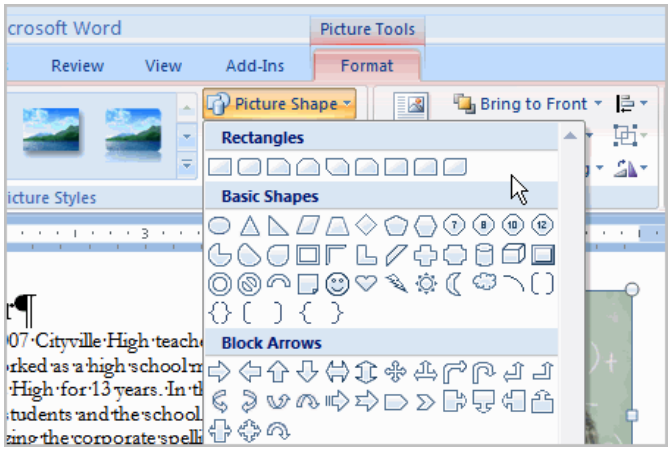


Figure 54: picture shapes options

- Left-click a shape to select it.

To apply a picture style:

- Select the picture.
- Select the **Format** tab.
- Click the **More** drop-down arrow to display all of the picture styles.
- Hover over a picture style to display a live preview of the style in the document.

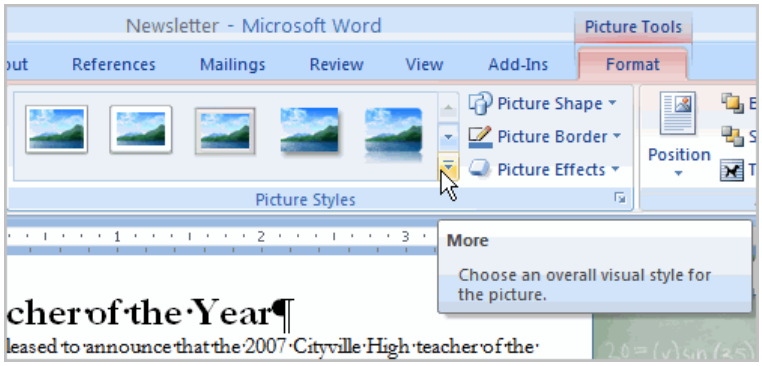


Figure 55: picture style options

- Left-click a style to select it.

To add a border to a picture manually:

- Select the picture.
- Select the **Format** tab.
- Left-click the **Picture Border** command, and select a color.

OR

- Select **Weight** from the menu and choose a line weight.

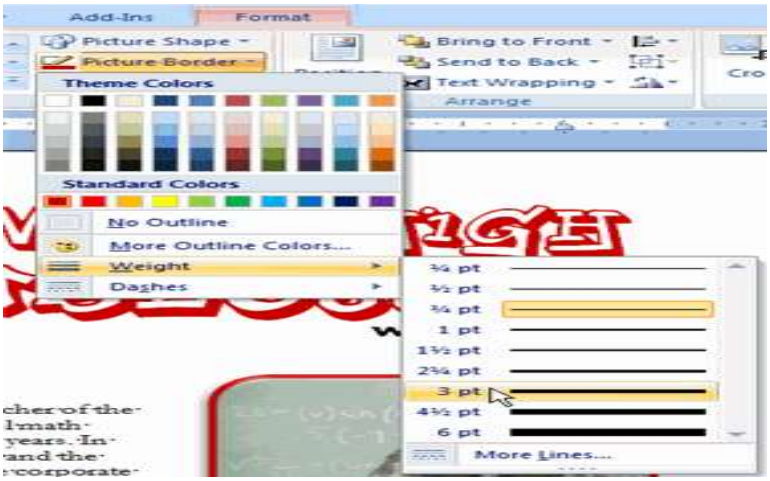
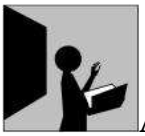


Figure 56: add a border to a picture

 **Activity 10**

Use the newsletter or any Word document you choose to complete this activity.

- **Insert** a picture into a document.
- **Crop** the image proportionally.
- Apply a **picture style** to the image.
- **Compress** the picture.



Mzuzu University

Word Self-Assessment Test

Instructions

- Answer all questions

Load Ms Word program and key in the text below that starts with ***Picture and Images****. Include all the spelling mistakes. After you have typed the text, perform the* ***Task A.***

Pictures and Images

Pictures give an immediate impression to a room, establishing the look and mood as much as the furnishings and furniture. The use of space, the arrangement pattern and the placement of pictures with refernc to other contents all exert a powerful influence on the overall character.

At the outset, it is recommended that you should decide what image you wish to convey – calm or busy, warm or cool, minimalist or luxurious. Consider areas of focus such as the space over a chimney breast, over a bed or over a sofa. Before making final decisions, and holes in the walls, take the opportunity of placing the pictures around the room so that you can see how they look, and you can also assess the effct on the pictures of light, adjacent fixtures and furnishings.

Some people like to group small pictures together, either following a theme or completely at random, and this approach can be very effectve, especially for alcoves and staircases.

TASK A

1. Type your Registration Number at the bottom of your document. **1 Mark**
2. Insert a page number in the Header section of the document. **1 Mark**
3. Insert the text © **Mzuzu University** in Footer section of the document. **3 Marks**
4. Save the file in the folder you have created using filename **Pictures**. **1 Mark**
5. Use a spell check facility to check your work for spelling errors. **1 Mark**
6. Move the last paragraph and paste it between the first and second paragraph so that it becomes the second paragraph. **1 Mark**
7. Fully justify your document. **1 Mark**
8. Using thesaurus, find the synonym for the word luxurious in the document and replace it with any suitable word. **2 Marks**
9. Centre, embolden and underline the heading (Use double underline style). **3 Marks**

10. Perform the following to the first paragraph only:
    (a) 1.5 line spacing.
    (b) Green font colour.
    (c) Emboss effect. **3 Marks**
11. Insert any clipart at the end of the document. **2 Marks**
12. Change the Top and Bottom margins to 3 cm **1 Mark**
13. Change the Left and Right margins to 2.5 cm **1 Mark**
14. Save the file in the folder you have created using filename **Zithunzi**.

Note: You should have two documents saved in the folder you have created: The first file called **Pictures** that you saved in step 4 and the second file called **Zithunzi** that you saved in step 14.

TASK B

Reproduce the text and graphics as shown on the attached page. The font colour for the text **Smoking is Bad for You** should be Red and the font size should be 36. You will find the required graphics in My Pictures. Change the page orientation to Landscape.

10 Marks

Unit Summary

In this unit we have learnt more about word processor – The Microsoft Word. We have seen how a document in word processor is created and how text can be added and formatted. The ribbon and its tools have been explored in his unit i.e. copy, cut and paste. We also have seen how you can use proofing tools to ensure your work is free of spelling and grammatical errors. The layout feature of the document has been covered as well. We have learnt also in this unit the use of a mail merge tool as well as creating, formatting and working with Tables.

References

i. https://support.office.com/en-us/article/word-for-windows-training-7bcd85e6-2c3d-4c3c-a2a5-5ed8847eae73
ii. https://www.javatpoint.com/ms-word-tutorial
iii. https://www.lynda.com/Word-training-tutorials/326-0.html
iv. https://edu.gcfglobal.org/en/word2007/
v. https://www.techonthenet.com/word/index.php

UNIT 3

THE INTERNET

Introduction

The Internet is a network made up of hundreds of thousands of computers all over the world, with millions of users connected electronically by communication lines (telephone lines, fibre-optic cables and other links). These computers can share and exchange information. The Internet is a network of networks.

The Internet is the largest computer network in the world, connecting millions of computers. A network is a group of two or more computer systems linked together.

The Internet is the hardware – the computers and the connections between them – and the software that let them communicate with each other. It is the base on which the Web is built.

It's no secret that more and more the Internet is becoming an integral part of our everyday lives. But if you are new to the online experience, it may be a bit overwhelming.

In the early days, most people just used the internet to search for information. Today's Internet is a constantly evolving tool that not only contains an amazing variety of information, but also provides new ways of accessing, interacting and connecting with people and content.

The main characteristic of Internet and the World Wide Web are:

(a) Users across the world can connect or access information irrespective of time and space factor.

(b) Point to point communication, rather than one to many broadcast communication.

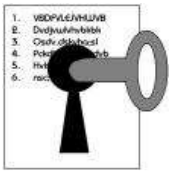
(c) Provides access to large number of databases.

(d) The facility of hyper linking from one server to another by clicking on a highlighted word which enables the user to directly switch to another data source, on the other side of the world.

(e) Instant and interactive community creation on a global base.

Areas of emphasis:

- Web Browsers
- The World Wide Web
- Searching Information
- Netiquette
- Electronic Mail



Key Words

- Internet, WWW, Netiquette, browser, ISP, URL, Email, CC, BCC, Attachment, et al.

Prerequisite knowledge

Ability to use the keyboard and type correctly as well as understanding the applications existing in Windows for accessing and browsing the Internet.



Learning outcomes

By the end of this unit, you should be able to:

- Understand the role of Internet as a source of information
- Able to browse the Internet using different we browsers
- Be able to use the Internet to search information
- the expected behaviour when using the Internet
- understand how to use the electronic mail

Resources needed

- a working computer with an Internet connection.

3.1.0 Getting started with the Internet

Once you've set up your computer, you may want to purchase **home Internet access** so you can send and receive email, browse the Web, stream videos, and more. You may even want to set up a **home wireless network**, commonly known as **Wi-Fi**, so you can connect multiple devices to the Internet at the same time. To achieve this you will need an Internet Service Provider to help you access the internet.

Types of Internet service

The type of Internet service you choose will largely depend on which **Internet service providers** (ISPs) serve your area, along with the types of service they offer. Here are some common types of Internet service.

- **Dial-up**: This is generally the slowest type of Internet connection, and you should probably avoid it unless it is the only service available in your area. Dial-up Internet uses your **phone line**, so unless you have multiple phone lines you will not be able to use your landline and the Internet at the same time.
- **DSL**: DSL service uses a **broadband connection**, which makes it much faster than dial-up. DSL connects to the Internet **via a phone line** but does not require you to have a landline at home. And unlike dial-up, you'll be able to use the Internet and your phone line at the same time.

- **Cable**: Cable service connects to the Internet **via cable TV**, although you do not necessarily need to have cable TV in order to get it. It uses a broadband connection and can be faster than both dial-up and DSL service; however, it is only available where cable TV is available.
- **Satellite**: A satellite connection uses broadband but does not require cable or phone lines; it connects to the Internet **through satellites orbiting the Earth**. As a result, it can be used almost anywhere in the world, but the connection may be affected by weather patterns. Satellite connections are also usually slower than DSL or cable.
- **3G and 4G**: 3G and 4G service is most commonly used with mobile phones, and it connects **wirelessly** through your ISP's network. However, these types of connections aren't always as fast as DSL or cable. They will also **limit the amount of data** you can use each month, which isn't the case with most broadband plans.

3.1.1 Internet Service Provider

Now that you know about the different types of Internet service, you can do some research to find out what ISPs are available in your area. If you're having trouble getting started, we recommend talking to friends, family members, and neighbors about the ISPs they use. This will usually give you a good idea of the types of Internet service available in your area.

Most ISPs offer several tiers of service with different Internet speeds, usually measured in **Mbps** (short for **megabits per second**). If you mainly want to use the Internet for **email** and **social networking**, a slower connection (around 2 to 5 Mbps) might be all you need. However, if you want to **download music** or **stream videos**, you'll want a faster connection (at least 5 Mbps or higher).

You'll also want to **consider the cost** of the service, including installation charges and monthly fees. Generally speaking, the faster the connection, the more expensive it will be per month.

Although **dial-up** has traditionally been the **least expensive** option, many ISPs have raised dial-up prices to be the **same as broadband**. This is intended to encourage people to switch to broadband. We do not recommend dial-up Internet unless it's your only option.

Hardware needed: *Modem*

Once you have your computer, you really don't need much additional hardware to connect to the Internet. The primary piece of hardware you need is a **modem**.

The type of Internet access you choose will determine the type of modem you need. **Dial-up** access uses a **telephone modem**, **DSL** service uses a **DSL modem**, **cable** access uses a **cable modem**, and **satellite** service uses a **satellite adapter**. Your ISP may give you a modem—often for a fee—when you sign a contract, which helps ensure that you have the **right type** of modem. However, if you would prefer to shop for a **better** or **less expensive** modem, you can choose to buy one separately. Basically, a modem is used for transmitting and receiving data over a communication channel, such as twisted-pair telephone lines, coaxial cables, and optical fibres. Currently the purpose of a modem is to convert a computer's data stream to analog format so that it can be transmitted over the analog telephone line.

Router



Figure 1 : Router

A **router** is a hardware device that allows you to connect **several computers** and **other devices** to a single Internet connection, which is known as a **home network**. Many routers are **wireless**, which allows you to create a **home wireless network**, commonly known as a **Wi-Fi network**.

You **don't necessarily need to buy a router** to connect to the Internet. It's possible to connect your computer directly to your modem using an Ethernet cable. Also, many modems include a **built-in router**, so you have the option of creating a Wi-Fi network without buying extra hardware.

3.1.2 Use of Internet

Internet is the transport vehicle for the information stored in files or documents on another computer. It carries together various information and services, such as electronic mail, online chat, file transfer, and the interlinked Web pages and other documents of the World Wide Web. The Internet itself does not contain information, it is a slight misstatement to say a "document was found on the Internet." It would be more correct to say it was found through or using the Internet. What it was found in (or on) is one of the computers linked to the Internet.

Every aspect of our day to day life is affected by the internet. Whether it is shopping, business, banking, communication, paying your bills, social gathering, party, learning, education etc. Internet is everywhere, knocking at our door, making our life easier and smooth. Moreover, when it comes to education and research internet is paving way for a great leap and sure library and information centers has no exception. The internet made the information on our finger tips. The libraries of the developed world has adopted the internet facilities to provide the fast and better library services to its patron but this is not the case with many developing nations and third world countries.

3.1.3 Web Browsers and their features

A browser is a computer program that resides on your computer enabling you to use the computer to view WWW (World Wide Web) documents and access the Internet taking advantage of text formatting, hypertext links, images, sounds, motion, blogs, and other

features. Firefox and Internet Explorer are among the leading "graphical browsers" in the world (meaning they facilitate the viewing of graphics such as images and video and more).

The main difference between web browsers is a thing called a layout engine. The browser interprets these instructions using its layout engine. Chrome & Safari use a layout engine called webkit (& Opera is moving to it too), Firefox's is called Gecko, IE's is called Trident.

The other way in which browsers differ is in the convenience features they offer for navigating and managing the Web and all the URLs (Uniform Resource Locators) you may want to keep track of. Netscape and Internet Explorer both offer the ability to e-mail documents, download them, print them, and keep track of where you've been and sites you want to "*bookmark*" or "*Add to Favorites*"

Browsers all rely on "plug-ins" to handle the fancier files you find on the Web. Plug-ins are sub-programs stored within a browser or elsewhere in your computer especially to support special types of files you may click on. If you click on a link, and your computer does not currently have the plug-in needed for the file you clicked on, you are usually prompted with an opportunity to get the plug-in. Most plug-ins are free, and easy and safe to install.

URLs and the address bar

URLs are the method by which all resources on the Web are addressed. Web resources are usually (but not necessarily) computer files, such as Web page documents. The URL specifies exactly where a resource can be found. All resources have unique URLs. The URL contains the following information:

- The type of service that should be used to get the resource.
- The name and address of the host computer storing the resource – this sometimes includes a country code.
- The port number used by the service we want on the host computer (usually omitted).
- The location (i.e. the directory path) and name of the resource. This information is presented in the following way:
- Type_of_service://host_name_and_address:port_number/directory_path/resource_name

Example:

Consider the following URL: http://www.mzuni.ac.mw/news/news.html

- The above URL begins with the letters ***http***. This tells us that we need to use the Hypertext Transfer Protocol (HTTP) to access this resource. Hypertext Markup Language (HTML) documents are usually accessed using this protocol, and HTML is a language in which Web pages are created.
- A colon and two forward slashes (://) separate the type of service used to retrieve the resource from the rest of the URL.
- The name and address of the host computer is www.mzuni.ac.mw, where www is the name of a World Wide Web server at mzuni (Mzuzu University), which is an academic institution in Malawi. This part of the URL is called the domain name.

- The port umber has been omitted, so a default port number will be assumed. The default port number for an HTTP server is 80, but this is rarely required.
- The next forward slash separates the domain name and port number (if included) from the resource name.
- The resource name may include a directory path specifying its exact location on the host computer. In the above URL, the resource is a Web document news.html, which is stored in the news directory. Hence the full path is news/news.html

Note:

- The last part of the domain name describes the type of domain. The convention is to have .ac or .edu for academic or education, .co or .com for commercial, .org for non-profit organizations, and so on. Others include .net for network providers, and .gov for government bodies.
- There are around 250 country codes that can be included in a domain name (such as .uk for United Kingdom, .mw for Malawi, .za for South Africa). As the Internet originated in the U.S.A., the county code for the U.S.A. (.us) is rarely used.
- The Web page document addressed in the above URL has the extension .html. Most Web pages created using HTML end with either .htm or .html.
- Domain names are used to identify one or more IP addresses. For example, the domain name microsoft.com represents about a dozen IP addresses. Domain names are used in URLs to identify particular Web pages. For example, in the URL http://www.mzuni.ac.mw/index.html, the domain name is mzuni.ac.mw

Using Uniform Resource Locators (URLs)

A URL indicates where the web page is stored on the Internet. URLs almost never use back slashes (\). All slashes are forward slashes (/). You need to type a URL exactly for your browser to locate the desired web page. Although URLs may contain spaces between characters, they usually do not.

Links

Whenever you see a word or phrase on a website that's **blue** or **underlined in blue**, it's probably a **hyperlink**, or **link** for short. You might already know how links work, even if you've never thought about them much before. Links are used to **navigate the Web**. When you click a link, it will usually take you to a different webpage. You may also notice that your cursor changes into a **hand icon** whenever you hover over a link.

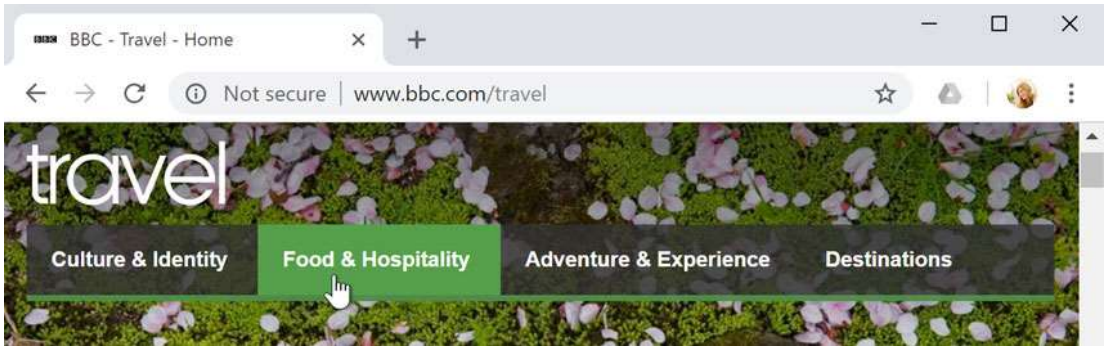


Figure 2: icons on web pages

If you see this icon, it means you've found a link. You'll find other types of links this way too. For example, many websites actually use **images** as links, so you can just **click the image** to navigate to another page.

Navigation buttons

The **Back** and **Forward** buttons allow you to move through websites you've **recently viewed**. You can also click and hold either button to see your recent history.



Figure 3 : navigation buttons on web pages

The **Refresh** button will **reload** the current page. If a website stops working, try using the Refresh button.



Figure 4: refresh/reload button on web pages

Tabbed browsing

Many browsers allow you to open links in a new **tab**. You can open as many links as you want, and they'll stay in the **same browser window** instead of cluttering your screen with multiple windows.

To open a link in a new tab, **right-click** the link and select **Open link in new tab** (the exact wording may vary from browser to browser).

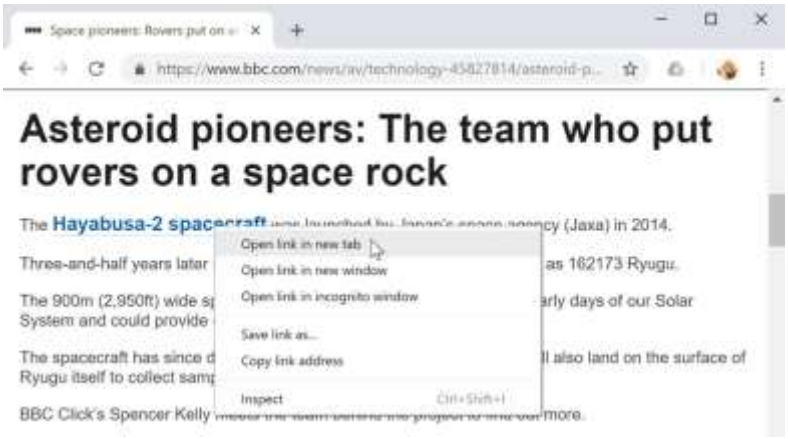


Figure 5: new tabs on web pages

*To **close** a tab, click the **X**.*



Figure 6: closing a tab on web pages

*To create a **new blank tab**, click the button to the right of any open tabs.*



Figure 7: new blank tabs on web pages

Bookmarks and history

If you find a website you want to view later, it can be hard to memorize the exact web address. **Bookmarks**, also known as **favorites**, are a great way to save and organize specific websites so you can revisit them again and again. Simply locate and select the **Star** icon to bookmark the current website.

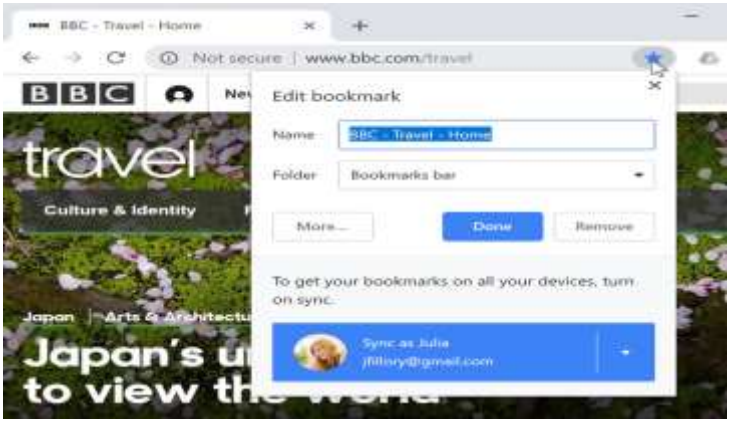


Figure 8: bookmarks and history

Your browser will also keep a history of every site you visit. This is another good way to find a site you visited previously. To view your history, open your browser settings—usually by clicking the icon in the upper-right corner—and select **History**.

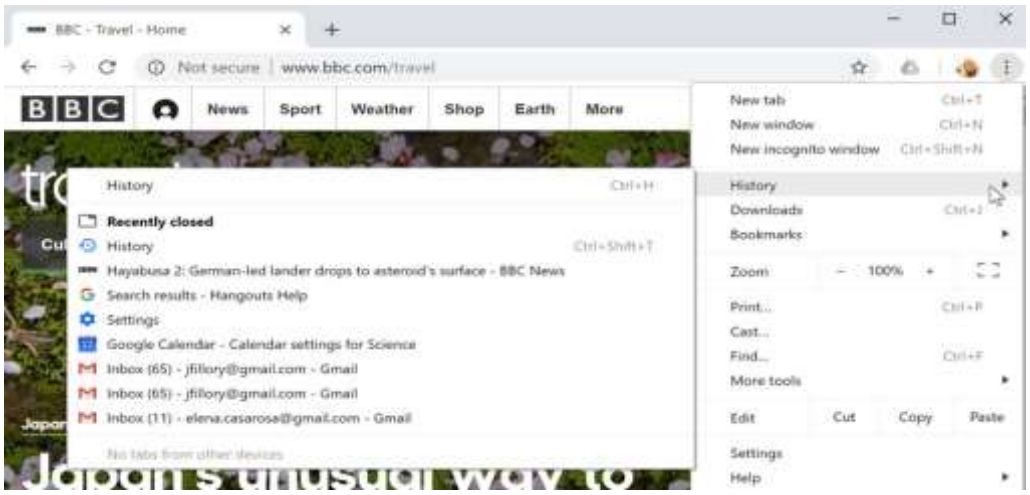


Figure 9: browsing history on web pages

Downloading files

Links don't always go to another website. In some cases, they point to a **file** that can be **downloaded**, or saved, to your computer.

If you click a link to a file, it may download automatically, but sometimes it just **opens within your browser** instead of downloading. To prevent it from opening in the browser, you can **right-click** the link and select **Save link as** (different browsers may use slightly different wording, like **Save target as**).



Figure 10: saving links

Saving images

Sometimes you may want to save an image from a website to your computer. To do this, right-click the image and select **Save image as** (or **Save picture as**).

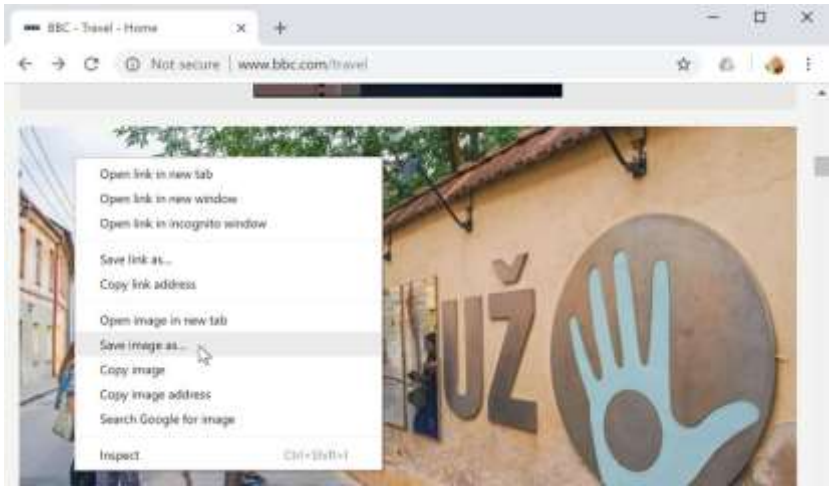


Figure 11: saving Images

3.1.4 Searching for information on the Web

The Web is growing rapidly and hundreds of thousands of pages of information are being added every day. It is also developing in an extraordinarily ad hoc and erratic way. Hence finding what you are looking for can be extremely difficult. The Web is a very large and complex place. Information is duplicated and distributed amongst many thousands of different servers across the world. The main problem with finding information, software or anything else on the Internet is:

- There is a huge amount of information on everything
- There is no central place that holds an index of everything on the web.

However, there are various indexes to some of the available information. These indexes, called **search engines**, provide a convenient and easy way to finding some (but by no means all) Web resources especially if you are researching an esoteric topic. At these sites you can search their databases that hold key information about pages and sites on the Web.

Search Engines

Search engines are programs called **software agents** that automatically traverse the Web, retrieving and indexing Web pages. Starting with the sites they know, they seek out all the links embedded in those pages, follow them, and then repeat the process for all the new sites they come across. Indexing software then compiles enormous databases of information, addresses and links. These databases are accessed when someone submits an enquiry.

Search engines vary in the way they collect and hold data, and in their degree of completeness. 100% coverage is impossible as pages are being added and changed constantly. If you do not find what you want using one search engine, it often worth trying another.

With billions of websites online today, there is **a lot** of information on the Internet. **Search engines** make this information easier to find. Let's look at the basics of using a search engine, as well as some techniques you can use to get **better search results**.

How to search the Web

There are many different search engines you can use, but some of the most popular include **Google**, Yahoo!, and Bing. To perform a search, you'll need to navigate to a search engine in your web browser, type one or more **keywords**—also known as **search terms**—then press **Enter** on your keyboard. In this example, we'll search for **recipes**.

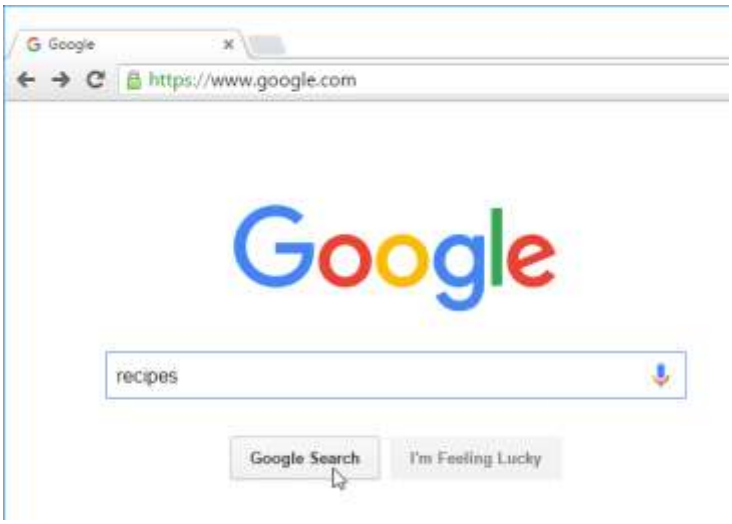


Figure 12: google search

After you run a search, you'll see a list of **relevant websites** that match your search terms. These are commonly known as **search results**. If you see a site that looks interesting, you can click a link to open it. If the site doesn't have what you need, you can simply return to the results page to look for more options.

Most browsers also allow you to perform a web search directly from your **address bar**, although some have a separate **search bar** next to the address bar. Simply type your search terms and press **Enter** to run the search.

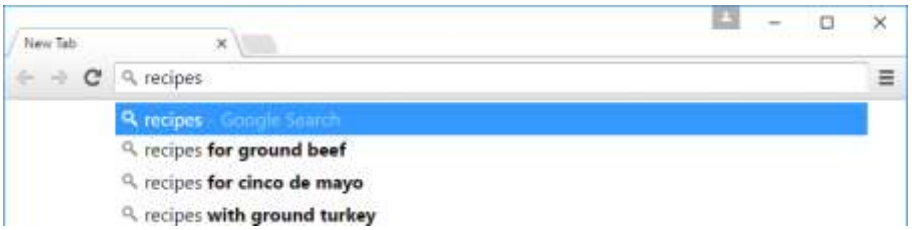


Figure 13: search options

Search suggestions

If you don't find what you're looking for on the first try, don't worry! Search engines are good at finding things online, but they're not perfect. You'll often need to **try different search terms** to find what you're looking for.

If you're having trouble thinking of new search terms, you can use **search suggestions** instead. These will usually appear as you're typing, and they're a great way to find new keywords you might not have tried otherwise. To use a search suggestion, you can click it with your mouse, or select it with the arrow keys on your keyboard.

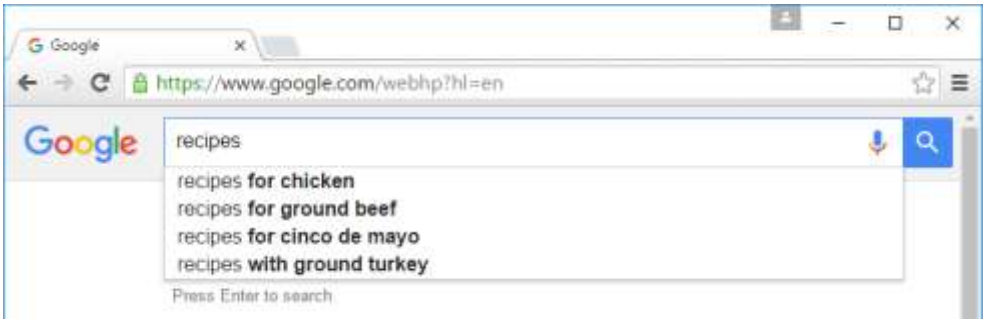


Figure 14: refining search options

Refining your search

If you're still having trouble finding exactly what you need, you can use some special characters to help refine your search. For example, if you want to **exclude a word** from a search, you can type a **hyphen** (**-**) at the beginning of a word. So if you wanted to find cookie recipes that don't include chocolate, you could search for **recipes cookies -chocolate**.

You can also search for **exact words** or **phrases** to narrow down your results even more. All you need to do is place **quotation marks** (" ") around the desired search terms. For example, if you search for **recipes** "**sugar cookies**", your search results will only include recipes for sugar cookies, instead of any cookies that happen to use sugar as an ingredient.

These techniques may come in handy in certain cases, but you probably won't need to use them with most searches. Search engines can usually figure out what you're looking for without these extra characters.

Content-specific searches

There may be times when you're looking for something more specific, like a **news article**, **picture**, or **video**. Most search engines have **links** at the top of the page that allow you to perform these unique searches.

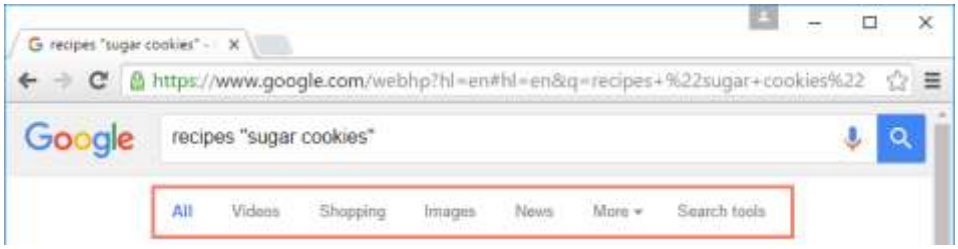


Figure 15: content specific search

In the example below, we've used the same search terms to look for **images** instead of websites. If you see an image you like, you can click to visit the website it originally came from.

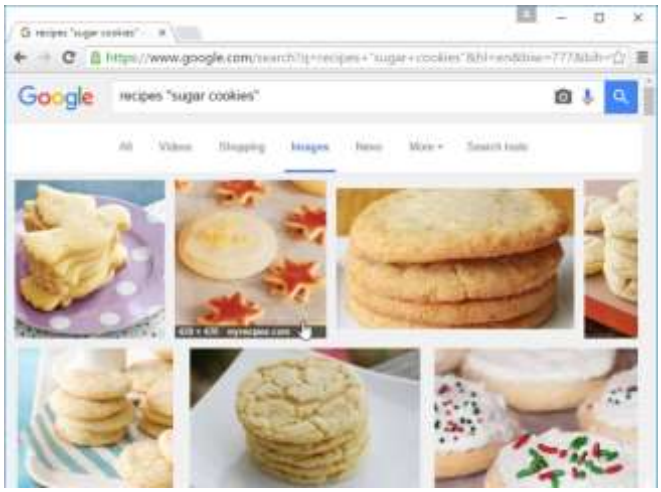


Figure 16: search by image

You can use the extra **search tools** to narrow down your results even more. These tools will change based on the type of content you're looking for, but in this example we can filter our images by **size**, **color**, **image type**, and more. So if you wanted to find cookies with pink frosting, you could search for images that are mostly pink.

3.1.5 Working offline

When you use offline files, you can access copies of your network files even when you can't connect to the network.

Working with offline files isn't as hard as it sounds. After you've turned the offline files feature on, you can choose the files (or folders) on your network that you want to work with while you're not connected. Windows creates a copy of each file on your PC, and the next time you connect, your offline files will automatically get synced with the files in the network folder.

Enabling offline files

When your PC is connected to the network, open File Explorer and find the network file or folder you want to make available offline.

Select the file or folder, tap or click the Home tab, tap or click Easy access, and then choose Always available offline. A small green sync icon ( ) will appear in the corner of the file or folder icon.

Whenever you're connected to the file or folder on the network, it will automatically sync with your offline copy. You can also sync it manually at any time. If you're working offline and

make a change to a file that someone else has also changed in the network location, there will be a sync conflict, and you'll be asked which version you want to keep.

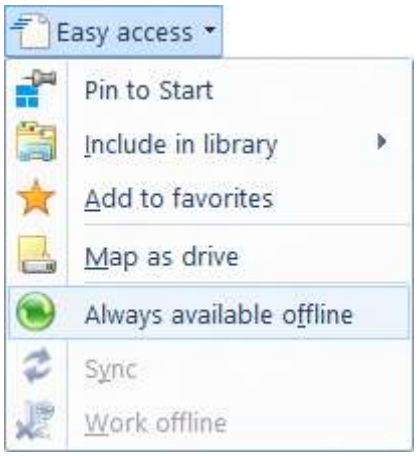


Figure 17: working offline

3.1.6 Net-etiquette

Netiquette is short for "Internet etiquette." Just like etiquette is a code of polite behaviour in society, netiquette is a code of good behaviour on the Internet. This includes several aspects of the Internet, such as email, social media, online chat, web forums, website comments, multiplayer gaming, and other types of online communication.

While there is no official list of netiquette rules or guidelines, the general idea is to respect others online.

The Internet provides a sense of anonymity since you often do not see or hear the people with whom you are communicating online. But that is not an excuse for having poor manners or posting incendiary comments. While some users may feel like they can hide behind their keyboard or smartphone when posting online, the fact is they are still the ones publishing the content. Remember – if you post offensive remarks online and the veil of anonymity is lifted, you will have to answer for the comments you made.

So good netiquette benefits both you and others on the Internet. Posting a positive comment rather than a negative one just might make someone's day.

3.1.7 Electronic Mail

E**mail** is a way to **send and receive messages** across the Internet. Almost everyone who uses the Internet has their own email account, usually called an **email address**. This is because you'll need an email address to do just about anything online, from online banking to creating a Facebook account.

This does not refer to a specific email client software. Rather, it provides a general overview of email functionality.

Most people use the Internet primarily to email and surf the World Wide Web.

Simply put, email, or "electronic mail," is the process of sending and receiving messages over the Internet. While email is similar to a traditional mail system, or "snail mail," as it's sometimes referred to, it's also very different.

After reading your email, you can save it, forward it to others, or delete it. If you want paper copies, you can print your email messages.

Email advantages

- *Productivity tools:* Email is usually packaged with a calendar, address book, instant messaging, and more for convenience and productivity.
- *Access to web services:* If you want to sign up for an account like Facebook or order products from services like Amazon, you will need an email address so you can be safely identified and contacted.
- *Easy mail management:* Email service providers have tools that allow you to file, label, prioritize, find, group, and filter your emails for easy management. You can even easily control spam, or junk email.
- *Privacy:* Your email is delivered to your own personal and private account with a password required to access and view emails.
- *Communication with multiple people:* You can send an email to multiple people at once, giving you the option to include as few as or as many people as you want in a conversation.
- *Accessible anywhere at any time:* You don't have to be at home to get your mail. You can access it from any computer or mobile device that has an Internet connection.

Understanding email addresses

To receive emails, you will need an email account and an email address. Also, if you want to send emails to other people, you will need to obtain their email addresses. It's important to learn how to write email addresses correctly because if you do not enter them exactly right, your emails will not be delivered or might be delivered to the wrong person.

Email addresses are always written in a standard format that includes a user name, the @ (at) symbol, and the email provider's domain. The user name is the name you choose to identify yourself. The email provider is the website that hosts your email account.

Some businesses and organizations use email addresses with their own website domain. For example Mzuzu University. users who are staff and students of the university have such emails like, bacs0120@mzuni.ac.mw or chingwalu.g@mzuni.ac.mw

About email Service providers

In the past, people usually received an email account from the same companies that provided their Internet access. For example, if Skyband provided your Internet connection, you'd have a skyband email address. While this is still true for some people, today it's increasingly common to use a free web-based email service, also known as webmail. Anyone can use these services, no matter who provides their Internet access.

If you've ever started a Gmail, Hotmail, Yahoo or other email account, chances are you've used webmail. With webmail, you read and send email through your web browser and a web-based interface. In fact, webmail allows you to access your email from any web browser, on any device, as long as you are connected to the Internet.

When you access your domain emails via a webmail interface, you will open your web browser and type in the address of your webmail login panel. For example, if you are logging into your university domain email account via webmail, you will type in *https://www.mzuni.ac.mw/webmail/* into your web browser.

But If you've used a program like Microsoft Outlook, Windows Live Mail, Mozilla Thunderbird or Apple Mail to manage your emails, then you've used an email client (Conventional).

An email client is a piece of software that is installed on your computer. You then use this software to download your emails from the server to your computer and from there you can read and send emails.

In other words, in order to use an email client to access your domain emails, you will first need to install the email client software on your computer.

If you have the Microsoft Office bundle installed on your computer, you will find Outlook as part of the bundle; or if you are using a Mac, you can easily find Apple Mail, the default Mac email application.

Both webmail and conventional ones (email clients) are used for sending and receiving emails. Webmail is portable and accessible everywhere – all you need is a web browser and an Internet connection.

But conventional (email clients) have their benefits, too.

With webmail, all of your email is backed up on the server, meaning you will still have access in the event your computer breaks down. On the other hand, with conventional (email clients), your email is downloaded onto your computer, allowing you to backup your emails onto your hard drive or cloud storage, if you are using one.

Conventional (email clients) also allows you to manage your emails offline because email is downloaded onto your computer. Of course you will still need an Internet connection to download the emails from your server to computer, but once downloaded, you can access them anytime you want, even when there is no Internet connectivity.

Not everyone needs offline access to their email, but if you need the ability to read your emails from a train or a long flight without Wi-Fi, then you will find an email client helpful.

Conventional (email clients) may be better for you if:

- You have multiple email addresses with different domains, and you need the ability to manage (including read and send) emails from each one as individual addresses but all in one place (ie. without having to log into each address separately).

- You need access to your emails offline.
- You prefer to back up your email regularly and/or you have limited email storage with your email provider. When you access your emails using an email client, you are essentially downloading your emails from the mailserver to your computer. As long as you didn't specify to save a copy of the email on the mailserver, the email will no longer be kept on the mailserver, thus freeing up server space.
- You need integration with desktop apps like Dropbox, Cloudapp, your calendar or other tools.

Webmail may be better for you if:

- You prefer a single inbox for each of your accounts. This way, you will need to log into each of the email accounts separately, but it keeps each account separate and easier to manage.
- You prefer web-based services to downloadable applications, or you don't want to or can't install software on your computer (ie. you are using a work or school computer).
- You feel more secure keeping your email online and off your computer. This is especially true if you are using a public or shared computer to access your emails.

E-mail attachments

Email is one of the quickest ways of sharing information with others. Learning to create, send, and open an email attachment is the next step to becoming an email pro.

Using your email client's attachment feature, you can attach and send different file types along with your email messages. When sending an email attachment, your email program changes (encodes) your file so it can be sent across the Internet.

To send an email attachment:

- Enter a message into an email message window. Click the Attachment button.
- A dialog box opens (depends on email client). Browse until you locate the file you want to attach.
- Select the file.
- Click OK or Insert or Choose (depends on email client).
- The dialog box closes. Look for the file icon or name in your email client's attachment area. The file should be attached.
- Click Send. You might get a message such as, "Encoding Attachment," meaning the attachment is being sent.

Email attachment etiquette

Attachments are an easy way to share files, photos, and more, but many people aren't aware of some of the most common attachment mistakes. Be sure to follow these basic rules when including attachments in your emails.

Mention what is included attachments

Never attach a file without mentioning it in the body of your email. Something as simple as I've attached a few photos to this email will help your recipients know what to expect. On the other hand, make sure the attachments you mention are actually included with the message—it's easy to focus on your message and forget to include the file itself. We recommend attaching any files before you start writing.

Consider file size and format

Avoid sending excessively large attachments or uncompressed photos, which can take a long time for your recipients to download. You can always ZIP or compress files to make them easier to send. Additionally, make sure your attachments don't need to be viewed in a specific application—use universal file types like .PDF, .RTF, and .JPG.

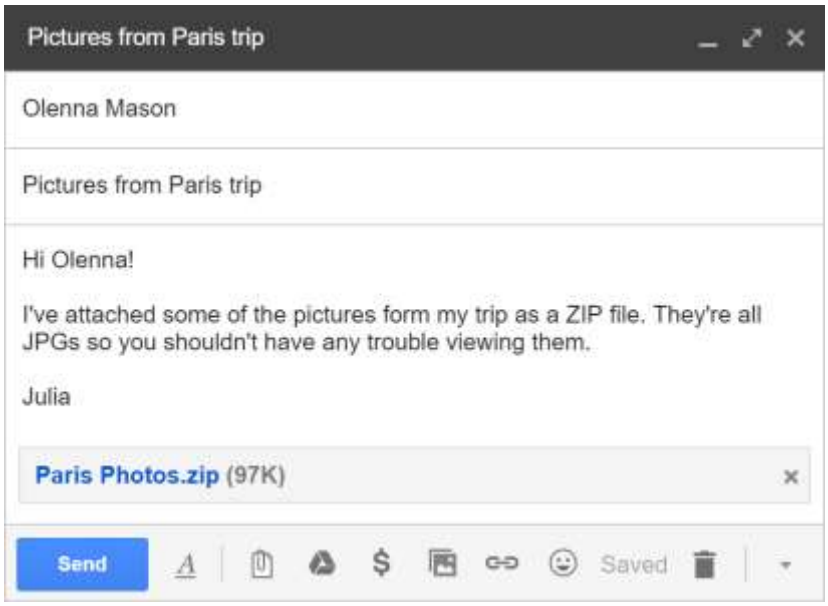


Figure 18: email attachment

Only include related files

If you need to send a lot of **different files** to the same person, consider sending the attachments through multiple emails. If you include several **unrelated files** in the same email, it can be difficult for your recipients to find the exact file they need.

While attachments are still an easy way to share files quickly over email, many people have begun using cloud-based storage services like Google Drive and Microsoft OneDrive to **share large files**. Rather than sharing the file itself, you can **share a link**, allowing others to access the file remotely.

Organising an address book

Once you've added people to your contacts list, you'll never have to worry about remembering an email address. Whenever you create a new email, you can select the **To** or **Add Recipients** button.

Then **choose recipients** from your **contacts list**. The contacts' email addresses will be added **automatically** to the message. You'll even be able to use the same contact information for other webmail services, like **instant messaging** and **calendar sharing**.

Most email services will also autocomplete or suggest an email address when you start typing, as long as it's an address that's already been saved in your contacts.

Each email window has the **TO** field in which you indicate the main recipients of the email you are sending. If there is a need to copy the same email to others then you can indicate their emails in the **CC** (*carbon copy)* field.

In cases of privacy where you want someone to get a copy of the email without other reciepients noticing then you indicate the persons email address in the **BCC** (*Blind Cabon Copy*) field.

Internet has come offer a lot of help to your studies as a student and you need to make good use of it.



Internet Self-Assessment Test

Instructions

- Answer all questions

1.Define the following terms:

(i) Attachment

(ii) CC

(iii) HTML

(iv) HTTP

2. What is the purpose of MODEM in dialup networking?

3. The Internet and the World Wide Web are two terms used to describe the same thing. Do you agree or disagree? Explain your answer.

4. What is the main function of a router on the Internet?

5. Describe what you understand by the term Case Sensitivity with regards to search engines.

6. What kind of data can you send by e-mail?

7. What are the two main differences of web browsers?

8. Given this email address Ekari@yahoo.co.uk

9. Given this address www.mzuni.ac.mw/programs/sciences.html

10. What are the two things to keep in mind about Uniform resources Locators (URL)



Unit Summary

In this unit, we have looked at the Internet and the World Wide Web. We have discussed how you can get on the Internet. We have discussed the role of browsers on the internet and how they differ from each other. Searching of information on the internet using search engines like Google, Yahoo etc. has been discussed with emphasis on various ways of doing the search. Netiquette issues have been looked at as they call for proper conduct when online. Finally we have looked at Electronic mail and how it works as well as how you can embed in an attachment.



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UNIT 4

SPREADSHEETS

Introduction

Spreadsheet is a piece of paper or a computer program used for accounting and recording data using rows and columns into which information can be entered. Microsoft Excel, a program in which you enter data into columns, is an example of a spreadsheet program.

Excel is a spreadsheet program that allows you to store, organize, and analyze information. While you may believe Excel is only used by certain people to process complicated data, anyone can learn how to take advantage of the program's powerful features. Whether you're keeping a budget, organizing a training log, or creating an invoice, Excel makes it easy to work with different types of data.

Areas of emphasis:

- Creating workbooks
- Formatting workbooks
- Working with calculations
- Graphs and charts



Key Words

- Spreadsheet, workbook, formula, worksheet, charts, et al.

Prerequisite knowledge

This unit assumes you are familiar with the basic operation of a personal computer. Specific skills required for this tutorial are:

- Use of the mouse and keyboard
- Opening and saving files on the hard disk and on USB memory stick (thumb drive)
- Managing files in general i.e. the difference between drive letters, hard disk, etc
- Running programs from the Windows Start menu
- Minimizing, maximizing and resizing windows

You should have a USB memory stick (thumb drive) or other portable memory device available to store the files that will be created in this unit.



Learning outcomes

By the end of this unit, you should be able to:

- Create a worksheet
- Use, edit, save, close and retrieve workbook
- Able to work with basic calculations
- Create charts and graphs

Resources needed

- a working computer with Microsoft Excel package installed.

4.1.1 Creating workbooks

A *spreadsheet* is a document that stores data in a grid of horizontal rows and vertical columns. *Spreadsheet* a computer program that calculates numbers and organizes information in columns and rows. A **spreadsheet** is a grid of rows and columns in which you enter text, numbers, and the results of calculations. GRID - a network of lines that cross each other to form a series of squares or rectangles. MS Excel is a computer program used to create electronic spreadsheets. Within excel users can organize data. create chart and perform calculations. **Microsoft Excel 2010** is the spreadsheet program in Microsoft Office 2010. Excel, a computerized spreadsheet is called a **worksheet**. The file used to store worksheets is called a **workbook**.

Microsoft Excel Window

Microsoft Excel is an electronic spreadsheet. You can use it to organize your data into rows and columns. You can also use it to perform mathematical calculations quickly.

Excel enables you to perform a wide variety of functions including sorting and selecting from lists. Excel includes a feature that will produce charts from numeric data.

Starting Excel

- Click on the Start button to open its pop-up menu
- Point to Programs. Click on Microsoft Excel
- You use the window to interact with Excel.

The Microsoft Excel window appears and your screen looks similar to the one shown below.

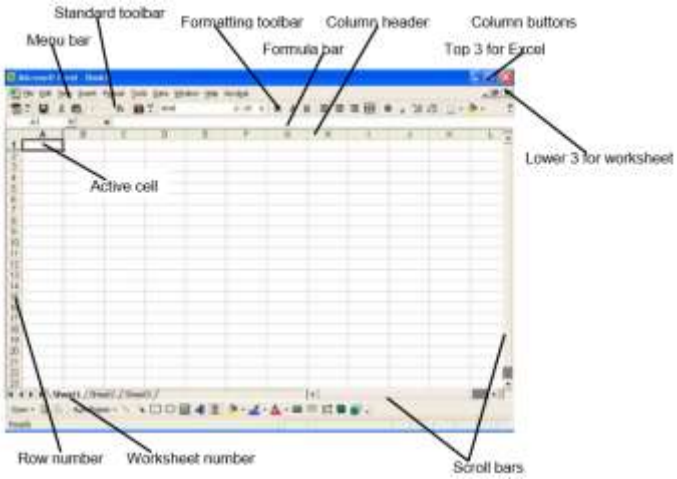


Figure 1: Excel window

4.1.2 Worksheet

Microsoft Excel consists of worksheets. Each worksheet contains columns and rows. Columns are lettered A to Z and then continuing with AA, AB, AC and so on; the rows are numbered

1 to1,048,576. The number of columns and rows you can have in a worksheet is limited by your computer memory and your system resources. The combination of a column coordinate and a row coordinate make up a cell address. For example, the cell located in the upper-left corner of the worksheet is cell A1, meaning column A, row 1. Cell E10 is located under column E on row 10. You enter your data into the cells on the worksheet.

When you start Excel, you open a file that is called a **workbook.** Each new workbook comes with three **worksheets** into which you enter data. The first workbook you've opened is called Book1. This title appears in the bar at the top of the window until you save the workbook with your own title. Sheet tabs appear at the bottom of the window. It's a good idea to rename the sheet tabs to make information on each sheet easier to identify. Worksheets are divided into columns, rows, and cells. Columns go from top to bottom on the worksheet, vertically. Each column has an alphabetical heading at the top. Rows go across the worksheet, horizontally. Each row also has a heading. Row headings are numbers, from 1 through 1,048,576.



Figure 2: the active cells

The Formula Bar

If the Formula bar is turned on, the cell address of the cell you are in, displays in the Name Box which is located on the left side of the Formula bar. Cell entries display on the right side of the Formula bar. If you do not see the Formula bar in your window, perform the following steps:

- Choose the View tab.
- Click Formula Bar in the Show/Hide group.

The Formula bar appears.

The status bar

The Status bar appears at the very bottom of the Excel window. It provides information as the sum, average, zoom, minimum, and maximum value of selected numbers. You can change what displays on the Status bar by right-clicking on the Status bar and selecting the options

you want from the Customize Status Bar menu. You click a menu item to select it. You click it again to deselect it. A check mark next to an item means the item is selected.

Move Around a Worksheet

By using the **Arrow keys**, you can move around your worksheet. You can use the **Down Arrow key** to move downward one cell at a time. You can use the **Up Arrow key** to move upward one cell at a time. You can use the **Tab key** to move across the page to the right, one cell at a time. You can hold down the **Shift key and then press the Tab key** to move to the left, one cell at a time. You can use the **right and left arrow keys** to move right or left one cell at a time.

The **Page Up and Page Down keys** move up and down one page at a time.

If you hold down the Ctrl key and then press the Home key, you move to the beginning of the worksheet.

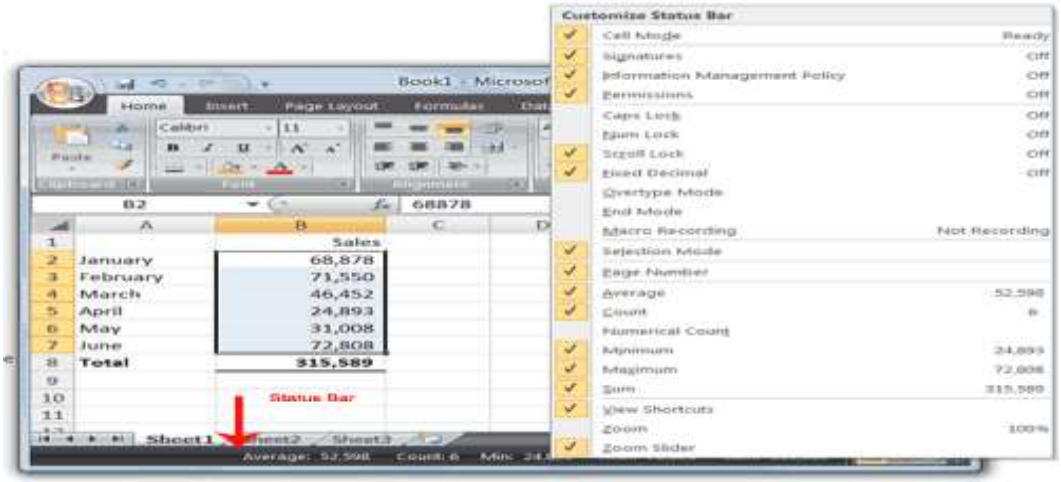


Figure 3: the status bar

Activity 1

Move A round the Worksheet

The Down Arrow Key

Press the down arrow key several times. Note that the cursor moves downward one cell at a time.

The Up Arrow Key

Press the up arrow key several times. Note that the cursor moves upward one cell at a time.

The Tab Key

Move to cell A1 Press the Tab key several times. Note that the cursor moves to the right one cell at a time.

The Shift+Tab Keys

Hold down the Shift key and then press Tab. Note that the cursor moves to the left one cell at a time.

The Right and Left Arrow Keys

Press the right arrow key several times. Note that the cursor moves to the right. Press the left arrow key several times. Note that the cursor moves to the left.

Page Up and Page Down

- Press the Page Down key. Note that the cursor moves down one page.
- Press the Page Up key. Note that the cursor moves up one page.
- Move the cursor to column J.
- Stay in column J and move the cursor to row 20.
- Hold down the Ctrl key while you press the Home key. Excel moves to cell A1.

 **Activity 2**

Go To Cells Quickly

The following are shortcuts for moving quickly from one cell in a worksheet to a cell in a different part of the worksheet.

1. Go to -- F5

The F5 function key is the "Go To" key. If you press F5 key, you are prompted for the cell to which you wish to go.

- Enter the cell address, and the cursor jumps to that cell.
- Press F5. The Go To dialog box opens.
- Type **J3** in the Reference field. Press Enter. Excel moves to cell J3.

2. Go to -- Ctrl+G

You can also use Ctrl+G to go to a specific cell.

- Hold down the Ctrl key while you press "g" (Ctrl+G).
- The Go To dialog box opens. Type **C4** in the Reference field.
- Press Enter. Excel moves to cell C4.

The name box

You can also use Name Box to go to a specific cell.

- Just type the cell you want to go to in the Name Box and then press Enter.
- Type **B10** in the Name box.
- Press Enter. Excel moves to cell B10

 **Activity 3 -** ***Select cells***

If you want to perform a function on a group of cells, you must first select those cells by highlighting them.

Go to cell A1.

- Press the F8 key. This anchors the cursor.
- Note that "Extend Selection" appears on the Status bar in the lower-left corner of the window. You are in the Extend mode.
- Click in cell E7. Excel highlights cells A1 to E7.
- Press Esc and click anywhere on the worksheet to clear the highlighting.

Alternative Method: Select Cells by Dragging

You can also select an area by holding down the left mouse button and dragging the mouse over the area. In addition, you can select noncontiguous areas of the worksheet by doing the following:

 **Activity 4 -** *Go to cell A1.*

- Hold down the Ctrl key. You won't release it until step 9. Holding down the Ctrl key enables you to select noncontiguous areas of the worksheet.
- Press the left mouse button.
- While holding down the left mouse button, use the mouse to move from cell A1 to C5.
- Continue to hold down the Ctrl key, but release the left mouse button.
- Using the mouse, place the cursor in cell D7.
- Press the left mouse button.
- While holding down the left mouse button, move to cell F10. Release the left mouse button.
- Release the Ctrl key. Cells A1 to C5 and cells D7 to F10 are selected.
- Press Esc and click anywhere on the worksheet to remove the highlighting.

You can use Excel to enter all sorts of data, professional or personal. You can enter two basic kinds of data into worksheet cells: numbers and text. You can use Excel to create budgets, record student grades or attendance, or list the products you sell.

 **Activity 5 -** ***Enter Data***

First, place the cursor in the cell in which you want to start entering data.

- Type some data, and then press Enter.
- If you need to delete, press the **Backspace** key to delete one character at a time.

- Place the cursor in cell A1.
- **John Jordan**. Do not press Enter at this time.

 **Activity 6 -** ***Delete Data***

- Backspace key erases one character at a time.
- Press the Backspace key until Jordan is erased.
- Press Enter. The name “John" appears in cell A1.

 **Activity 7 -** ***Edit a Cell***

After you enter data into a cell, you can edit the data by pressing F2 while you are in the cell you wish to edit.

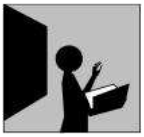
Change "John" to "Jones."

- Move to cell A1.
- Press F2.
- Use the Backspace key to delete the "n" and the "h."
- Type **nes**.
- Press Enter.

 **Activity 8 -** ***Alternate Method: Editing a Cell by Using the Formula Bar***

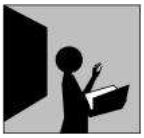
You can also edit the cell by using the Formula bar. You change "Jones" to "Joker" in the following exercise.

- Move the cursor to cell A1.
- Click in the formula area of the Formula bar.
- Use the backspace key to erase the "s," "e," and "n."
- Type **ker**.
- Press Enter.

 **Activity 9 -** ***Alternate Method: Edit a Cell by Double-Clicking in the Cell***

You can change "Joker" to "Johnson" as follows:

- Move to cell A1.
- Double-click in cell A1.
- Press the End key. Your cursor is now at the end of your text.
- Use the Backspace key to erase "r," "e," and "k."
- Type **hnson**. Press Enter.

 **Activity 10 -** ***Text Wrapping***

When you type text that is too long to fit in the cell, the text overlaps the next cell. If you do not want it to overlap the next cell, you can wrap the text.

- Move to cell A2.
- Type **Text too long to fit**.
- Press Enter.
- Return to cell A2.
- Choose the Home tab.
- Click the Wrap Text button . Excel wraps the text in the cell.

 **Activity 11 -** ***To save your file:***

- Click the Office button or File Tab. A menu appears.
- Click Save. The Save As dialog box appears.
- Go to the directory in which you want to save your file.
- Type **Lesson1** in the File Name field.
- Click Save. Excel saves your file.

 **Activity 12-** ***Close Excel***

Click the Office button or File Tab. A menu appears. Click Close. Excel closes

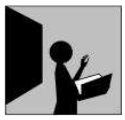
If you have amended a spreadsheet since the last time you saved your work you will now be asked to confirm whether you want to exit Excel without saving. In this case, select **Yes** to exit, or **No** to return to the spreadsheet. You have now exited from Excel and the Desktop window is displayed on your screen.

4.1.3 Entering Excel Formulas and Formatting Data

A major strength of Excel is that you can perform mathematical calculations and format your data. Let us see how to perform basic mathematical calculations and how to format text and numerical data.

In Microsoft Excel, you can enter numbers and mathematical formulas into cells. Whether you enter a number or a formula, you can reference the cell when you perform mathematical calculations such as addition, subtraction, multiplication, or division. When entering a mathematical formula, start the formula with an equal sign. Use the following to indicate the type of calculation you wish to perform:

- + Addition
- -Subtraction
- Multiplication
- / Division
- ^ Exponential

 **Activity 13 -** ***Addition***

- Type **Add** in cell A1.
- Press Enter. Excel moves down one cell.
- Type **1** in cell A2.
- Press Enter. Excel moves down one cell.
- Type **1** in cell A3.
- Press Enter. Excel moves down one cell.
- Type **=A2+A3** in cell A4.
- Click the check mark on the Formula bar. Excel adds cell
- A1 to cell A2 and displays the result in cell A4. The formula displays on the Formula bar.

Note: *Clicking the check mark on the Formula bar is similar to pressing Enter. Excel records your entry but does not move to the next cell.*

 **Activity 14 -** ***Subtraction***

- Press F5. The Go To dialog box appears.
- Type **B1** in the Reference field.
- Press Enter. Excel moves to cell B1.
- Type **Subtract**.
- Press Enter. Excel moves down one cell.
- Type **6** in cell B2.
- Press Enter. Excel moves down one cell.
- Type **3** in cell B3.
- Press Enter. Excel moves down one cell.
- Type **=B2-B3** in cell B4.

Click the check mark on the Formula bar. Excel subtracts cell B3 from cell B2 and the result displays in cell B4. The formula displays on the Formula bar.

 **Activity 15 -** ***Multiplication***

- Hold down the Ctrl key while you press "g" (Ctrl+g). The Go To dialog box appears.
- Type **C1** in the Reference field.
- Press Enter. Excel moves to cell C1
- Type **Multiply**.

- Press Enter. Excel moves down one cell.
- Type **20** in cell C2.
- Press Enter. Excel moves down one cell.
- Type **5** in cell C3.
- Press Enter. Excel moves down one cell.
- Type **=C2*C3** in cell C4.

Click the check mark on the Formula bar. Excel multiplies C1 by cell C2 and displays the result in cell C3. The formula displays on the Formula bar.

 **Activity 16 -*Division***

- Press F5.
- Type **D1** in the Reference field.
- Press Enter. Excel moves to cell D1.
- Type **Divide**.
- Press Enter. Excel moves down one cell.
- Type **250** in cell D2.
- Press Enter. Excel moves down one cell.
- Type **50** in cell D3.
- Press Enter. Excel moves down one cell.
- Type **=D2/D3** in cell D4.

Click the check mark on the Formula bar. Excel divides cell D2 by cell D3 and displays the result in cell D4. The formula displays on the Formula bar.

When creating formulas, you can reference cells and include numbers. All of the following formulas are valid:

=A2/B2

=A2+12-B3

=A2*B2+12

=24+53

When you perform mathematical calculations in Excel, be careful of precedence (order of importance). Calculations are performed from left to right, with multiplication and division performed before addition and subtraction.

 **Activity 17 - Calculations**

- Move to cell A7.
- Type **=3+3+12/2*4**.
- Press Enter.

Note: Microsoft Excel divides 12 by 2, multiplies the answer by 4, adds 3, and then adds another 3. The answer, 30, displays in cell A7

To change the order of calculation, use parentheses (brackets). Microsoft Excel calculates information in parentheses first.

 **Activity 18**

- Double-click in cell A7.
- Edit the cell to read **=(3+3+12)/2*4**.
- Press Enter.

Note: Microsoft Excel adds 3 plus 3 plus 12, divides the answer by 2, and then multiplies the result by 4. The answer, 36, displays in cell A7.

Other numbers and how to enter them

- To enter fractions, leave a space between the whole number and the fraction. For example, **1 1/8**.
- To enter a fraction only, enter a zero first, for example, **0 1/4**. If you enter **1/4** without the zero, Excel will interpret the number as a date, January 4.
- If you type **(100)** to indicate a negative number by parentheses, Excel will display the number as -100.

 **Activity 19 -** ***Automatic Calculations***

By default, Microsoft Excel recalculates the worksheet as you change cell entries. This makes it easy for you to correct mistakes and analyze a variety of scenarios. Make the changes described below and note how Microsoft Excel automatically recalculates:

- Move to cell A2.
- Type **2**.
- Press the right arrow key. Excel changes the result in cell A4. Excel adds cell A2 to cell A3 and the new result appears in cell A4.
- Move to cell B2.
- Type **8**.
- Press the right arrow key. Excel subtracts cell B3 from cell B3 and the new result appears in cell B4.
- Move to cell C2.
- Type **4**.
- Press the right arrow key. Excel multiplies cell C2 by cell C3 and the new result appears in cell C4.
- Move to cell D2.
- Type **12**.
- Press the Enter key. Excel divides cell D2 by cell D3 and the new result appears in cell D4

4.1.4 Functions

Use functions (prewritten formulas) to add up values, calculate averages, and find the smallest or largest value in a range of values. A function is a prewritten formula. A formula is an equation that performs operations on worksheet data. Functions are predefined formulas that perform calculations by using specific values, called arguments, in a particular order, or structure. You can find these Functions on a Home Tab under Editing Group. As prewritten formulas, functions simplify the process of entering calculations.

Using functions, you can easily and quickly create formulas that might be difficult to build for yourself. Functions differ from regular formulas in that you supply the value but not the operators, such as +, -, *, or /. For example, you can use the SUM function to add. Excel has many functions. Some examples of functions are:

- AVERAGE – calculates the average (finds the mean average)
- MAX –finds the largest number in a range.
- MIN – find the smallest number in a range.
- SUM - adds values or ranges of cells.

When using a function, remember the following:

- Use an equal sign to begin a formula.
- Specify the function name.
- Enclose arguments within parentheses.

Arguments are values on which you want to perform the calculation. For example, arguments specify the numbers or cells you want to add. Use a comma to separate arguments.

Example of a function

=SUM(2,13,A1,B2:C7)

In this function:

- The equal sign begins the function.
- SUM is the name of the function.
- 2, 13, A1, and B2:C7 are the arguments.
- Parentheses enclose the arguments.
- Commas separate the arguments.

NOTE: After you type the first letter of a function name, the AutoComplete list appears. You can double-click on an item in the AutoComplete list to complete your entry quickly. Excel will complete the function name and enter the first parenthesis.

 ***Activity 20 - Open Microsoft Excel.***

- Type 12 in cell B1.
- Press Enter.
- Type 27 in cell B2.
- Press Enter.

- Type 24 in cell B3.
- Press Enter.
- Type =SUM(B1:B3) in cell A4.
- Press Enter. The sum of cells B1 to B3, which is 63, appears.

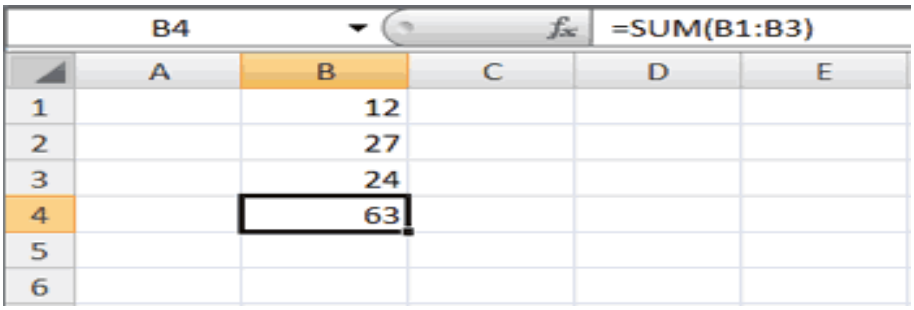


Figure 5 : using formulars

AutoSum

You can use the AutoSum button **Σ** on the Home Tab to automatically add a column or row of numbers. When you press the AutoSum button **Σ**, Excel selects the numbers it thinks you want to add. If you click the check mark on the Formula bar or press the Enter key, Excel adds the numbers. If Excel's guess as to which numbers you want to add is wrong, you can select the cells you want.

 **Activity 21 - *AutoSum***

The following steps illustrates AutoSum:

- Go to cell F1.
- Type **3.**
- Press Enter. Excel moves down one cell.
- Type **3**.
- Press Enter. Excel moves down one cell.
- Type **3**.
- Press Enter. Excel moves down one cell to cell F4.
- Choose the Home tab.
- Click the AutoSum button **Σ** in the Editing group. Excel selects cells F1 through F3 and enters a formula in cell F4.
- Press Enter. Excel adds cells F1 through F3 and displays the result in cell F4.

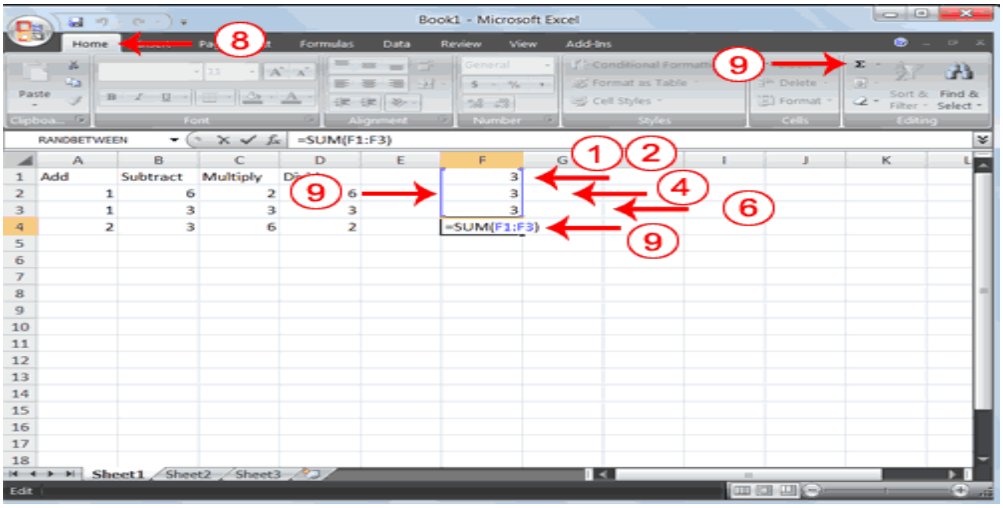


Figure 6: Inserting formulars

Activity 22 - *Calculate an Average*

You can use the AVERAGE function to calculate the average of a series of numbers. Do the following exercise:

- Move to cell A6.
- Type **Average**. Press the right arrow key to move to cell B6.
- Type **=AVERAGE(B1:B3)**.
- Press Enter. The average of cells B1 to B3, which is 21, appears.

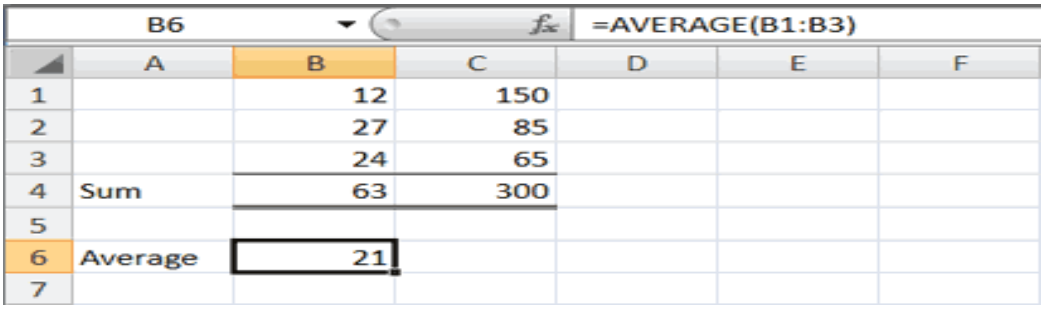


Figure 7: The average formula

Activity 23 - *Calculate an Average with the AutoSum Button*

In Microsoft Excel, you can use the AutoSum button to calculate an average. Do the following exercise:

- Move to cell C6.
- Choose the Home tab.
- Click the down arrow next to the AutoSum button .
- Click Average.
- Select cells C1 to C3.
- Press Enter. The average of cells C1 to C3, which is 100, appears.

 **Activity 24 -** ***Find the Lowest Number***

You can use the MIN function to find the lowest number in a series of numbers. Do the following:

- Move to cell A7.
- Type **Min**.
- Press the right arrow key to move to cell B7.
- Type = **MIN(B1:B3)**.
- Press Enter. The lowest number in the series, which is 12, appears

Note: You can also use the drop-down button next to the AutoSum button to calculate minimums, maximums, and counts.

 **Activity 25 -** ***Find the Highest Number***

You can use the MAX function to find the highest number in a series of numbers. Do the following:

- Move to cell A8.
- Type **Max.**
- Press the right arrow key to move to cell B8.
- Type = **MAX(B1:B3).**
- Press Enter. The highest number in the series, which is 27, appears

 **Activity 26 -** ***Count the Numbers in a Series of Numbers***

You can use the count function to count the number of numbers in a series. Do the following:

- Move to cell A9.
- Type Count.
- Press the right arrow key to move to cell B9.
- Choose the Home tab.
- Click the down arrow next to the AutoSum button . .
- Click Count Numbers. Excel places the count function in cell C9 and takes a guess at which cells you want to count.
- The guess is incorrect, so you must select the proper cells.
- Select B1 to B3.
- Press Enter. The number of items in the series, which is 3, appears.

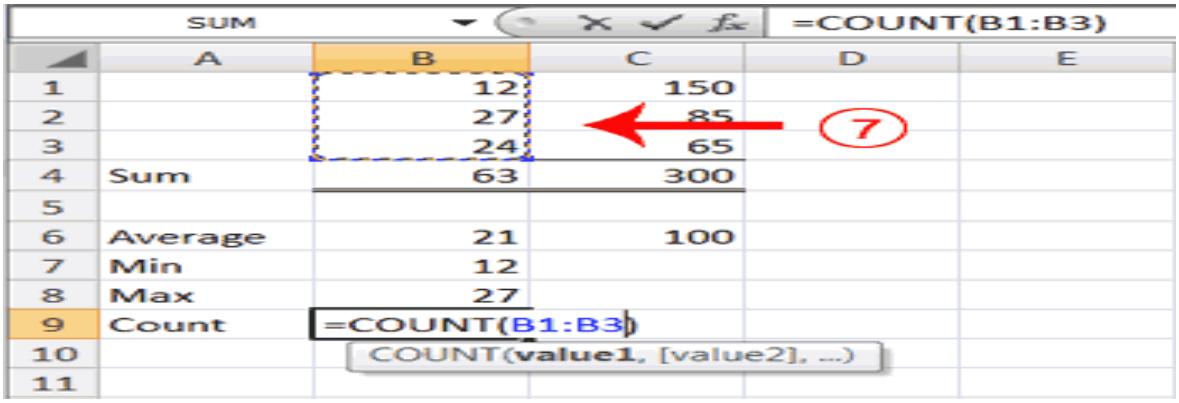


Figure 8: formulars

Align Cell Entries

When you type text into a cell, by default your entry aligns with the left side of the cell. When you type numbers into a cell, by default your entry aligns with the right side of the cell. You can change the cell alignment. You can center, left-align, or right-align any cell entry.

Activity 27 - ***Align Centre***

To center cells A1 to D1:

- Select cells A1 to D1.
- Choose the Home tab.
- Click the Center button in the Alignment group. Excel centers each cell's content.

Cut, copy and paste

Contents of a cell can be copied/moved to another location in the same or different worksheet by using the copy, cut and paste features. Keyboard shortcuts can also be used to do the copy, cut and paste actions.

- Ctrl +C
- Ctrl +X
- Ctrl +V

Insert and Delete Columns and Rows

You can insert and delete columns and rows. When you delete a column, you delete everything in the column from the top of the worksheet to the bottom of the worksheet. When you delete a row, you delete the entire row from left to right. Inserting a column or row inserts a completely new column or row.

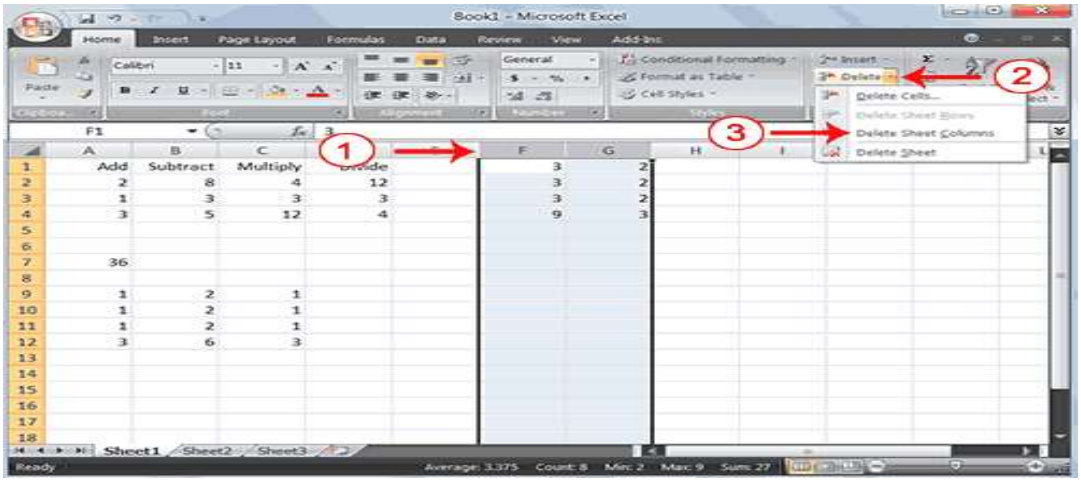


Figure 9: inserting column

Activity 28 - ***To delete columns F and G:***

- Click the column F indicator and drag to column G.
- Click the down arrow next to Delete in the Cells group. A menu appears.
- Click Delete Sheet Columns. Excel deletes the columns you selected.
- Click anywhere on the worksheet to remove your selection.

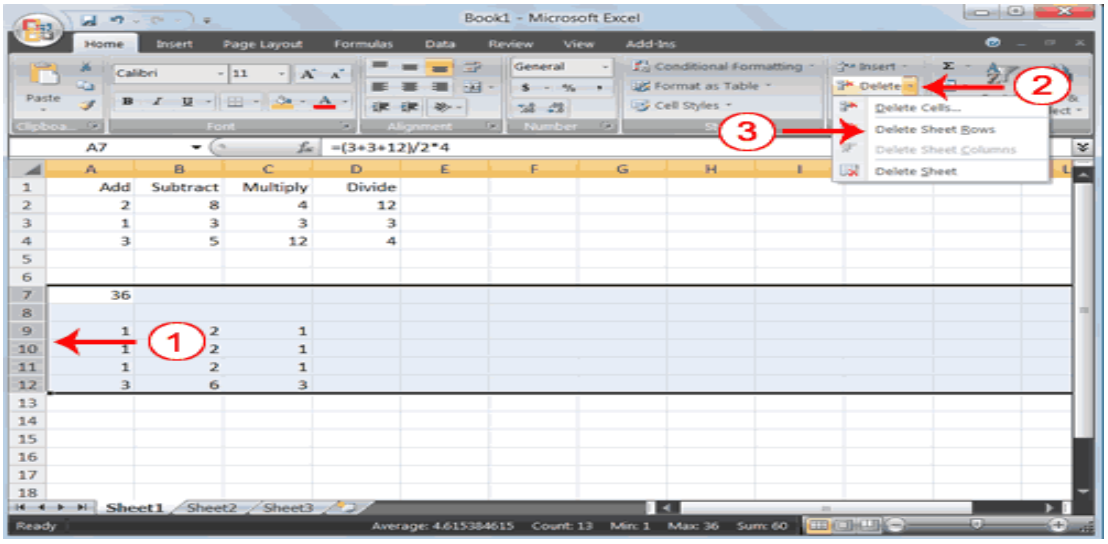


Figure 10: inserting rows

Activity 29 - ***To delete rows 7 through 12:***

- Click the row 7 indicator and drag to row 12.
- Click the down arrow next to Delete in the Cells group. A menu appears.
- Click Delete Sheet Rows. Excel deletes the rows you selected.
- Click anywhere on the worksheet to remove your selection.



Activity 30 - ***To insert a column:***

- Click on A to select column A.
- Click the down arrow next to Insert in the Cells group. A menu appears.
- Click Insert Sheet Columns. Excel inserts a new column. Click anywhere on the worksheet to remove your selection.



Activity 31 - ***To insert rows:***

- Click on 1 and then drag down to 2 to select rows 1 and 2.
- Click the down arrow next to Insert in the Cells group. A menu appears.
- Click Insert Sheet Rows. Excel inserts two new rows.
- Click anywhere on the worksheet to remove your selection.

4.1.5 Quick Ways to Enter Data

There are two time-savers you can use to enter data in Excel: **AutoComplete** and **AutoFill**

AutoComplete: Type a few letters in a cell, and Excel can fill in the remaining characters for you. AutoComplete detects when a value you are entering is similar to previously entered values. AutoComplete works for text or for text with numbers. It does not work for numbers only, for dates, or for times.

AutoFill: Type one or more entries in an intended series, and then extend the series. With AutoFill, you can quickly enter the months of the year, the days of the week, multiples of 2 or 3, or other data in a series. You type one or more entries, and then extend the series.

Sometimes Excel can't calculate a formula because the formula contains an error. If that happens, you'll see an error value in a cell instead of a result. Here are three common error values:

- #### The column isn't wide enough to display the contents of the cell. To fix the problem, you can increase column width, shrink the contents to fit the column, or apply a different number format.
- **#REF!** A cell reference isn't valid. Cells may have been deleted or pasted over.
- **#NAME?** You may have misspelled a function name or used a name that Excel doesn't recognize.

Cell references

Worksheets are divided into columns, rows, and cells. On the worksheet, the rectangular area where a row and a column intersect is known as a cell. **Cell references** identify individual cells or cell ranges in columns and rows. Cell references tell Excel where to look for values to use in a formula. Examples of cell references.

- A10 – the cell in column A and row 10
- A10,A20 - cell A10 and cell A20
- A10:A20 - the range of cells in column A and rows 10 through 20
- B15:E15 = the range of cells in row 15 and columns B through E
- A10:E20 - the range of cells in columns A through E and rows 10 through 20.

4.1.6 Using Reference Operators

Reference operators refer to a cell or a group of cells. There are two types of reference operators:

1. Range

- A range reference refers to all the cells between and including the reference.
- A range reference consists of two cell addresses separated by a colon.
- The reference A1:A3 includes cells A1, A2, and A3.
- The reference A1:C3 includes cells A1, A2, A3, B1, B2, B3, C1, C2, and C3.

2. Union

- A union reference includes two or more references.
- A union reference consists of two or more numbers, range references, or cell addresses separated by a comma.
- The reference A7,B8:B10,C9,10 refers to cells A7, B8 to B10, C9 and the number 10.
- Cell references
- You can type cell references directly into cells, or you can enter cell references by clicking cells, which avoids typing errors.

Three types of cell references:

- **Relative**,
- **Absolute and**
- **Mixed.**

Relative references automatically change as they're copied down a column or across a row. Using this method the formula adjust itself relatively to the current cell where it has been copied.

Absolute references are fixed. They don't change if you copy a formula from one cell to another. Absolute references have dollar signs ($) like this: A1. You use absolute cell references to refer to cells that you don't want to change as the formula is copied.

M**ixed reference** has either an absolute column and a relative row, or an absolute row and a relative column. As a mixed reference is copied from one cell to another, the absolute reference stays the same but the relative reference changes.

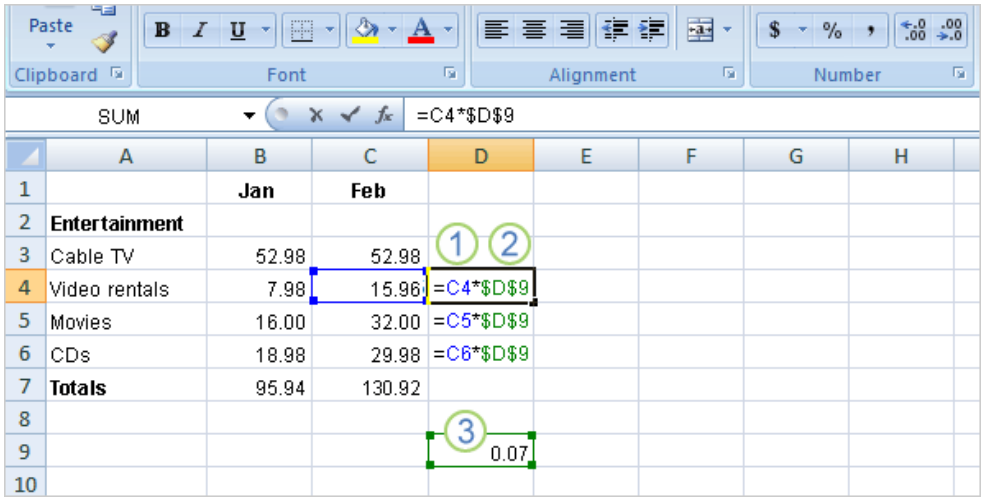


Figure 11: Cell referencing

Activity 32 - Say you receive some entertainment coupons offering a 7 percent discount for video rentals, movies, and CDs. How much could you save in a month by using the discounts? You could use a formula to multiply those February expenses by 7 percent.

- So start by typing the discount rate **.07** in the empty cell D9, and then type the formula in cell D4.
- Then in cell D4, type **=C4***. Remember that this relative cell reference will change from row to row.
- Enter a dollar sign (**$**) and **D** to make an absolute reference to column D, and **$9** to make an absolute reference to row 9. Your formula will multiply the value in cell C4 by the value in cell D9.
- Cell D9 contains the value for the 7 percent discount.
- You can copy the formula from cell D4 to D5 by using the fill handle.
- As the formula is copied, the relative cell reference changes from C4 to C5, while the absolute reference to the discount in D9 does not change; it remains as D9 in each row it is copied to.

4.1.7 Charts

A **chart**, or **graph**, is a visual representation of a set of data. With a chart, you can transform worksheet data to show comparisons, patterns, and trends. So instead of having to analyze columns of worksheet numbers, you can see at a glance what the data means. A chart gets your point across—fast.

Creating a Column chart

- Start by creating the worksheet below exactly as shown.
- After you have created the worksheet, you are ready to create your chart.

	A	B	C	D	E
1		Toy Sales			
2					
3	**Products**	**Region 1**	**Region 2**	**Region 3**	
4	**Dolls**	2478	2640	2388	
5	**Trucks**	2031	2173	2790	
6	**Puzzles**	1918	2722	2795	
7	**Total**	6427	7535	7973	
8					

Figure 12: Chart data

Activity 33 - *Do the following:*

- Select cells A3 to D6. You must select all the cells containing the data you want in your chart. You should also include the data labels.
- Choose the Insert tab.
- Click the Column button in the Charts group. A list of column chart sub-types types appears.
- Click the Clustered Column chart sub-type. Excel creates a Clustered Column chart and the Chart Tools context tabs appear.

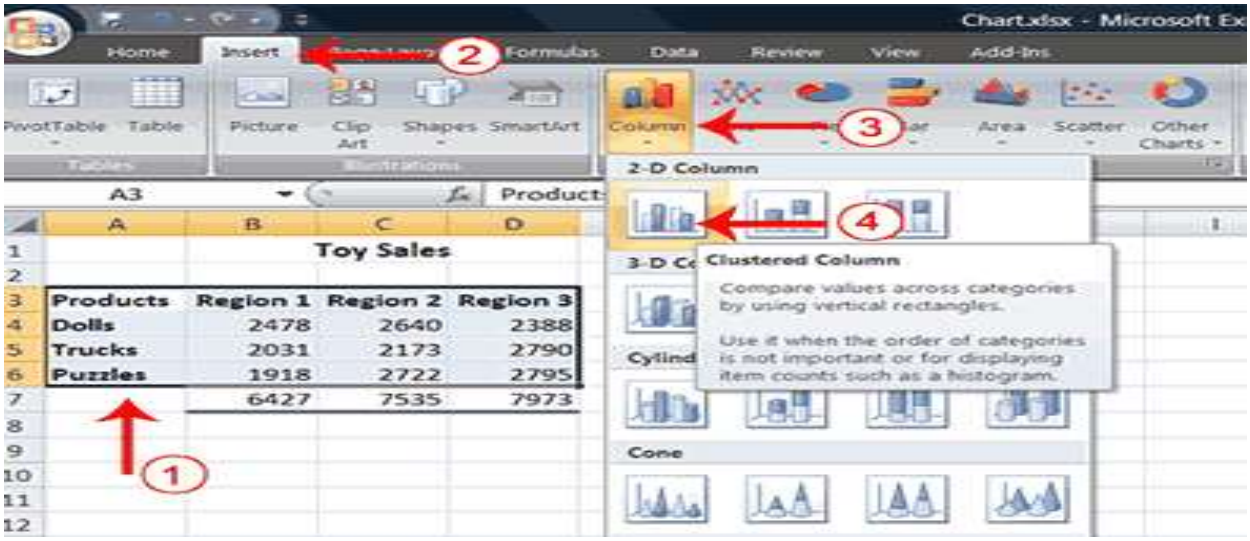


Figure 13: chart types

Apply a Chart Layout

Context tabs are tabs that only appear when you need them, Called Chart Tools. There are three chart context tabs: Design, Layout, and Format.

The tabs become available when you create a new chart or when you click on a chart. You can use these tabs to customize your chart.

You can determine what your chart displays by choosing a layout. For example, the layout you choose determines whether your chart displays a title, where the title displays, whether

your chart has a legend, where the legend displays, whether the chart has axis labels and so on.

Activity 34 - Excel provides several layouts from which you can choose.

Do the following:

- Click your chart. The Chart Tools become available.
- Choose the Design tab.
- Click the Quick Layout button in the Chart Layout group.
- A list of chart layouts appears.
- Click Layout 5. Excel applies the layout to your chart.

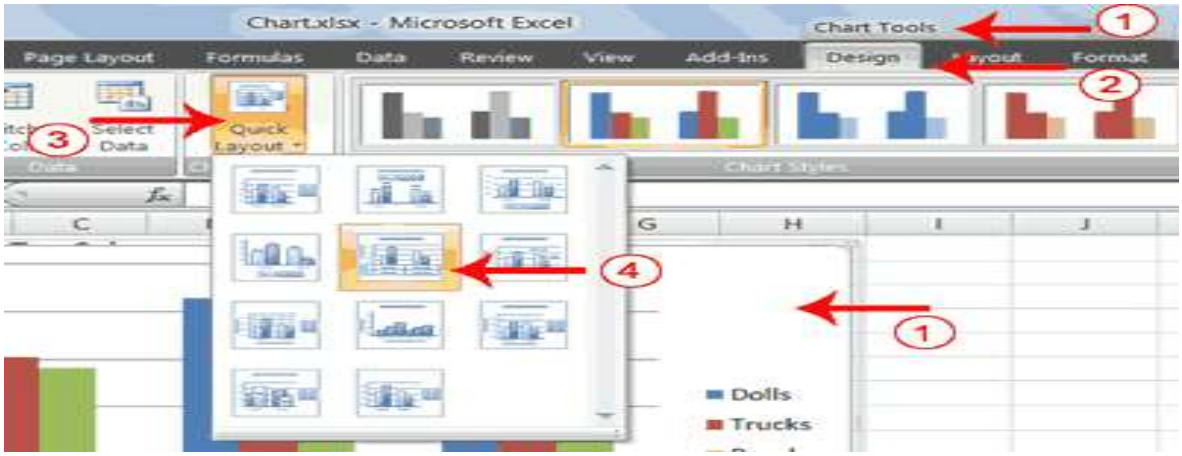


Figure 14: charts layout

Add Labels to a chart

- When you apply a layout, Excel may create areas where you can insert labels.
- You use labels to give your chart a title or to label your axes.
- When you applied layout 5, Excel created label areas for a title and for the vertical axis.

Do the following:

- Select Chart Title. Click on Chart Title and then place your cursor before the C in Chart and hold down the Shift key while you use the right arrow key to highlight the words Chart Title.
- Type Toy Sales. Excel adds your title.
- Select Axis Title. Click on Axis Title. Place your cursor before the A in Axis. Hold down the Shift key while you use the right arrow key to highlight the words Axis Title.
- Type Sales. Excel labels the axis.
- Click anywhere on the chart to end your entry.

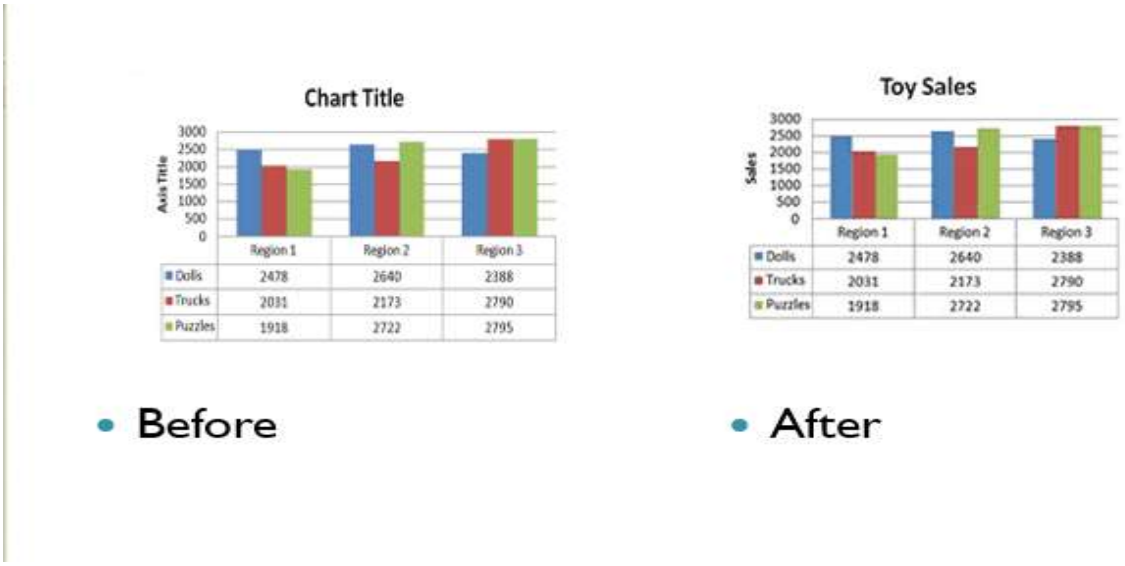


Figure 15: Chart labels

Switch Data in the chart

If you want to change what displays in your chart, you can switch from row data to column data and vice versa.

Do the following:

- Click your chart. The Chart Tools become available.
- Choose the Design tab.
- Click the Switch Row/Column button in the Data group.
- Excel changes the data in your chart.

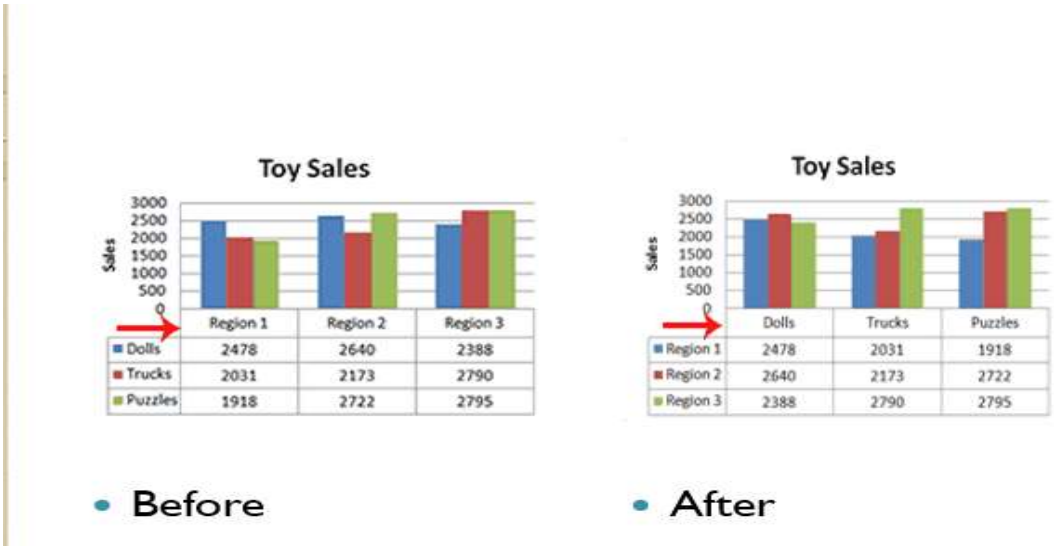


Figure 16: labels on charts

Move a Chart to a Chart Sheet

By default, when you create a chart, Excel embeds the chart in the active worksheet. However, you can move a chart to another worksheet or to a chart sheet. A chart sheet is a sheet

dedicated to a particular chart. By default Excel names each chart sheet sequentially, starting with Chart1. You can change the name.

Activity 35 - Do the following:

- Click your chart. The Chart Tools become available.
- Choose the Design tab.
- Click the Move Chart button in the Location group. The Move Chart dialog box appears.
- Click the New Sheet radio button.
- Type Toy Sales to name the chart sheet. Excel creates a chart sheet named Toy Sales and places your chart on it.

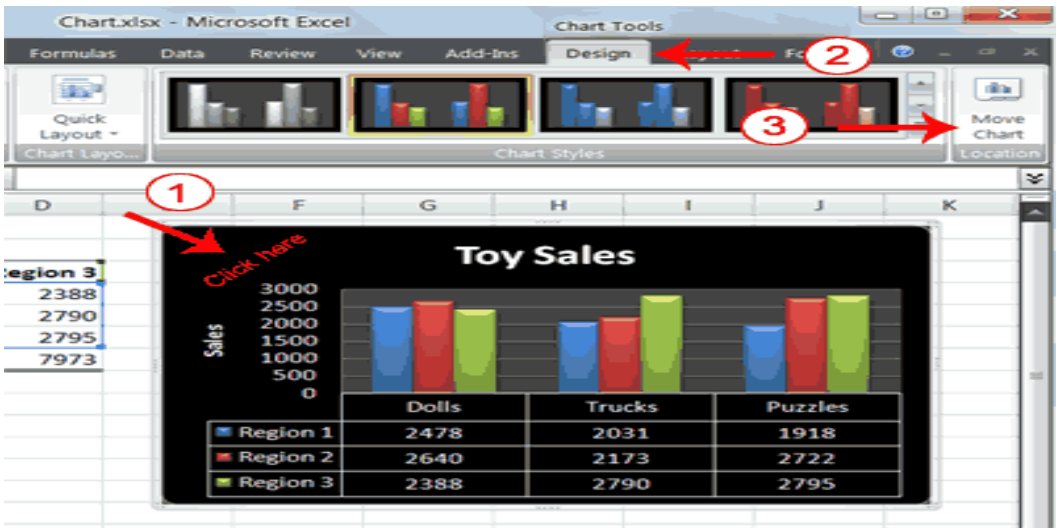


Figure 17: Naming charts

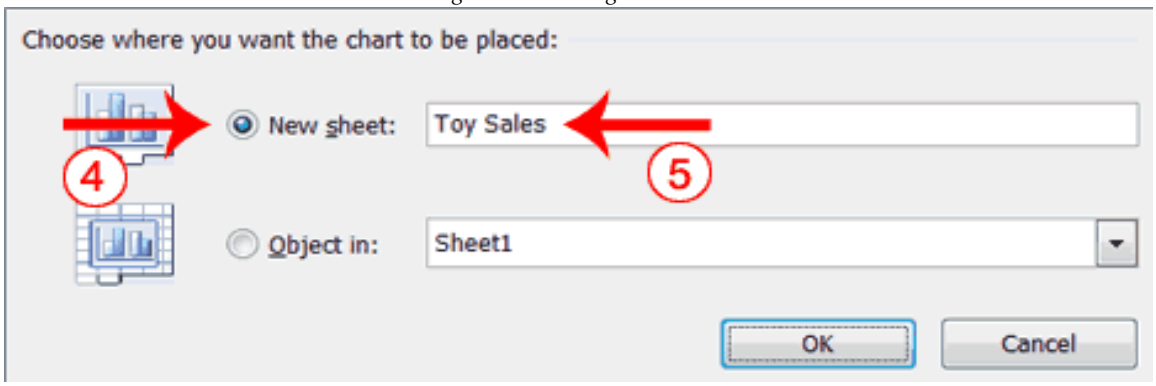


Figure 18 : Placing charts

Change the Chart Type

You can change the chart type from a column chart to a bar chart etc.

Activity 36 - Do the following:

- Click your chart. The Chart Tools become available.
- Choose the Design tab.
- Click Change Chart Type in the Type group. The Chart Type dialog box appears.
- Click Bar.
- Click Clustered Horizontal Cylinder.
- Click OK. Excel changes your chart type.

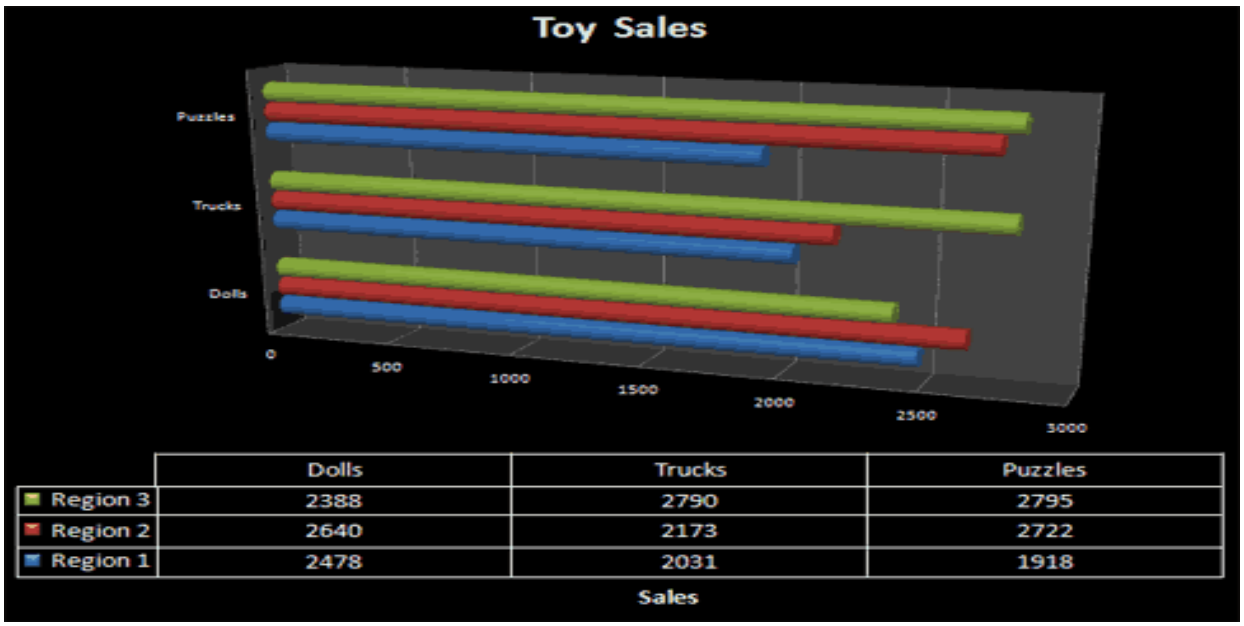


Figure 19: Complete chart

The IF Function in Excel

It is used when you want to test for more than one values. The IF Function looks like this:

IF(logical_test, value_if_true, value_if_false,)

The three items between the round brackets of the word IF are the arguments that the IF function needs.

Here's what they mean:

logical_test

The first argument is what you want to test for.

For example: Is the number in the cell greater than 80?

value_if_true

This is what you want to do if the answer to the first argument is YES. (for example: Award an A grade,).

value_if_false

This is what you want to do if the answer to the first argument is NO. (for example: Award a FAIL grade.)

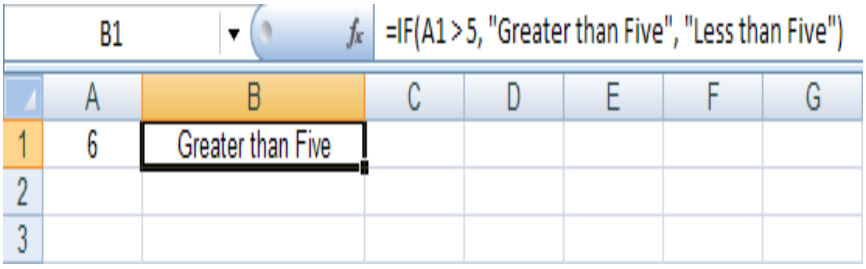


Figure 20: …IF statement

Activity 37 - Do the following:

- Widen the B column a bit, as we'll be putting a message in cell B1
- Now click in cell A1 and type the number 6
- Type the following in the formula bar of B1 (The right angle bracket after A1 means "Greater Than".)
- =IF(A1 > 5, "Greater than Five", "Less than Five")
- The right angle bracket (>) is a Conditional Operator.
- Hit the enter key on your keyboard

Here is what we're saying in the IF function above:

logical_test: Is the value in cell A1 greater than 5?

value_if_true: If the answer is Yes, display the text "Greater than Five"

value_if_false: If the answer is NO, display the text "Less than Five“

NOTE: You first tell Excel what you want to check the cell for, then what you want to do if the answer is YES, and finally what you want to do if the answer is NO. You separate each part with a comma.



Mzuzu University

Excel Self-Assessment Test

Instructions

- Answer all questions

1. Launch Microsoft Excel. Create a **spreadsheet** and enter data as shown below including all the formatting. Save your workbook as **Shop**.

	A	B	C	D	E
1	**Khumbo Investments**				
2					
3	**Shop Owner:**	*Type your Registration Number here*		Commission Rate:	
4				5%	
5					
6	**Item No**	**Selling Price**	**Cost Price**	**Commission**	**Any Profit?**
7	001	MWK 4,500.00	MWK 4,000.00		
8	002	MWK 500.00	MWK 550.00		
9	003	MWK 9,000.00	MWK 1,000.00		
10	004	MWK 1,500.00	MWK 2,200.00		
11	005	MWK 5,000.00	MWK 6,000.00		
12					
13	**Average**				
14	**Highest**				
15	**Lowest**				
16					

8 Marks

2. Type your Registration Number in cell B3. **1 Mark**

3. Insert formulae to calculate the average, highest and lowest Cost Price, in cells C13, C14 and C15 respectively. Replicate the formulae in cells B13, B14 and B15. **5 Marks**

4. Calculate the commission on each item using the formula **Commission = Selling Price *Commission rate**. The commission rate is found in cell D4. Use absolute referencing in your calculation. **5 Marks**

5. Use the **IF function** in cells E7 to E11 to indicate if there was profit on each item given that **Profit = Selling Price – Cost Price.** Output values should be Yes or No. **5 Marks**

6. Rename Sheet1 to Shop. Save your spreadsheet and Exit **1 Mark**



Unit summary

In this unit, we have discussed on how we can work with Microsoft excel and create a worksheet. We have explained how to use, edit, save, close and retrieve a workbook. There also basic calculations that can be done on the worksheet. We also shared on how to create charts and graphs in excel.



References

i. https://support.office.com/en-us/article/excel-for-windows-training-9bc05390-e94c-46af-a5b3-d7c22f6990bb
ii. https://www.guru99.com/excel-tutorials.html
iii. https://www.excel-easy.com/
iv. https://www.excel-easy.com/basics.html
v. https://digital.com/blog/excel-tutorials/
vi. https://trumpexcel.com/learn-excel/
vii. https://www.javatpoint.com/excel-tutorial

UNIT 5

PRESENTATIONS

Introduction

PowerPoint is a computer program that allows you to create and show slides to support a presentation. You can combine text, graphics and multi-media content to create professional presentations.

Areas of emphasis:

- Creating presentations
- Understanding slides
- Slide show
- Graphics in slides



Key Words

- Spreadsheet, Slide, Formatting, Transition, Animation, Slide Master, Templates, et al.

Prerequisite knowledge

This unit assumes you are familiar with the basic operation of a personal computer. Specific skills required for this tutorial are:

- Use of the mouse and keyboard
- Opening and saving files on the hard disk and on USB memory stick (thumb drive)
- Managing files in general i.e. the difference between drive letters, hard disk,etc
- Running programs from the Windows Start menu
- Minimizing, maximizing and resizing windows

You should have a USB memory stick (thumb drive) or other portable memory device available to store the files that will be created in this unit.



Learning outcomes

By the end of this unit, you should be able to:

- Create a Presentation
- Format slides
- Insert and use graphics
- Slide transitions and Animations
- Using the slide show

Resources needed

- a working computer with Microsoft powerpoint package installed.

5.1.0 Creating presentations

PowerPoint files are called presentations. Whenever you start a new project in PowerPoint, you'll need to create a new presentation, which can either be blank or from a template. You'll also need to know how to open an existing presentation.

Advantages of using PowerPoint:

- Easy to design a creative presentation as it provides so many options to insert text, images, charts, graphs, icons, drawings, illustrations, and videos.
- A complicated or long paragraph can be reduced using bullet points.
- Allows to create something visually appealing using the standard templates and themes.

Whenever you create a new presentation in PowerPoint, you'll need to know how to save in order to access and edit it later. You can even export and share presentations directly from PowerPoint.

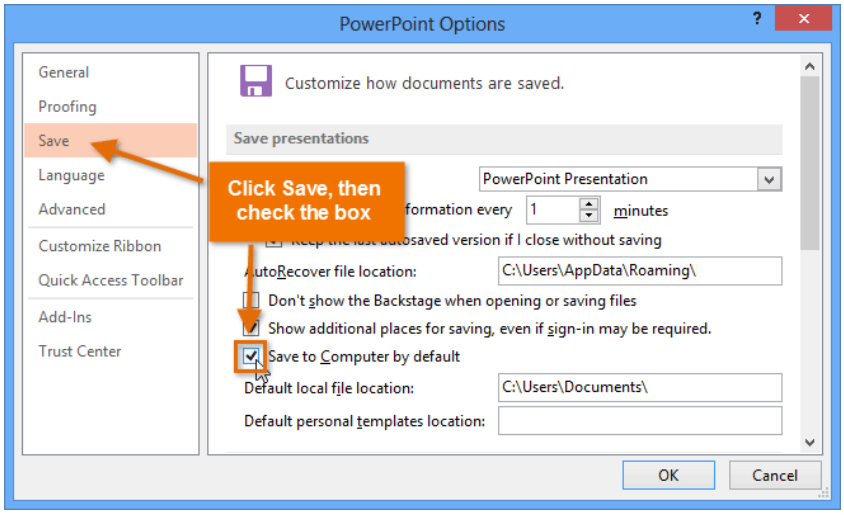


Figure 1 : PowerPoint options dialog box

PowerPoint presentations are made up of a series of slides. Slides contain the information you will present to your audience. This might include text, pictures, and charts. Before you start creating presentations, you'll need to know the basics of working with slides and slide layouts.

Understanding slides and slide layouts

When you insert a **new slide**, it will usually have **placeholders**. Placeholders can contain different types of content, including **text** and **images**. Some placeholders have **placeholder text**, which you can replace with your own text. Others have **thumbnail icons** that allow you to insert pictures, charts, and videos.



Figure 2 : PowerPoint thumbnail icons

Slides have different **layouts** for placeholders, depending on the type of information you want to include. Whenever you create a new slide, you'll need to choose a slide layout that fits your content.

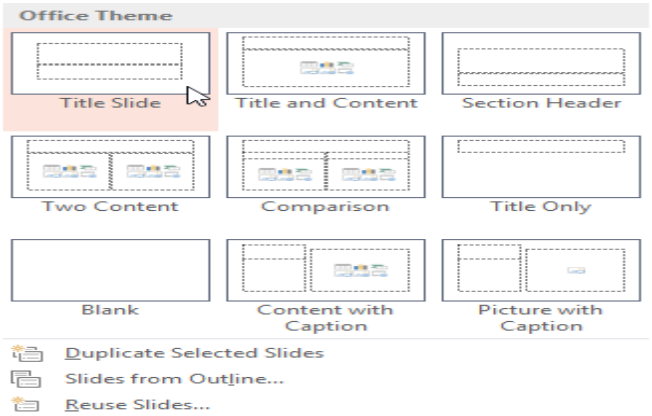


Figure 3 : Placeholder layout

To insert a new slide:

Whenever you start a new presentation, it will contain **one slide** with the **Title Slide** layout. You can insert as many slides as you need from a variety of layouts.



Activity 1

1. From the **Home** tab, click the **bottom half** of the **New Slide** command.
2. Choose the desired **slide layout** from the menu that appears.
3. The new slide will appear. Click any **placeholder** and begin typing to add text. You can also click an **icon** to add other types of content, such as a **picture** or a **chart**.

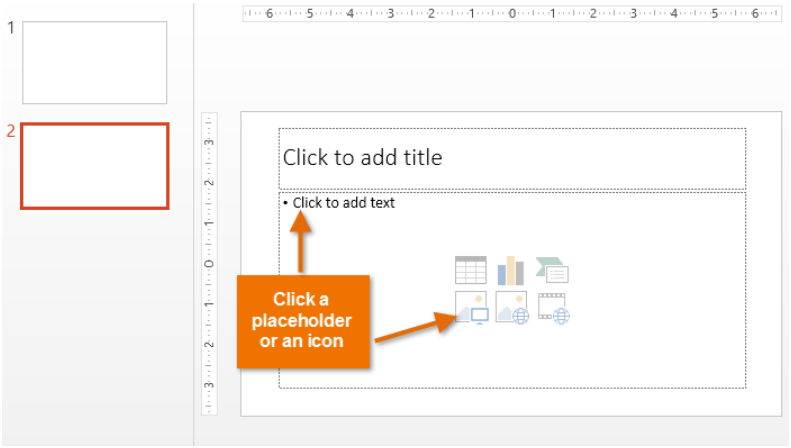


Figure 4 : Creating New Slides

To change the layout of an existing slide, click the **Layout** command, then choose the desired layout.

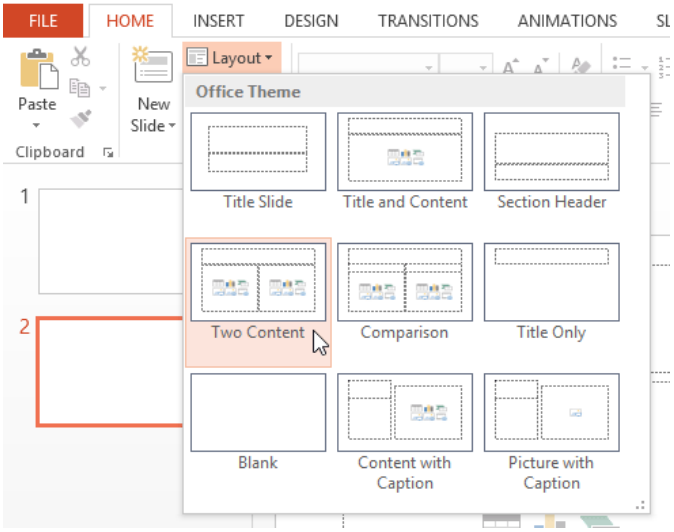


Figure 5 : layout command

To quickly add a slide that uses the same layout as the selected slide, click the top half of the **New Slide** command.



Figure 6 : New Slide option

5.1.1 Organizing slides

PowerPoint presentations can contain as many slides as you need. The **Slide Navigation pane** makes it easy to **organize** your slides. From there, you can **duplicate**, **rearrange**, and **delete** slides in your presentation.



Figure 7 : Slide Navigation pane

Working with slides

- **Duplicate slides:** If you want to copy and paste a slide quickly, you can **duplicate** it. To duplicate slides, select the slide you want to **duplicate**, right-click the mouse, and choose **Duplicate Slide** from the menu that appears. You can also duplicate **multiple slides** at once by selecting them first.

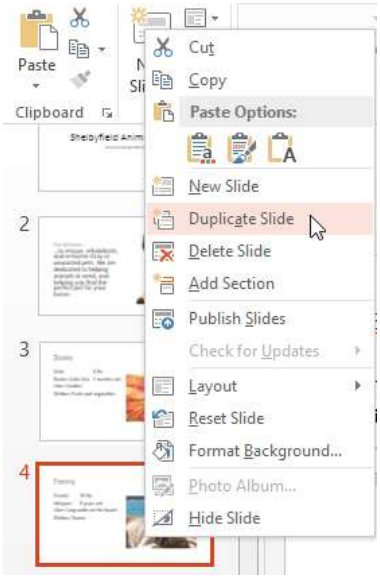


Figure 8 : Duplicate slide

- **Move slides:** It's easy to change the **order** of your slides. Just click, hold, and drag the **desired slide** in the Slide Navigation pane to the desired position.
- **Delete slides:** If you want to remove a slide from your presentation, you can **delete** it. Simply select the slide you want to delete, then press the **Delete** or **Backspace** key on your keyboard.

To copy and paste slides:

If you want to create several slides with the same layout, you may find it easier to **copy and paste** a slide you've already created instead of starting with an empty slide.

1. Select the **slide** you want to copy in the Slide Navigation pane, then click the **Copy** command on the **Home** tab.
2. In the Slide Navigation pane, click just below a slide (or between two slides) to choose a paste location. A **horizontal insertion point** will appear.
3. Click the **Paste** command on the **Home** tab.
4. The slide will appear in the selected location.

Customizing slides

To change the slide size:

By default, all slides in PowerPoint 2013 use a **16 by 9**—or **widescreen**—aspect ratio. You might know that widescreen TVs also use the 16-by-9 aspect ratio. Widescreen slides will work best with widescreen monitors and projectors. However, if you need your presentation to fit a standard **4-by-3** screen, it's easy to change the slide size to fit.

- To change the slide size, select the **Design** tab, then click the **Slide Size** command. Choose the desired slide size from the menu that appears, or click **Custom Slide Size...** for more options.

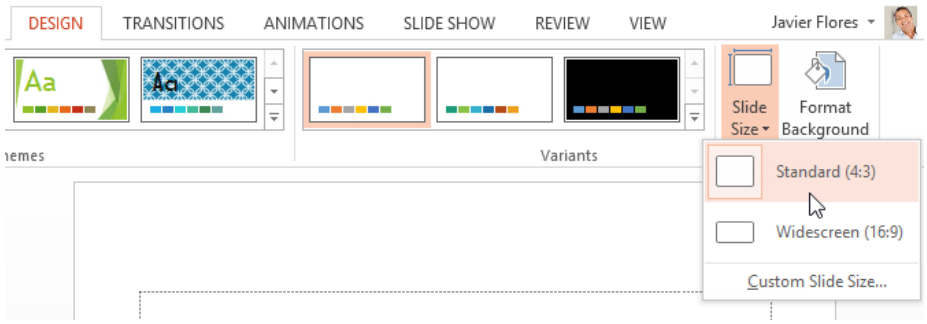


Figure 9 : Slide sizing pane

To format the slide background:

By default, all slides in your presentation use a **white background**. It's easy to change the background style for some or all of your slides. Backgrounds can have a **solid**, **gradient**, **pattern**, or **picture** fill.

1. Select the **Design** tab, then click the **Format Background** command.
2. The **Format Background** pane will appear on the right. Select the desired fill options. In our example, we'll use a **Solid fill** with a **light gold** color.

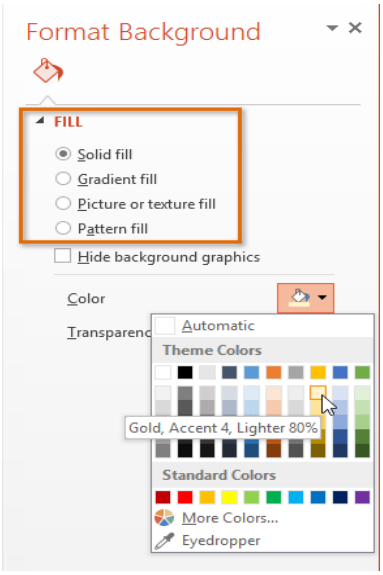


Figure 10 : Format Background pane

3. The background style of the selected slide will update.
4. If you want, you can click **Apply to All** to apply the same background style to all slides in your presentation.

Customizing slide layouts

Sometimes you may find that a slide layout doesn't exactly fit your needs. For example, a layout might have too many—or too few—placeholders. You might also want to change how the placeholders are arranged on the slide. Fortunately, PowerPoint makes it easy to adjust slide layouts as needed.



Activity 2 - Adjusting placeholders

- **To select a placeholder:** Hover the mouse over the edge of the placeholder and click (you may need to click on the text in the placeholder first to see the border). A selected placeholder will have a **solid line** instead of a dotted line.
- **To move a placeholder:** Select the placeholder, then click and drag it to the desired location.
- **To resize a placeholder:** Select the placeholder you want to resize. **Sizing handles** will appear. Click and drag the **sizing handles** until the placeholder is the desired size. You can use the corner sizing handles to change the placeholder's **height** and **width** at the same time.
- **To delete a placeholder:** Select the placeholder you want to delete, then press the **Delete** or **Backspace** key on your keyboard.

To add a text box:

Text can be inserted into both **placeholders** and **text boxes**. Inserting **text boxes** allows you to add to the slide layout. Unlike placeholders, text boxes always stay in the same place, even if you change the theme.

Activity 3

1. From the **Insert** tab, select the **Text Box** command.

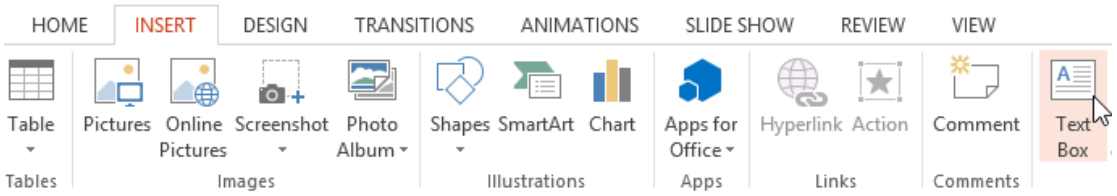


Figure 11 : Insert tab

2. Click, hold, and drag to draw the text box on the slide.



Figure 12 : drawing a text box

3. The text box will appear. To add text, simply click the text box and begin typing.

Using blank slides

If you want even more control over your content, you may prefer to use a **blank slide**, which contains no placeholders. Blank slides can be customized by adding your own text boxes, pictures, charts, and more.

Activity 4

To insert a blank slide, click the bottom half of the **New Slide** command, then choose **Blank** from the menu that appears.

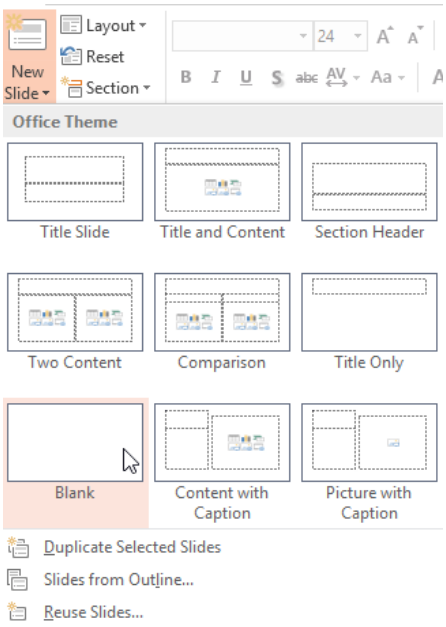


Figure 13 : New Slide command

While blank slides offer more flexibility, keep in mind that you won't be able to take advantage of the predesigned layouts included in each theme.

To play the presentation:

Once you've arranged your slides, you may want to **play** your presentation. This is how you will present your slide show to an audience.

1. Click the **Start From Beginning** command on the Quick Access toolbar to see your presentation.

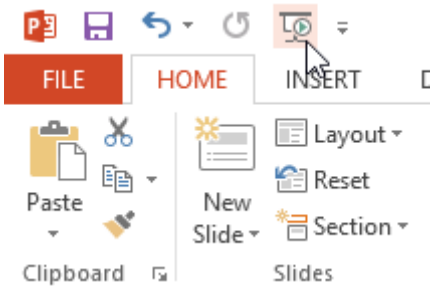


Figure 14 : slide show

2. The presentation will appear in full-screen mode.
3. You can advance to the next slide by **clicking your mouse** or pressing the **spacebar** on your keyboard. Alternatively, you can use the **arrow keys** on your keyboard to move forward or backward through the presentation.
4. Press the **Esc** key to exit presentation mode.

You can also press the **F5** key at the top of your keyboard to start a presentation.

Activity 5

1. Create a **new PowerPoint presentation.**
2. Insert a **new slide** with the **Title and Content** layout.
3. Try adding some **text** to the title placeholder.
4. Change the **layout** of a slide. If you are using the example, change the layout of slide 3 to **Section Header**.
5. **Copy and paste** a slide, then **move** it to a new location.
6. Apply a **theme** to your presentation. If you're using the example, apply the **Integral** theme.
7. Insert a **blank** slide, then insert a **text box** on the slide.
8. Change the slide size from **Widescreen** to **Standard**.

5.1.2 Formatting presentations

A theme is a predefined combination of colors, fonts, and effects. Different themes also use different slide layouts. You've already been using a theme, even if you didn't know it: the default Office theme. You can choose from a variety of new themes at any time, giving your entire presentation a consistent, professional look.

Theme elements

Every PowerPoint theme—including the default Office theme—has its own **theme elements**. Those elements are:



Figure 14 : Themes

1. Select the **desired theme**.
2. The theme will be applied to your **entire presentation**.

Try applying a few different themes to your presentation. Some themes will work better than others, depending on your content.

Theme Colors: There are **10 theme colors**, along with darker and lighter variations, available from every **Color** menu.

Theme Fonts: There are **two theme fonts** available at the top of the **Font** menu under Theme Fonts.

Theme Effects: These affect the present **shape styles**. You can find shape styles on the **Format** tab whenever you select a shape or SmartArt graphic.



Figure 14 : Shape styles

Why use themes?

If you're using a theme, you'll probably find that your presentation looks pretty good. All of the colors will work well together, which means you won't have to spend as much time formatting your presentation. But there's another great reason to use theme elements: When you switch to a different theme, **all of these elements will update** to reflect the new theme. You can drastically change the look of your presentation in a few clicks.



Figure 15 : theme elements

In our examples above, you can see the effect of applying different themes to the same slide—each theme uses its own **fonts** and **colors**. But you may have also noticed that the font and colors of the logo in the bottom-right remained **unchanged**: This is because they're **Standard Colors** and **Fonts** rather than theme elements. Colors and fonts will only update if you're using **Theme Fonts** or **Theme Colors**.

Themes and slide layouts

As you can see from the two different **Title Slides** above, themes can also change various **slide layouts**. Some themes, like the **Wisp** theme in the example below, even include **additional** layouts.

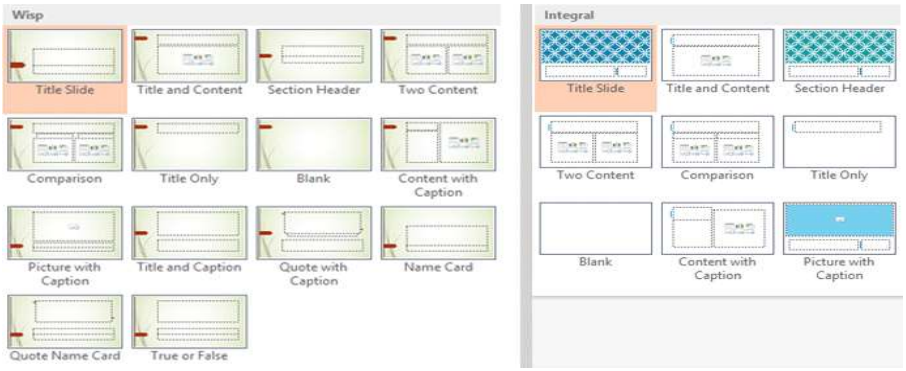


Figure 16 : theme types

If you use a **unique** slide layout—such as **Quote with Caption** or **Name Card**—and then switch to a theme that does not include that layout, it may give unexpected results.

Once you've applied a theme, you can also select a **variant** for that theme from the **Variants** group. Variants use **different theme colors** while preserving a theme's overall look. Some variants also use different **textures**, as in the example below.

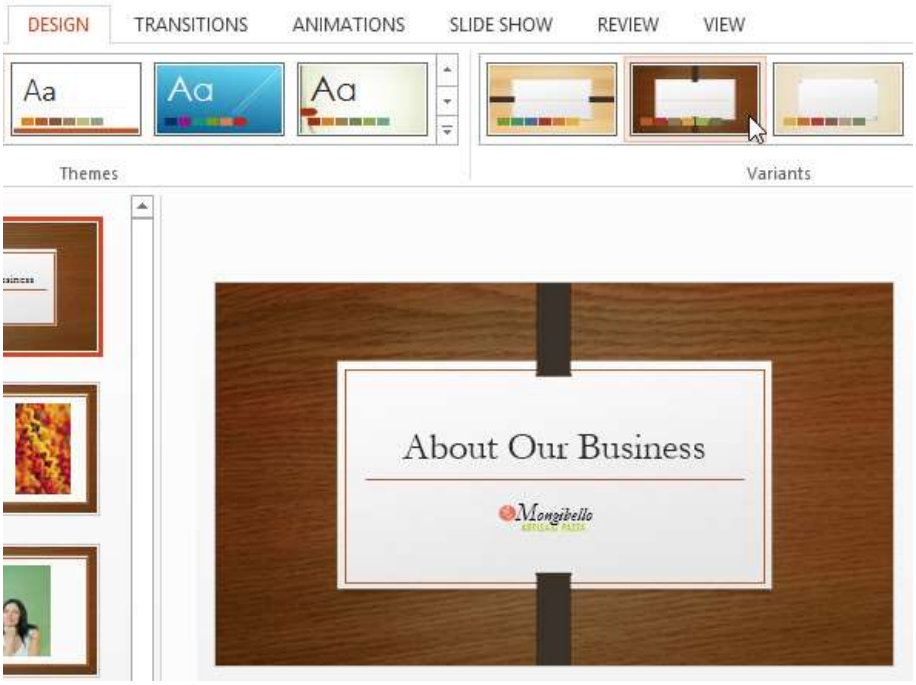


Figure 16 : theme variants



Activity 6

1. Open an **existing PowerPoint presentation**.
2. Try applying several different **themes**. If you are using the example, try the **Organic** theme.
3. Try selecting a theme **variant**.

5.1.3 Transitions

If you've ever seen a PowerPoint presentation that had special effects between each slide, you've seen slide transitions. A transition can be as simple as fading to the next slide or as flashy as an eye-catching effect. PowerPoint makes it easy to apply transitions to some or all of your slides, giving your presentation a polished, professional look.

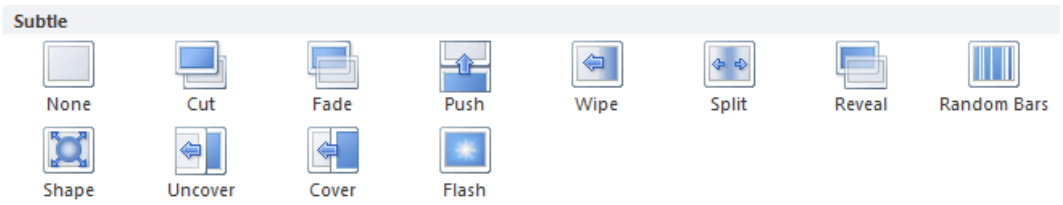
Transitions are motion effects that when in Slide Show view add movement to your slides as you advance from one slide to another. There are many transitions to choose from, each one of which allows you to control the speed and even add sound.

Applying transitions

If you've ever seen a PowerPoint presentation that had **special effects** between each slide, you've seen **slide transitions**. A transition can be as simple as fading to the next slide or as complex as a flashy, eye-catching effect. This means you can choose transitions to fit the style of any presentation.

There are **three categories** of unique transitions to choose from, all of which can be found on the **Transitions** tab:

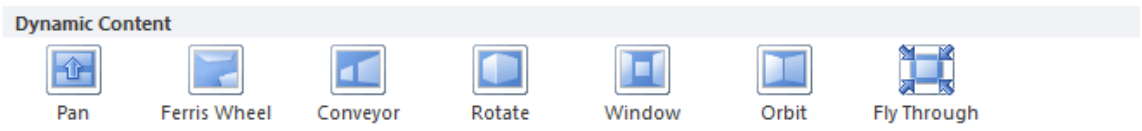
- **Subtle** (slight transitions)



- **Exciting** (strong transitions)



- **Dynamic Content** (strong transitions that affect only the content, such as text or images)



Activity 7 - ***To apply a transition:***

1. Select the **slide** you want to modify.
2. Click the **Transitions** tab.
3. Locate the **Transition to This Slide** group. By default, **None** is applied to each slide.

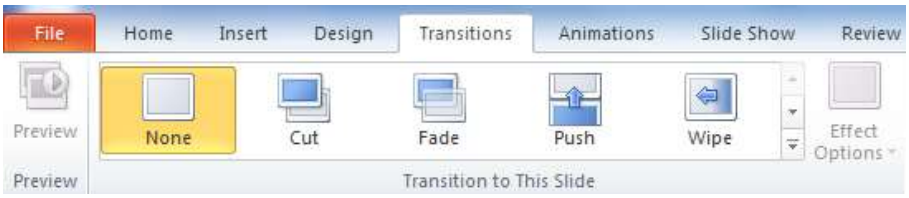


Figure 17 : transition tab

4. Click the **More** drop-down arrow to display all of the transitions.
5. Click a **transition** to apply it to the selected slide. This will automatically preview the transition as well.

When working with transitions, the **Apply To All** command in the **Timing** group can be used at any time to make your presentation uniform. Use this command with caution. Not only does it apply the **same transition** to every slide, but it also applies the settings in the **Timing** group, which you may not want to be the same throughout your presentation.

To preview a transition:

You can **preview** the transition for a selected slide at any time, using either of these two methods:

- Click the **Preview** command on the **Transitions** tab.

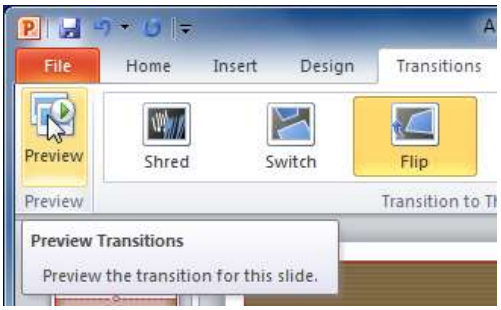


Figure 18 : transition preview

- Click the star **Play Animations** icon. The icon appears on the **Slides** tab in the **left pane** beside any slide that includes a transition.

Activity 8 - *Modifying transitions*

1. Select the slide that includes the **transition** you want to modify.
2. In the **Duration** field in the **Timing** group, enter the amount of time you want the transition to take. In this example, we will specify the length as 2 seconds, or 02.00.

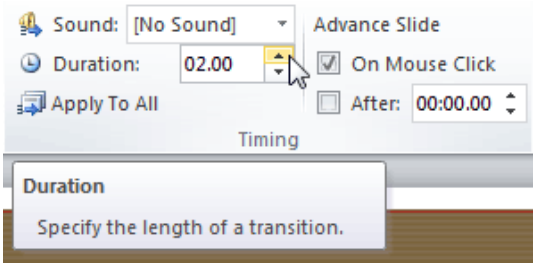


Figure 19 : timing transitions

Activity 9 - *To add sound:*

1. Select the slide that includes the **transition** you want to modify.
2. Click the **Sound** drop-down menu in the **Timing** group.
3. You will hear the sound and see a live preview of the transition as you hover over each sound.

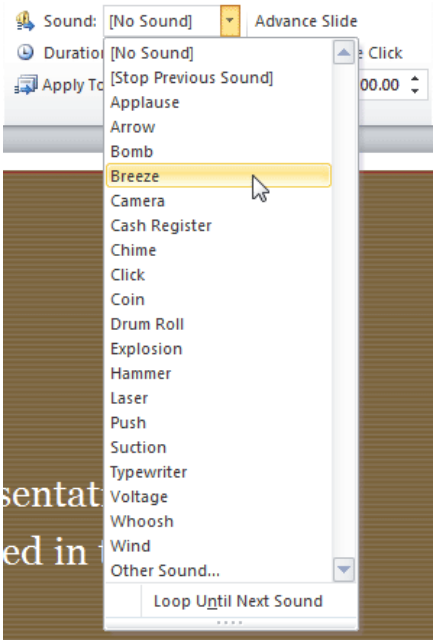


Figure 20 : adding sounds

4. Click a **sound** to apply it to the selected slide.

To remove a transition:

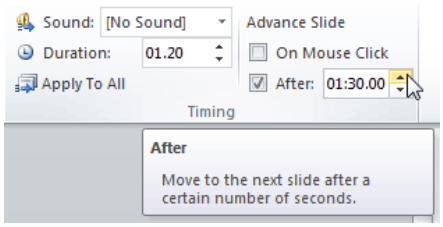
1. Select the **slide** you want to modify.
2. Choose **None** from the gallery in the **Transition to This Slide** group.
3. Repeat this process for each slide you want to modify.

To remove transitions from **all slides**, select a slide that uses **None**, then click the **Apply to All** command.

In Slide Show view, you would typically advance to the next slide by clicking your mouse or pressing Enter on your keyboard. Using the **Advance Slides** settings in the **Timing** group, you can set your presentation to advance on its own instead and display each slide for a specific amount of time. This is useful for unattended presentations, such as at a tradeshow booth.

 **Activity 10**

1. Select the slide you want to modify.
2. Locate the **Timing** group on the **Transitions** tab.
3. Under **Advance Slide**, uncheck the box next to **On Mouse Click**.
4. In the **After** field, enter the amount of time you want to display the slide. In this example, we'll advance the slide automatically after 1 minute 30 seconds, or 01:30:00.



5. Select another slide, and repeat the process until all the desired slides have the appropriate timing.

5.1.4 Editing the slide master

You may have noticed that when you select a different theme in PowerPoint, it rearranges the text on your slides and adds shapes to the background. This is because each theme has built-in slide layouts and background graphics. You can edit these layouts with a feature called Slide Master view. Once you learn how to use Slide Master view, you'll be able to customize your entire slide show with just a few clicks

Slide Master view is a special feature in PowerPoint that allows you to quickly modify the slides and slide layouts in your presentation. From there, you can edit the **slide master**, which will affect **every slide** in the presentation. You can also modify individual **slide layouts**, which will change any slides using those layouts.

For example, let's say you find a theme you like but you don't like the slide layouts. You could use Slide Master view to customize the layouts exactly the way you want them to look.

In Slide Master view, the **Slide Master** tab will appear first on the Ribbon, but you'll still be able to access commands on different tabs as normal.

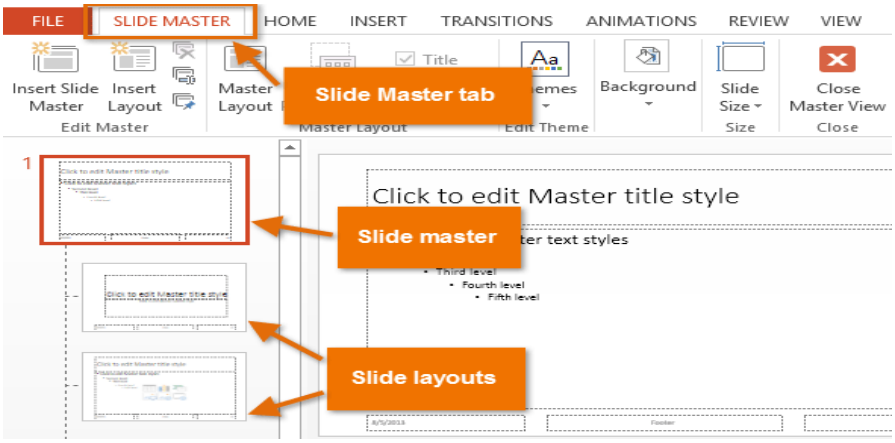


Figure 21 : slide master

Using Slide Master view

Whether you're making **significant changes** to your slides or just a few **small tweaks**, Slide Master view can help you create a consistent, professional presentation without a lot of effort. You could use Slide Master view to change just about anything in your presentation, but here are some of its most common uses:

- **Modify backgrounds:** Slide Master view makes it easy to customize the **background** for all of your slides at the same time. For example, you could add a watermark or logo to each slide in your presentation, or you could modify the background graphics of an existing PowerPoint theme.
- **Rearrange placeholders:** If you find that you often rearrange the placeholders on each slide, you can save time by rearranging them in Slide Master view instead. When you adjust one of the layouts in Slide Master view, all of the slides with that layout will change.
- **Customize text formatting:** Rather than changing the text color on each slide individually, you could use the Slide Master to change the text color on all slides at once.
- **Create unique slide layouts:** If you want to create a presentation that looks different from regular PowerPoint themes, you could use Slide Master view to create your own layouts. Custom layouts can include your own **background graphics** and **placeholders**.

Some overall presentation changes—like customizing the **theme fonts** and **theme colors**—can be made quickly from the **Design** tab.

To make changes to all slides:

If you want to change something on all slides of your presentation, you can edit the Slide Master. In our example, we'll add a **logo** to every slide.

1. Select the **View** tab, then click the **Slide Master** command.
2. The presentation will switch to **Slide Master view**, and the **Slide Master** tab will be selected on the Ribbon.
3. In the left navigation pane, scroll up and select the **first slide**. This is the **slide master**.

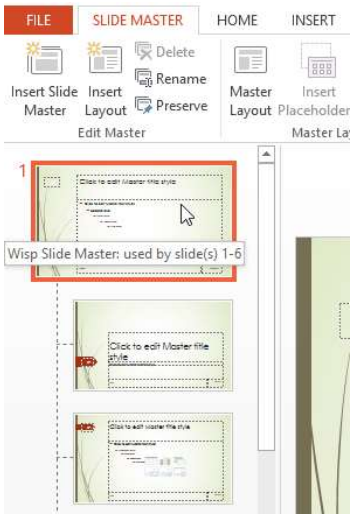


Figure 22 : slide master view

4. Use the desired **tabs** on the Ribbon to make changes to the Slide Master. In our example, we'll **insert a picture** in the upper-left corner of the slide.
5. **Move**, **resize**, or **delete** other slide objects as needed. In our example, we'll resize the **background graphic** to fit below our newly inserted logo.
6. When you're finished, click the **Close Master View** command on the **Slide Master** tab.
7. The change will appear on all slides of the presentation.

When you make a change to the Slide Master, it's a good idea to review your presentation to see how it affects each slide. You may find that some of your slides don't look exactly right.

If you're inserting a **background graphic**, you may want to send it **behind** all other objects on the slide. Right-click the object, select **Send to Back**, then click **Send to Back**.

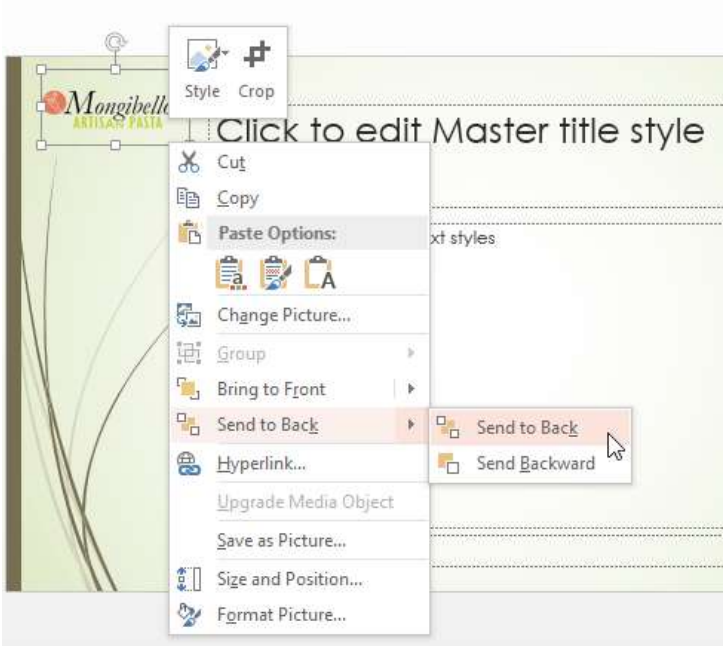


Figure 23 : editing slide master

Customizing text formatting

You can also customize the **text formatting** from Slide Master view, including the **font**, **text size**, **color**, and **alignment**. For example, if you wanted to change the font for every **title placeholder** in your presentation, you could modify the **master title style** on the Slide Master.



Figure 24: customizing text in a slide master

Each title placeholder is **connected** to the master title style on the Slide Master. As you can see in the example below, modifying the font on the Slide Master changed the title font on each slide.

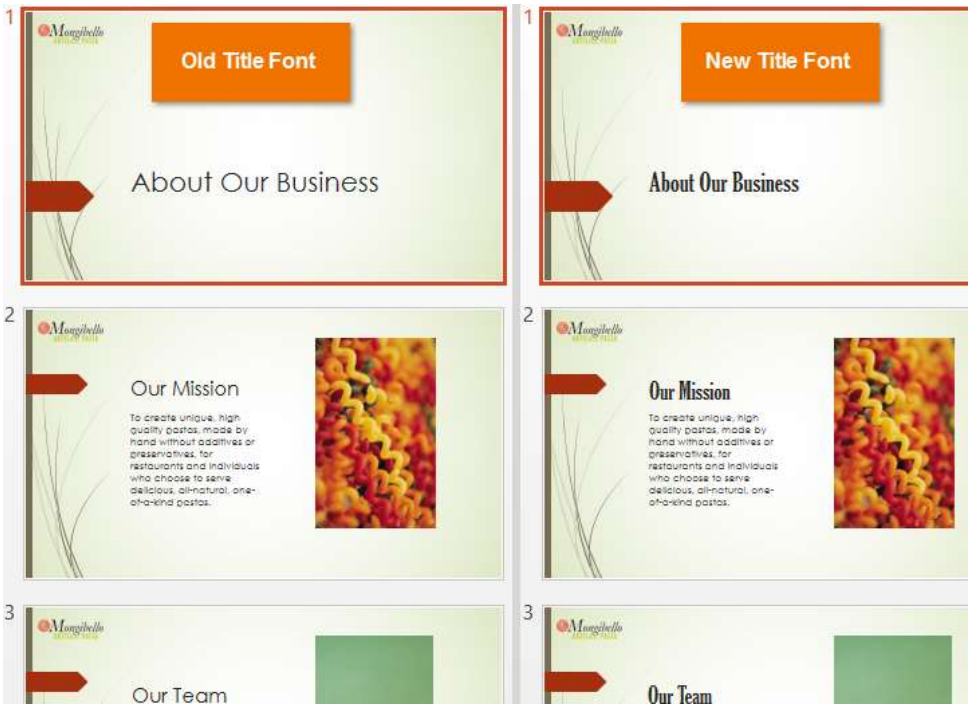


Figure 25 : Modified slide master

Rather than customizing individual placeholders, you could change the **theme fonts** for a presentation. From the **Slide Master** tab, click the **Fonts** command in the **Background** group, then select the desired fonts.

Creating new slide layouts

One of the most powerful features of Slide Master view is the option to create **new slide layouts**. This is an easy way to add interesting and unique slide layouts to an existing theme. You could even use this feature to design an entirely **new theme**, as in the example below.

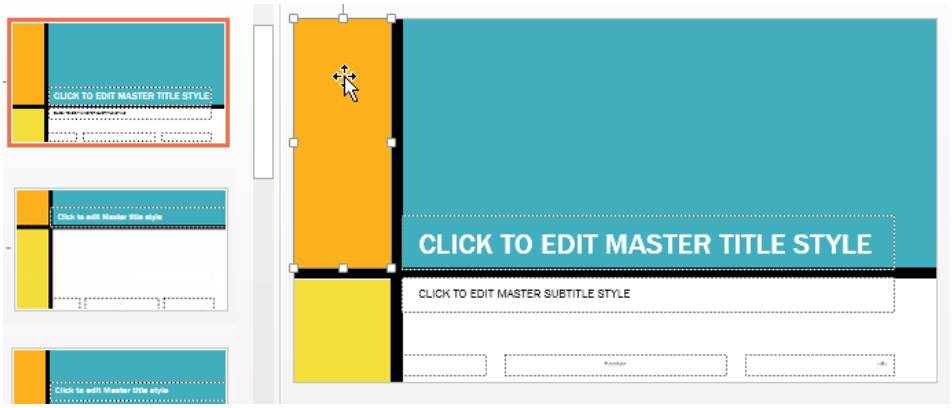


Figure 26: slide layouts

To insert a new slide layout:

1. Navigate to **Slide Master view**.
2. From the **Slide Master** tab, click the **Insert Layout** command.
3. A new slide layout will appear.
4. The layout will include **Title** and **Footer** placeholders by default. Click the **Title** and **Footers** boxes in the **Master Layout** group to toggle these placeholders on and off.

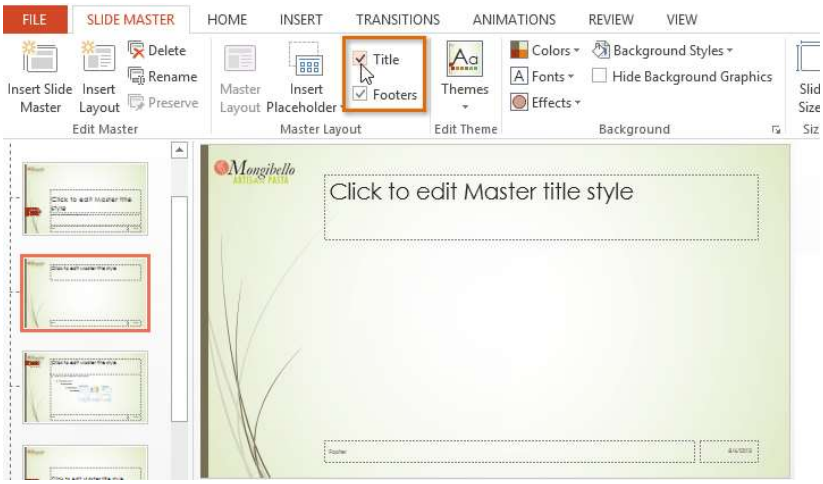


Figure 27 : New slide layout

5. Click the **bottom half** of the **Insert Placeholder** command, then select the desired placeholder type. We recommend using the **Content** placeholder most of the time because it can be used for any type of slide content.
6. Click and drag to draw the placeholder on the slide. You can add more placeholders if desired.
7. Using the other tabs on the Ribbon, you can also add background graphics, shapes, or pictures to the slide layout.

PowerPoint also allows you to add new placeholders to an **existing** slide layout. In the example below, we're adding a **Picture** placeholder to the **Section Header** layout.

5.1.5 Inserting Graphics

Adding a video to your presentation can help emphasize certain points and provide an example. Once you add a video, you can **edit** it and **format** its appearance.

To insert a video from a file on your computer:

1. From the **Insert** tab, click the **Video** drop-down arrow and select **Video from File**.

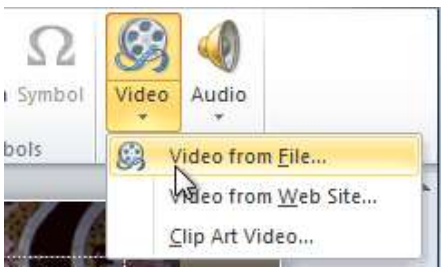


Figure 28 : Inserting a video

2. Locate and select the desired video file, then click **Insert**.
3. The video will be added to the slide.

Editing and formatting videos

The **Playback** tab has several options you can use to edit your video. For example, you can **trim** your video so it will only play an excerpt, add a **fade in** and **fade out**, and add **bookmarks** that allow you to jump to specific points in the video.

Most of the features on the Playback tab can only be used with videos that are inserted from a file. They will not work with embedded videos.

To trim the video:

1. From the **Playback** tab, click the **Trim Video** command. The Trim Video dialog box will appear.



Figure 29 : Trim video dialog box

2. Use the **green** and **red** handles to set the start and end times.
3. To preview the video file, click the **Play** button.
4. Adjust the green and red handles again if necessary, then click **OK**.

To add a fade in and fade out:

1. On the **Playback** tab, locate the **Fade In** and **Fade Out** fields.
2. Type the desired values, or use the **up** and **down** arrows to adjust the times.

Video options

There are other options you can set to control how your video plays. These are found in the **Video Options** group on the **Playback** tab.

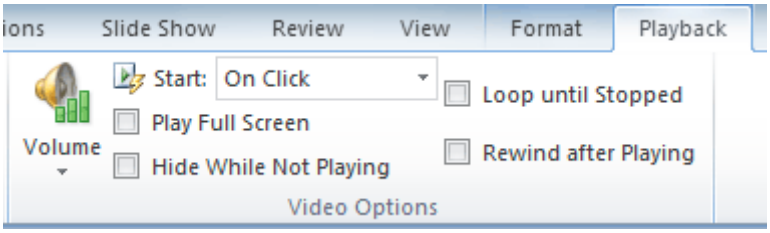


Figure 30 : Video options group

- **Volume**: This changes the audio volume in the video.
- **Start**: This controls whether the video file starts **automatically** or when the mouse is **clicked**.
- **Play Full Screen**: This lets the video fill the entire screen while it's playing.
- **Hide While Not Playing**: This hides the video when it's not playing.
- **Loop Until Stopped**: This causes the video to repeat until it is stopped.
- **Rewind After Playing**: This causes the video to return to the beginning when it's finished playing.

Working with images

Adding images to your presentations makes them more interesting and engaging. Pictures, clip art, and screenshots can be inserted into PowerPoint to help you effectively communicate your ideas to your audience.

Adding **clip art** and **pictures** to your presentation can be a great way to **illustrate important information** or **add decorative accents** to existing text. You can insert images from your computer, search Microsoft's large selection of clip art to find the image you need, or add a screenshot of your own. Once an image has been inserted, you can resize and move it to the location you want.

To insert an image from a file:

1. Select the **Insert** tab.
2. Click the **Picture** command in the **Images** group. The Insert Picture dialog box appears.
3. Select the desired image file, then click **Insert**.



Figure 31 : insert image

4. The picture will appear in your slide.



Figure 32 : image in a slide

You can also select the **Insert Picture from File** command in a **placeholder** to insert images.



Figure 33 : insert image placeholder

To locate clip art:

1. Select the **Insert** tab.
2. Click the **Clip Art** command in the **Images** group.
3. The clip art options appear in the **task pane** to the right of the document.
4. Enter keywords in the **Search for:** field that are related to the image you want to insert.
5. Click the drop-down arrow in the **Results should be:** field.
6. Deselect any types of media you do not want to see.

7. If you want to also search for clip art on Office.com, place a check mark next to **Include Office.com content**. Otherwise, it will just search for clip art on your computer.
8. Click **Go**.

To insert clip art:

1. Review the results from a clip art search in the **Clip Art** pane.
2. Select the desired image.
3. The clip art will appear in your slide.

You can also select the **Insert Clip Art from File** command in a **placeholder** to insert clip art.

Inserting screenshots

Screenshots are pictures that capture the visible windows and items displayed on your computer screen. They may include an open window of a website, items on your desktop, or an open program, like the PowerPoint images displayed in this tutorial. These images can be useful for explaining or displaying computer programs, functions, and websites. PowerPoint allows you to capture an image of an entire window or a **screen clipping** of part of a window.

To insert screenshots of a window:

1. Select the **Insert** tab.
2. Click the **Screenshot** command in the **Images** group.
3. The **Available Windows** from your desktop will appear. Select the window you want to capture as a screenshot.
4. The screenshot will appear in your slide.

To insert a screen clipping from a window:

1. Select the **Insert** tab.
2. Click the **Screenshot** command, then select **Screen Clipping**.
3. A faded view of your current desktop will appear, and your cursor will turn into a cross shape  .
4. Click, hold, and drag on the area of the window you want to capture.
5. The screen clipping will appear in your slide.

Resizing and moving images

To resize an image:

1. Click the image.
2. Position your mouse over any one of the **corner sizing handles**. The cursor will become a pair of directional arrows  .
3. Click, hold, and drag your mouse until the image is the desired size.

4. Release the mouse. The image will be resized.

The side sizing handles change the image's size but do not keep the same proportions. If you want to keep the image's proportions, always use the corner handles.

To move an image:

1. Click the image. The cursor will turn into a cross with arrows  .
2. While holding down the mouse button, **drag the image** to the desired location.
3. Release the mouse button. The box will be moved.

To rotate the image, click and drag on the green circle located at the top of the image.

 **Activity 11**

1. Open an **existing PowerPoint presentation**.
2. Insert an **image** from your files.
3. Insert a **clip art** image. If you are using the example, search for a healthy image to add to slide 2.
4. Capture and insert a **screenshot** or screen clipping into a slide.
5. **Resize** and **move** an image.

5.1.6 Organisation chart

Whether for business or a family tree, it's easy to create an organizational chart using SmartArt in Microsoft PowerPoint. Let's get started.

Head to the "Insert" tab and then click "SmartArt." In the Choose a SmartArt Graphic window that opens choose the "Hierarchy" category on the left. On the right, click an organization chart layout, such as "Organization Chart." When you're done, click "OK."



Figure 34 : SmartArt graphic

Click a box in the SmartArt graphic, and then type your text.

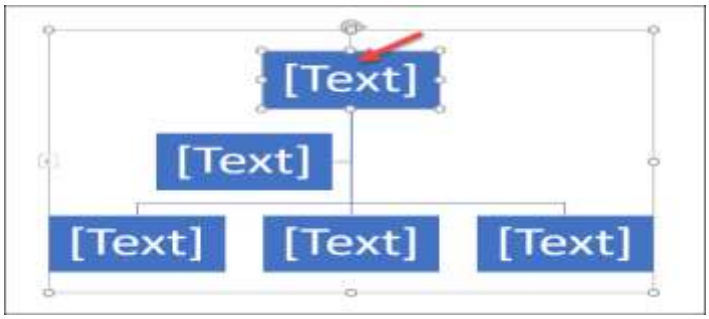


Figure 35 : SmartArt graphic add text

Type the text you want to replace the placeholder text. Click on each additional text box in the SmartArt graphic and then type your text in those, as well.

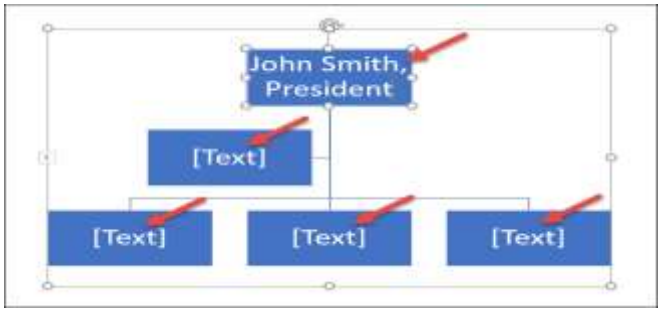


Figure 36 : SmartArt making of a chart

Here's an example of what your organizational chart might look like so far:

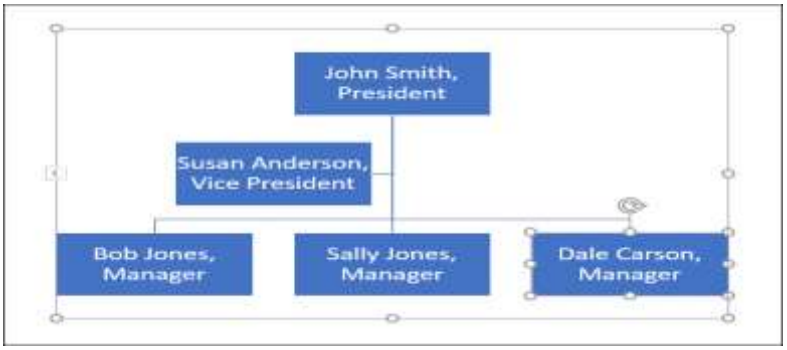


Figure 37 : Organization chart

As an alternative, you can also type text in a text pane instead of directly in the boxes. If the "Type Your Text Here" pane is not visible, click the control on the edge of the SmartArt graphic.

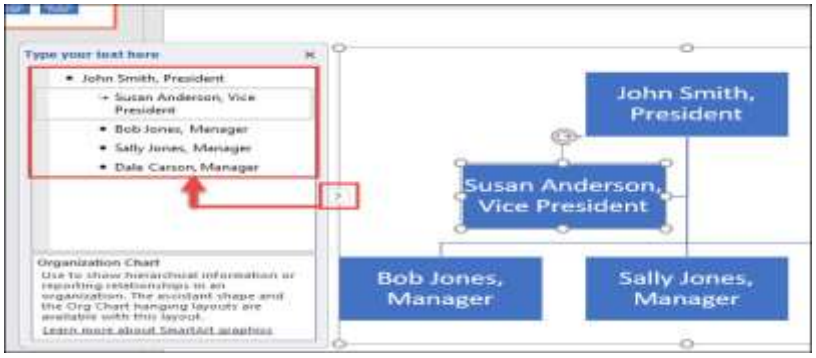


Figure 38 : SmartArt Graphic control edge

To insert a new box, click the existing box that is located closest to where you want to add the new box. On the Design tab, click "Add Shape." Type your new text directly into the new box or via the text pane.

5.1.7 Running a slide show

Once your slide show is completed, you will need to learn how to present it to an audience. PowerPoint offers several tools and features that can help make your presentation smooth, engaging, and professional.

Presenting a slide show

To present your slide show, you'll need to know how to start it. PowerPoint allows you to start your slide show from the first slide or from any slide within the slide show. Once your slide show has started, you'll need to know how to advance through the slides.

To start a slide show:

1. Select the **Slide Show** tab.
2. Click the **From Beginning** command in the **Start Slide Show** group to start the slide show with the first slide.

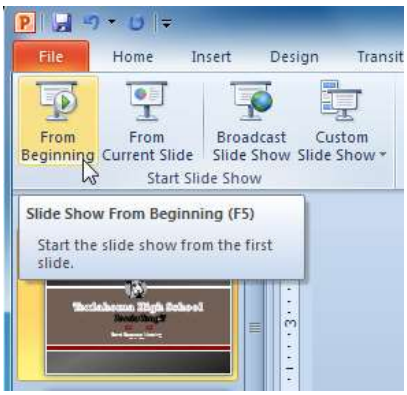


Figure 39 : Slide show tap

You can also start the slide show from the slide you prefer by selecting the slide and clicking on **From Current Slide** from the Start Slide Show group. This option is convenient if you only want to view or present certain slides.

Another option for starting the slide show is to select **Slide Show** view at the bottom of the window.

To advance and reverse slides:

1. Hover your mouse over the bottom-left of the screen. A menu will appear.
2. Click the **right arrow** to advance slides and the **left arrow** to reverse slides.

To stop or end a slide show:

To end a slide show, hover and select the **menu box options** command and click **End Show**. You can also press the **Esc** key at the top left of your keyboard to end the show.

Presentation tools and features

PowerPoint provides convenient tools and features you can use while you are presenting your slide show. Features include changing your mouse pointer to a pen or highlighter to draw attention to items in your slides. In addition, you can jump around to slides in your presentation or access items from your computer desktop if needed.

To access the pen or highlighter:

1. Hover and click the **pen menu option** in the bottom-left of your screen.
2. Select **Pen** or **Highlighter** based on your preference.
3. Use the pointer to **draw on or mark your slides**.

From the same menu, you can also **change the color** of the pen or highlighter. Keep in mind that light color choices are best for the highlighter.

To jump to a nonadjacent slide:

1. Hover and click the **menu box option** in the bottom-left of your screen.
2. Select **Go to Slide**, and choose the slide you want to jump to in your presentation.

To access the desktop:

Sometimes you may need to access the Internet or other files and programs on your computer during your presentation. PowerPoint allows you to access your desktop task bar without ending your presentation.

1. Hover and click the **menu box option** in the bottom-left of your screen.
2. Select **Screen**, then click **Switch Programs**.

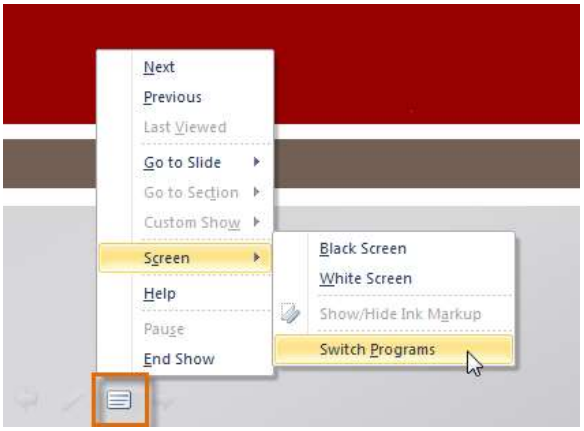


Figure 40 : switching to a desktop

3. Your computer's **task bar** will appear. Choose a program you want to switch to.

Menu access options:

You can also access any of the above menu items by **right-clicking** anywhere on the screen during your slide show.

Keyboard shortcuts

- Switch between the pen pointer and mouse pointer by pressing **Ctrl+P** (pen) or **Ctrl+M** (mouse) on your keyboard.
- Press **E** on the keyboard to erase any ink markings while using the pen or highlighter.

Slide show setup options

PowerPoint has various options for setting up and playing a slide show. For example, you can set up an unattended presentation that can be displayed at a kiosk and make your slide show repeat with continuous looping.

To access slide show setup options:

1. Select the **Slide Show** tab.
2. Click the **Set Up Show** command.



Figure 41 : Set up slide show tab

3. The Set Up Show dialog box will appear. Click the buttons in the interactive below to learn about the various options available for setting up and playing a slide show.
4. Click **OK** to apply the settings to the slide show.

Activity 12

1. Open an **existing PowerPoint presentation**.
2. Practice starting a slide show and using the menu options to **advance** and **reverse** through slides.
3. Change your mouse pointer to a **pen** or **highlighter**, and practice marking a slide. If you are using the example, add markings to the chart on slide 8.
4. **Erase** or **discard** any markings you have made on slides.
5. Practice going to nonadjacent slides in your slide show. If you are using the example, go to slides 3 and 10.
6. Change the **ink color** and add **continuous looping** to your slide show with the Set Up Slide Show command.

5.1.8 Design Template

A **template** is a **predesigned presentation** you can use to create a new slide show quickly. Templates often include **custom formatting** and **designs**, so they can save you a lot of time and effort when starting a new project.

To create a new presentation from a template:

1. Click the **File** tab to access **Backstage view**.
2. Select **New**. You can click a suggested search to find templates or use the **search bar** to find something more specific. In our example, we'll search for **Business** presentations.

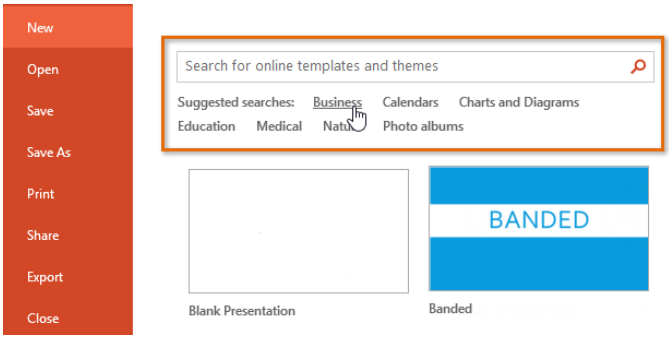


Figure 42 : Design template

3. Select a **template** to review it.
4. A **preview** of the template will appear, along with **additional information** on how the template can be used.
5. Click **Create** to use the selected template.
6. A new presentation will appear with the **selected template**.

It's important to note that not all templates are created by Microsoft. Many are created by third-party providers and even individual users, so some templates may work better than others.

5.1.9 Integration with other applications.

With PowerPoint we will look at an integration with Excel. If you have already created a chart in Excel, you can **embed** and **link** it to your PowerPoint presentation. When you embed an Excel chart in PowerPoint, any updates you make to the original Excel chart will **automatically update** in your presentation, as long as the files remain in the same location. This helps the data stay **in sync**, so you won't have incorrect or out-of-date information in your chart.

To embed a chart from Excel:

1. In PowerPoint, select the **Insert** tab.
2. Click the **Object** command in the **Text** group.
3. A dialog box will appear. Select **Create from file**, then click **Browse**.
4. Locate and select the desired Excel file, then click **Insert**.
5. Check the box next to **Link to file** if you want to **link the data** to the Excel chart. This will enable your chart to update itself when changes are made to the Excel chart.
6. Click **OK**. The chart will now appear in your presentation.

To **edit** an embedded chart, double-click the chart. An Excel spreadsheet containing the chart's source data will appear. After you have finished editing, be sure to **save** the chart in Excel.

Once you have embedded a chart, be careful not to delete or move the original Excel file. If the location of either the PowerPoint presentation or the Excel file changes, you may have to embed the chart again in order for it to display correctly.

By default, PowerPoint presentations are saved in the **.pptx** file type. However, there may be times when you need to use **another file type**, such as a **PDF** or **PowerPoint 97-2003 presentation**. It's easy to **export** your presentation from PowerPoint in a variety of file types:

- **PDF:** Saves the presentation as a **PDF document** instead of a PowerPoint file
- **Video:** Saves the presentation as a video
- **Package for CD:** Saves the presentation in a folder along with the Microsoft PowerPoint Viewer, a special slide show player anyone can download
- **Handouts:** Prints a handout version of your slides
- **Other file type:** Saves in other file types, including **PNG** and **PowerPoint 97-2003**

To export a presentation:

Let us save the presentation as a **PowerPoint 97-2003** file.

1. Click the **File** tab to access **Backstage view**.
2. Click **Export**, then choose the desired option. In our example, we'll select **Change File Type**.
3. Select a **file type**, then click **Save As**.

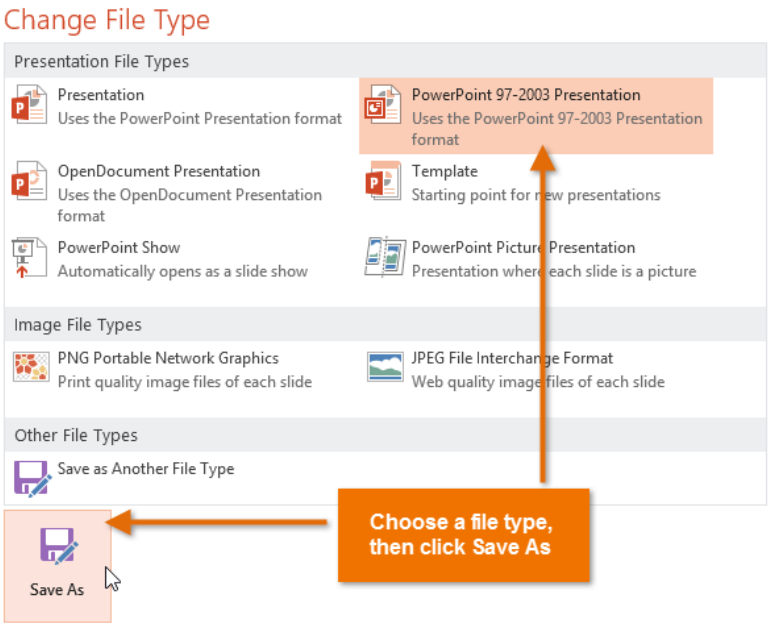


Figure 43: Change file type

4. The **Save As** dialog box will appear. Select the **location** where you want to export the presentation, type a **file name**, then click **Save**.

You can also use the **Save as type:** drop-down menu in the **Save As** dialog box to save presentations in a variety of file types. Be careful to choose a file type others will be able to open.

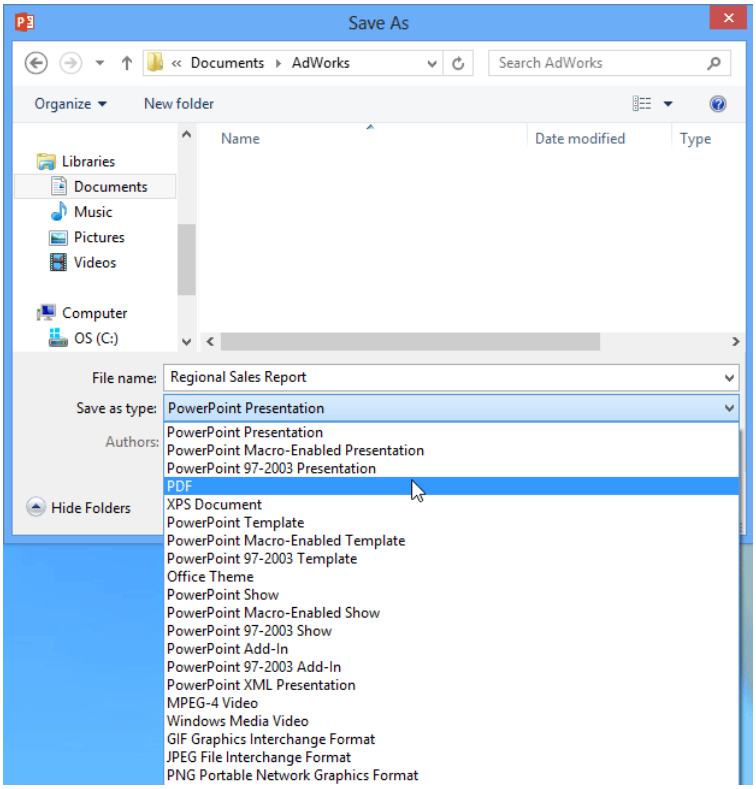


Figure 43: save as type dialogbox



Mzuzu University

Powerpoint Self-Assessment Test

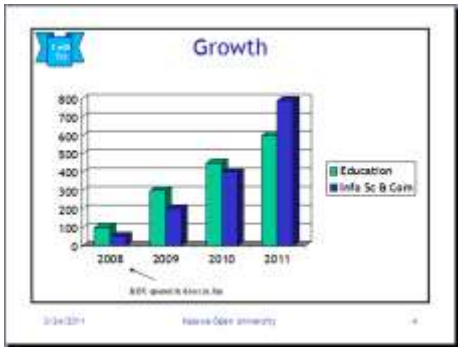
Instructions

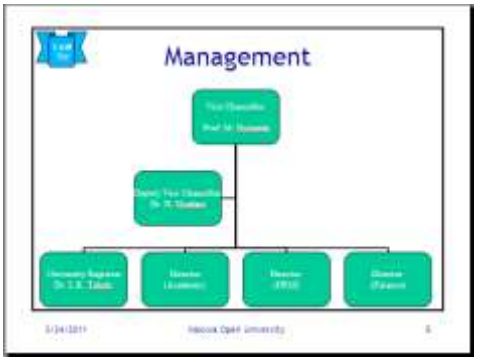
- Answer all questions

1. Create a five-slide presentation using the *Blank presentation* option. Use the content as shown on the slides below for your presentation.
2. Save the presentation in the folder you have created as Kalowa Open University.









20 Marks

3. Replace the text **Write your Registration number here** on Slide 1 with your Registration Number.
1 Mark
4. Choose a suitable background for your presentation. DO NOT use preformatted slide templates. **2 Marks**
5. Include slide transition and animation of your choice in your presentation. Make all slides advance both on mouse click and automatically after 00:05.
5 Marks

Unit Summary

In this unit we have learnt on how we can create presentations using Microsoft Powerpoint and understanding the ribbon tools associated with powerpoint. We have also seen how slides are formatted and how you can insert and use graphics. Slide animations and transitions have been discussed in much detail up to running a slide show.

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iv. http://www.electricteacher.com/tutorial3.htm
v. https://slidesgo.com/slidesgo-school/powerpoint-tutorials
vi. https://www.slideshare.net/ibcschools123/powerpoint-tutorial
vii. https://www.webucator.com/tutorial/learn-microsoft-powerpoint/index.cfm

UNIT 6

DATABASES

Introduction

A *database* is simply a collection of related information. For example, if you gathered together all your photographs, you would have a database of photographs. If you collected all your photographs that included your dog, you would have either a more targeted database or a subset of your larger database.

If your database is small (for example, your home insurance policies), you can probably manage the information manually. In such instances, you might use a traditional management method such as a card file or a simple list on a piece of paper. However, as the database becomes larger, your management task becomes more difficult. For example, it would be virtually impossible to manually manage the customer database of a large corporation. This is where your computer and a *database management system* (DBMS) come in handy. DBMS software (such as Access) lets you manage large amounts of information quickly and easily.

Areas of emphasis:

- Creating tables
- Designing forms
- Working with queries
- Generating Reports



Key Words

- Database, Access Database, Queries, Tables, Reports, Primary Key, Design view ,Datasheet view, Data types, et al.

Prerequisite knowledge

This unit assumes you are familiar with the basic operation of a personal computer. Specific skills required for this tutorial are:

- Use of the mouse and keyboard
- Opening and saving files on the hard disk and on USB memory stick (thumb drive) for example
- Managing files in general i.e. the difference between drive letters, hard disk, etc.
- Running programs from the Windows Start menu
- Minimizing, maximizing and resizing windows

You should have a USB memory stick (thumb drive) or other portable memory device available to store the files that will be created in this unit.

Learning outcomes

By the end of this unit, you should be able to:

- Create an access database
- Understand primary and secondary key
- Design Forms
- Create reports
- Work with queries

Resources needed

- This unit is based on Access 2016. But, all features will work in Access 2010, 2013, 2016, 2019, 365 and most in 2007.
- You will need Microsoft Access installed on your computer to follow along and complete the activities.

6.1.0 Access Terminology

Access database is defined as a collection of related objects. These objects are Tables, Forms, Queries, Reports, Modules and Macros. They all relate and make access database work.

Access provides a powerful yet comfortable environment in which you can manage your information. Using Access, you can tame even the most difficult data-management tasks. This unit is the first step in your journey. Here you will learn basic terms, as well as how to start and exit Access. You will also learn how to get help and even catch a glimpse of what you can do with Access.



Figure 1: The access window

Database

In Access, a database consists not just of information, but also the tables into which the information is organized. Access databases also contain related queries, forms, reports, and programming instructions. Because these terms deserve further definition, they are covered in the following sections.

Access 2016 uses the Ribbon to organize commands, just like in Access 2013 and 2010. Access uses a **tabbed Ribbon system** instead of traditional menus. **The Ribbon** contains **multiple tabs**, each with several **groups of commands**. For example, the Clipboard group on the Home tab contains commands such as Cut, Copy, and Paste.

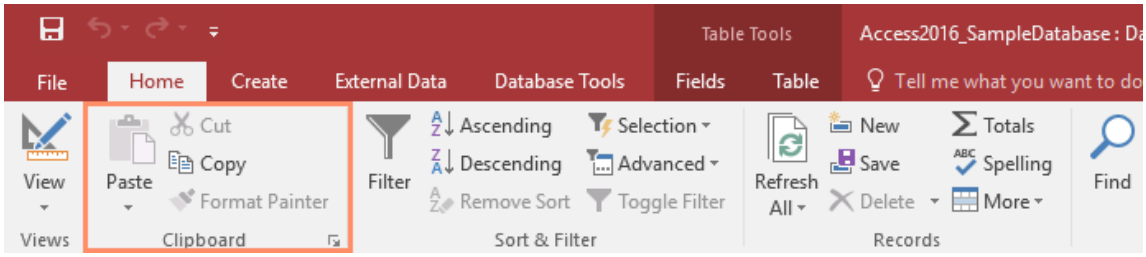


Figure 2: The access ribbon

Some groups also have a small arrow in the bottom-right corner that you can click for even more options.

The **Navigation pane** is a list containing every object in your database. For easier viewing, the objects are organized into groups by type. You can **open**, **rename**, and **delete** objects using the Navigation pane.

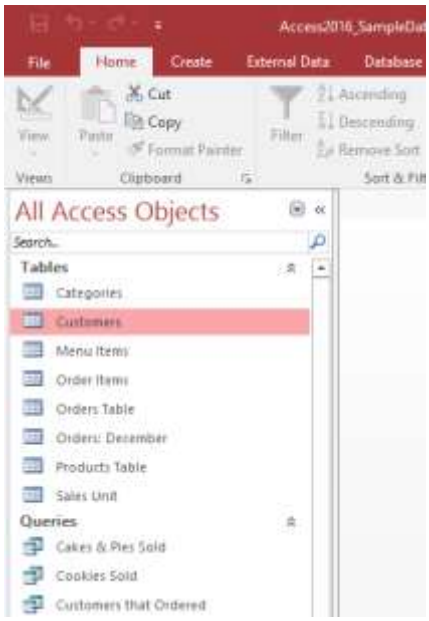


Figure 3: The navigation bar

To minimize and maximize the Navigation pane:

The Navigation pane is designed to help you manage all of your objects; however, if you feel that it takes up too much of your screen space, you can **minimize** it.

Each Access database consists of multiple objects that let you interact with data. Databases can include forms for entering data, queries for searching within it, reports for analyzing it, and tables for storing it. Whenever you work with your database, you are working with many of these objects at once. Fortunately, Access makes managing these objects pretty easy.

6.1.1 Objects in Access

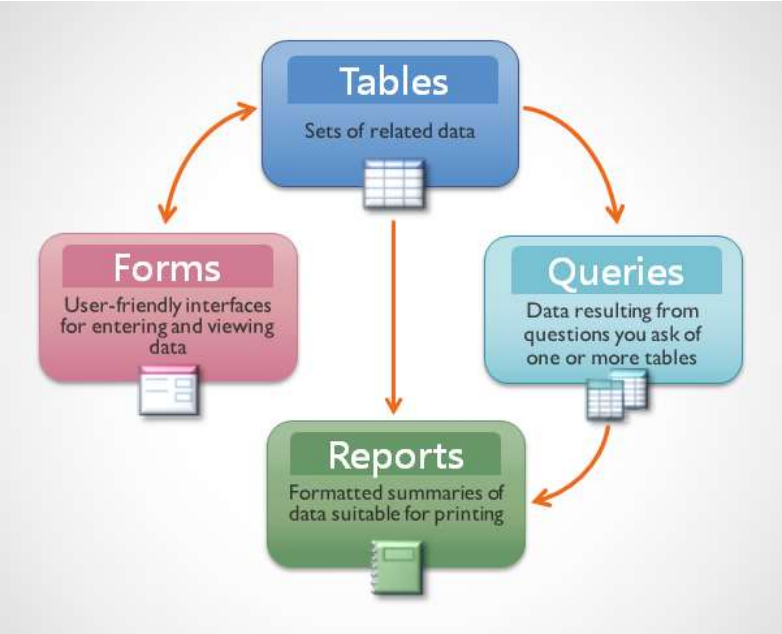


Figure 4: objects in Access

Forms allow you to both **add** data to tables and **view** data that already exists. Reports **present** data from tables and also from queries, which then **search for** and **analyze** data within these same tables.

There are two ways in which you can start the Ms Access program.

- The first one is to click start by clicking the windows button then to programs then Microsoft office suite and click Access.
- You can also double-click on the Ms Access shortcut icon on the desktop of your computer if there is one already on it.

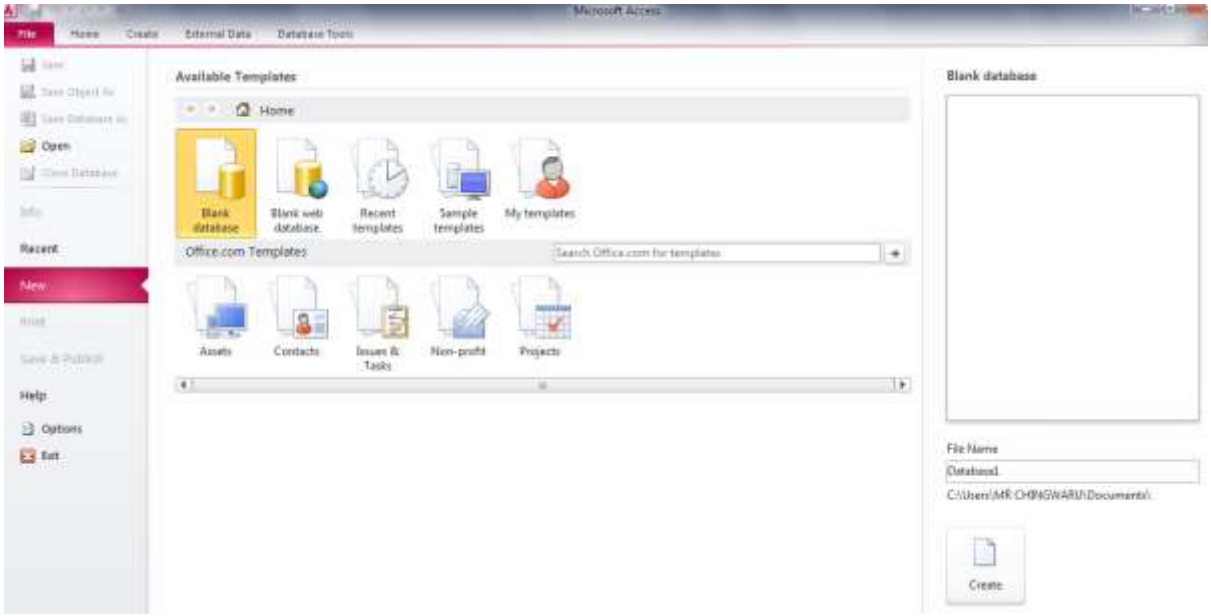


Figure 5: create database window

When access open it present you with options of creating the database using a blank database (from a scratch) or from an existing template. To learn and master working with Access you will have to opt for the blank database. When you click on the blank database option, you are also prompted to give your new database a name and decide where you want it to be stored on your computer. By default MS Access will point to the documents folder as a destination where it will store the database.

When you click Create, Access will create a new blank database and will open up the table which is also completely blank.



Figure 6: new blank database and table

Once the database is created, you can open it by double-clicking it in folder that it is stored. You can also re-open the MS Access program from the start menu the Microsoft office suite and click Access, when the program loads, click on file then open, a dialog box will appear and you will locate to where the database is stored. Click open on the dialog box and the database will open.

6.1.2 Table

In Access, *tables* contain the actual information in your database. There can be more than one table in a database. The information in each table can relate to information in other tables in your database. For example, you might have one table that contains a record of all the door locks in your building. In the same database, another table might have a list of all the keys for those locks. Still another table might contain the names of all the people who have the keys. All three tables contain related information, so they belong to the same database.

All data is stored in tables, which puts tables at the heart of any database. You might already know that tables are organized into vertical columns and horizontal rows.

The rows and columns are referred to as records and fields. A field is more than just a column; it's a way of organizing information by the type of data it is. Every piece of information within a field is of the same type. For example, every entry in a field called First Name would be a name, and every entry in field called Street Address would be an address.

Likewise, a record is more than just a row; it's a unit of information. Every cell in a given row is part of that row's record.

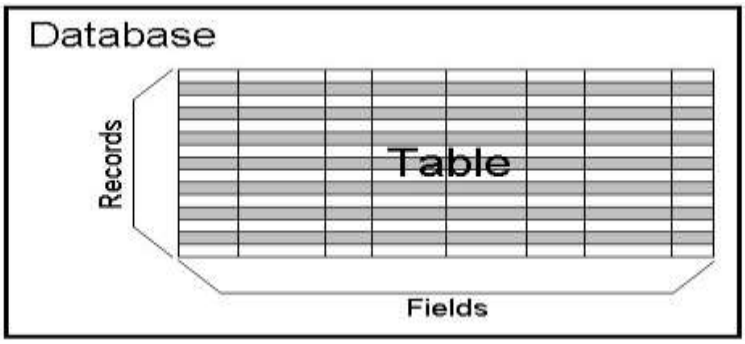


Figure 7: How fields, records, tables, and databases are related.

Your first step in creating a database is to determine the information you need. Next, you will use Access to design a table to hold the information. You use Design view to specify the framework used for each of your table entries. Data types are used to specify how the information will be stored in the fields. Every field in a table has properties and these properties define the field's characteristics and behavior. The most important property for a field is its data type. A field's data type determines what kind of data it can store. The data type determines the kind of the values that users can store in any given field. Each field can store data consisting of only a single data type. Here are some of the most common data types you will find used in a typical Microsoft Access database.

Type of Data	Description	Size
Short Text	Text, including numbers which does not need calculation. (e.g., Mobile numbers).	Up to 255 characters.

Long Text	This data type is used for lengthy text or alphanumeric data.	Maximum 63, 999 characters.
Number	Numeric data type used for storing mathematical calculations.	1, 2, 4, 8, and 16 bytes.
Date/Time	Store Date/time for the years 100 through 9999.	8 bytes.
Currency	It allows you to store currency values and numeric data with one to four decimal places.	8 bytes.
Auto Number	Assign a unique number or assigned by Microsoft Access when any new record is created. Usually used as the primary key	Four bytes (16 bytes if it is set as a Replication ID).
Yes/No	It only stores logical values Yes and No.	1 bit
Attachment	It stores files, such as digital photos. Multiple files can be attached per record.	Up to 2 GB Data can be stored.
OLE objects	OLE objects can store audio, video, other Binary Large Objects.	Up to 2 GB data can be stored.
Hyperlink	Text or combinations of text and numbers stored. That text is used as hyperlink address.	Each part of a Hyperlink data type allows you to store a maximum 2048 characters.
Calculated	Helps you to create an expression that uses data from one or more fields.	You can create an expression which uses data from one or more fields.

Table 1: data types in Access

After you design your table or tables, you will use Datasheet view to enter and examine (view) your data.

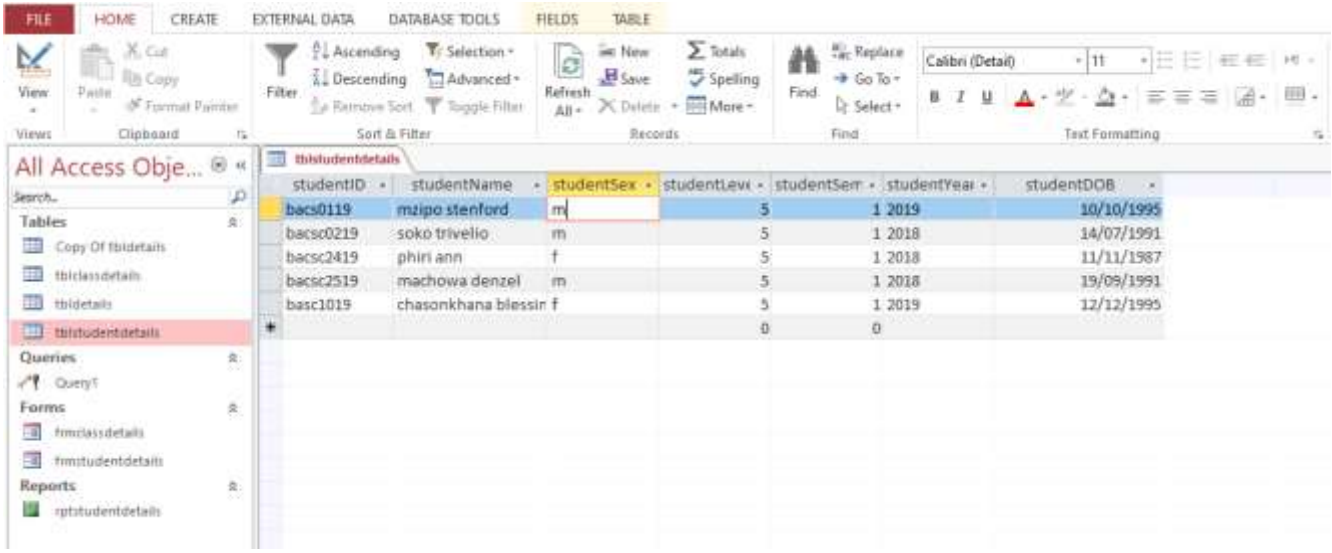


Figure 8: datasheet view of a Table

Notice how each record spans several fields. Even though the information in each record is organized into fields, it belongs with the other information in that record. See the number at

the left of each row? It's the StudentID number that identifies each record. The StudentID number for a record refers to every piece of information contained on that row.

Tables are good for storing closely related information. Let's say you own a bakery and have a database that includes a table with your customers' names and information, like their phone numbers, home addresses, and email addresses. Because these pieces of information are all details on your customers, you'd include them all in the same table. Each customer would be represented by a unique record, and each type of information about these customers would be stored in its own field. If you decided to add any more information—say, a customer's birthday—you would simply create a new field within the same table.

6.1.3 Primary Key and Secondary (Foreign) Keys in Access

The use of Primary key and Secondary (also known as Foreign key) in the table design is very important. First, let's discuss each of these concepts and what these terms really mean:

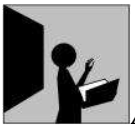
- The term ***Primary Key*** **(PK)** refers to a field (or combination of fields) whose values uniquely identify each record in that table. In Access, it is common to define a Primary key field in each table, usually as a field that is of **AutoNumber** data type, to serve as the PK. This **AutoNumber** field will automatically create a unique number for each record as it is created, so that those records can be identified from the other records in the table.
- The term Secondary or ***Foreign Key*** **(FK)** refers to a field that in a table that stores values from a Primary key field in another table, to relate the two records to each other.

For example, assume that you have some *contacts* for which you want to store some information, and that information happens to include the *phone numbers* for each contact. You realize that some *contacts* have more *phone numbers* than others and you want to store the data in your database application as efficiently as possible. So, you decide to create an Access database application that has a **Contacts** table and a **PhoneNumbers** table to store this data. In your database application, you need a way to identify any number of phone number records that relate to each contact record. If the PK in the **Contacts** table is an **AutoNumber** type field, then the FK in the **PhoneNumbers** table should store the **ContactID** (**AutoNumber**) from the **Contacts** table and will need to be the type of **Long Integer** (**Number**). This means that a record in the **PhoneNumbers** table has the potential to contain the **ContactID** value of the contact record in the related **Contacts** table.

The first step to store data in the database is to create a Table where data will reside. Post creation of the table, we can keep inserting the rows in the table.

There are two ways to create Database in MS ACCESS

- Create a Table from Datasheet View
- Create a Table from Design View



Activity 1 Create Table – Datasheet View



Figure 9: creating a table in datasheet view

- Step 1) First Click Create tab. Then from Tables group, click Table.
- Step 2) System will display the default table created with 'Table1' name.
- Step 3) To Rename Column, double click on Column Header and enter the new column Name.

Note that the Data type of *Course_ID* is '**AutoNumber**.' Hence this is also the Unique Key of the table.

- Step 4) You can Add Column by clicking on any category from the 'Add & Delete' group. Alternatively, you can also add a column by clicking on 'Click to Add'
- Step 5) Column will be added with the default name as 'Field1.'
- Step 6) Click on Header and rename as *'Course_Dept*.'
- Step 7) Column will be added with the default name as 'Field1.'
- Step 8) Click on Header and rename as *'Course_Lecturer*.'
- Step 9) Column will be added with the default name as 'Field1.'
- Step 10) Click on Header and rename as *'Course_Faculty*.'
- Step 11) Column will be added with the default name as 'Field1.'
- Step 12) Click on Header and rename as *'Course_Campus*.'
- Step 13) Press 'Ctrl + S' and Enter the new table name to save the table. Tables are saved with a name beginning letters ***tbl*** save as ***tblcourse_details***



Activity 2 **Create Table – Design View**

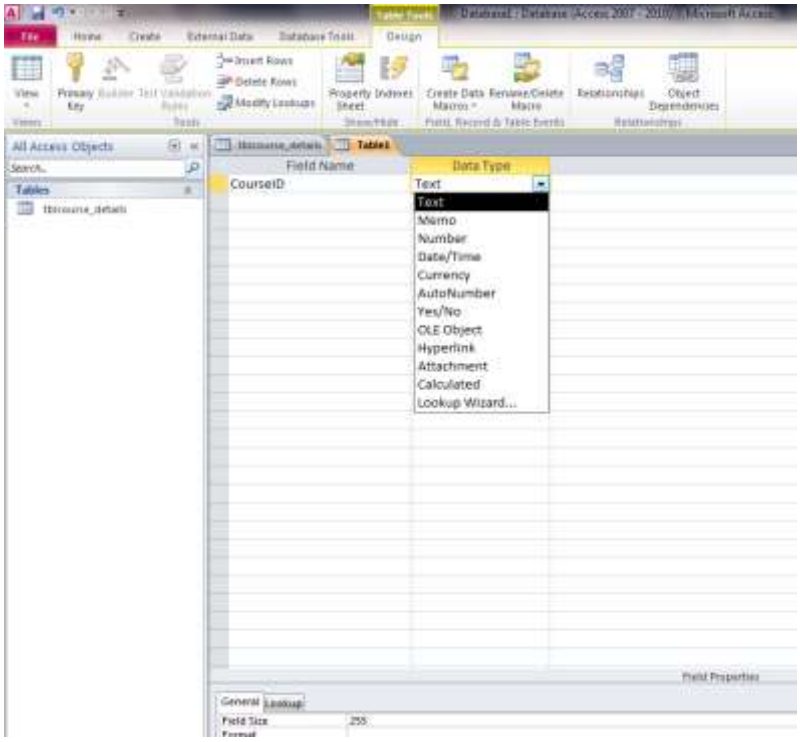


Figure 10: creating a table in design view

- Step 1) First Click Create tab. Then from Tables group, click Table Design.
- Step 2) Table design Dialog box appears. For each Field enter Filed Name, Data Type and Description. The fields are *Course_Lab_Hrs, Course_Lec_Hrs, Course_Grading, Course_PassMark*
- Steps 3) To Add *Course_ID* as Primary Key, select it and Click on 'Primary Key.' *Course_Id* will be Preceded by KEY ICON:
- Steps 4) Press 'Ctrl+S.' Enter the Table Name ***tblcourse_info*** and Click OK

How to Switch views between Datasheet and Design

To switch views, between the datasheet (spreadsheet view) and the design view, click the 'View' button in the top-left hand corner of the Access Ribbon (shown in the Home/Help/Design Tabs). And Click on the View you need to display.

Adding Data

You have created tables' ***tblCourse_detail*** and ***tblCourse_info***

There are two ways to add data. One is Datasheet view and other from Form. We will learn to add data from Datasheet view in this section. There is no need to save the data manually. Once you click other rows, Access automatically saves the data. The only thing you need to

take care while adding data is, you should enter the correct **data type**, i.e., if Column is defined as Number we can add only number and not text.

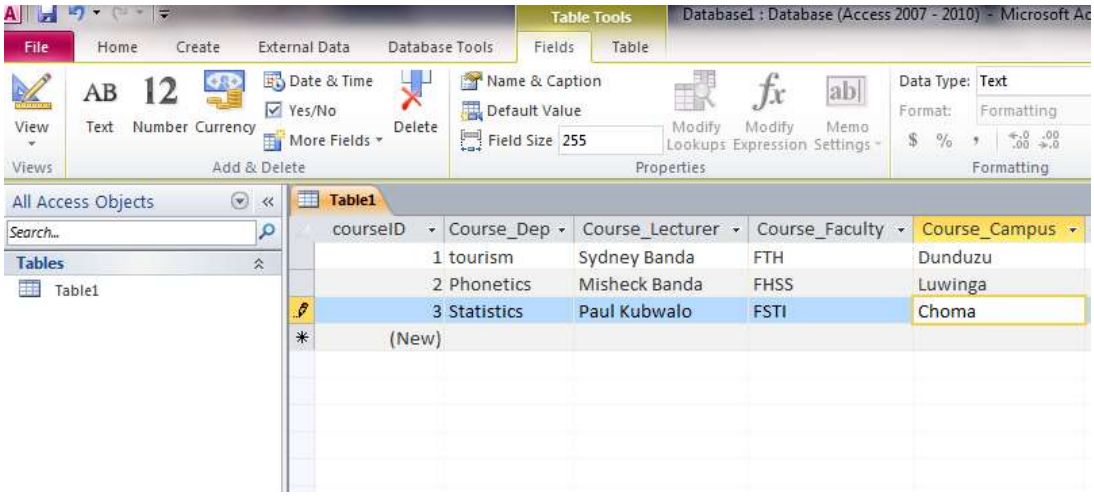


Figure 11: adding records in datasheet view

Let's add some data in your table by opening the Access database you have created earlier.

- Step 1) Select the 'tblcourse_details' table
- Step 2) Select the Datasheet view option in the ribbon and add some data by entering the values in It. Updated Data will be Autosaved.

Now let's say You want to delete 3rd row with Satistics as Department name.

- Step 3) Select the row by clicking on the leftmost column and Right Click on the row. The options menu will appear with the 'Delete Record' Option
- Step 4) Popup Window will appear to confirm the deletion of the record. Click 'OK'

6.1.4 Summarizing Tables:

You now know how you can use the development tools built into Access (such as the Table Wizard) to quickly and easily set up tables. You also know how to create your own database and enter information in tables.

Datasheet View: Displays in the view, which allows you to enter raw data into your database table. The Datasheet lets you directly enter information in a table and edit existing information.

Use a primary key in your table to help Access efficiently order the table and quickly find records. Primary key too makes your field to be unique.

You can use the Find command to search for information in your table. This command lets you build search criteria using wildcard characters, if desired. You can use the Replace

command to replace information in your table. This command is similar to the Find command, but also allows the replacements.

To delete a record from the Datasheet, select the record and press the **DELETE** key or choose the Delete option from the Edit menu.

Access saves your changes on a record-by-record basis. When you move from one record to another, Access automatically saves any changes to the older record to disk. Access provides you with great flexibility not just in entering data, but also in working with tables as a whole. In addition, you now know how to copy, rename, and delete entire tables.

Design view: Displays the view, which allows you to enter fields, data types, and descriptions into your database table. Design view lets you change an existing table's structure. Access presents information in the Design view window in much the same way as it does in the Datasheet window.

Data types specify the type of information that is stored in a field. Using the correct data types helps Access to efficiently store information in a table.

As you design your table, you can and should add comments to your fields. These comments help you remember the purpose of the fields and document any additional information about them. Comments do not affect the way Access functions.

Each field possesses properties or characteristics which indicate how Access should process it. The types of properties available for a field depend on the data type of the field.

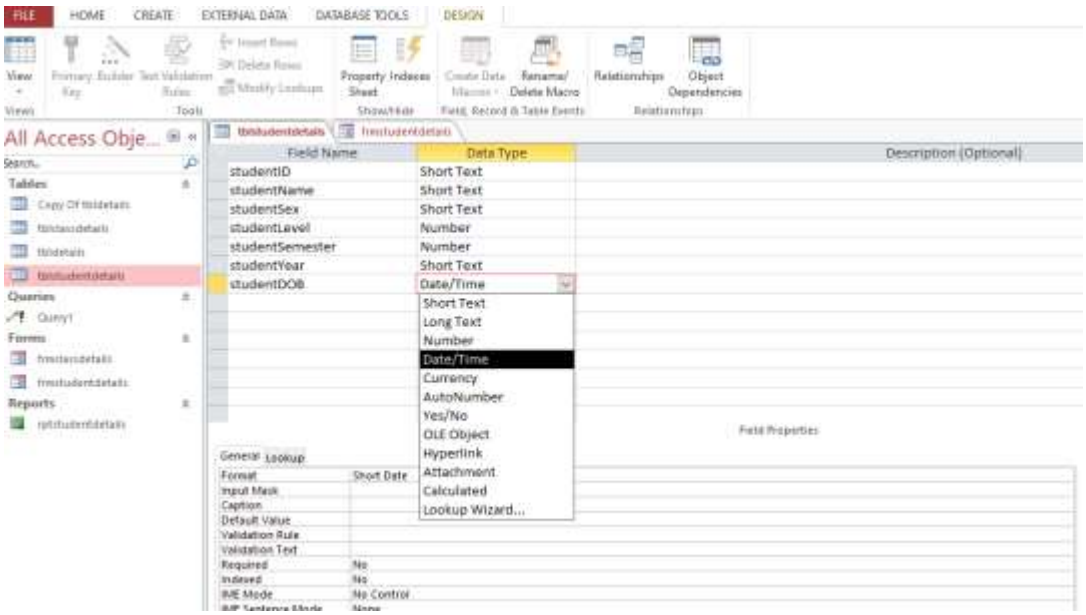


Figure 12: assigning data types in design view

Designing a table requires some planning. You need to examine what data you are going to manage, as well as the data's characteristics. As you input fields in the Design view window, you may need to change the data types and properties which Access presents. The process of refining your table layout is iterative in nature. It involves working alternately in both the Datasheet and Design view windows, repeatedly switching from one window to the other until your design is correct.

As you work in the Datasheet, you should look for ways to improve the layout of your table. Switch to the Design view window to make your changes, and then back to the Datasheet to test those changes.

Although Access provides many different tools (such as Wizards), you don't need to use these tools to create a great database. Instead, you can manually create and refine tables that will work just as well. And as you work with Access in this way, you will become more aware of its capabilities and how to use them.

6.1.5 Queries

When you work with large databases, there will be times when you will want to work with specific pieces of your data. For example, if you have a company database, you may want to view the names of all your customers who reside in New York. On such occasions, you would form a query.

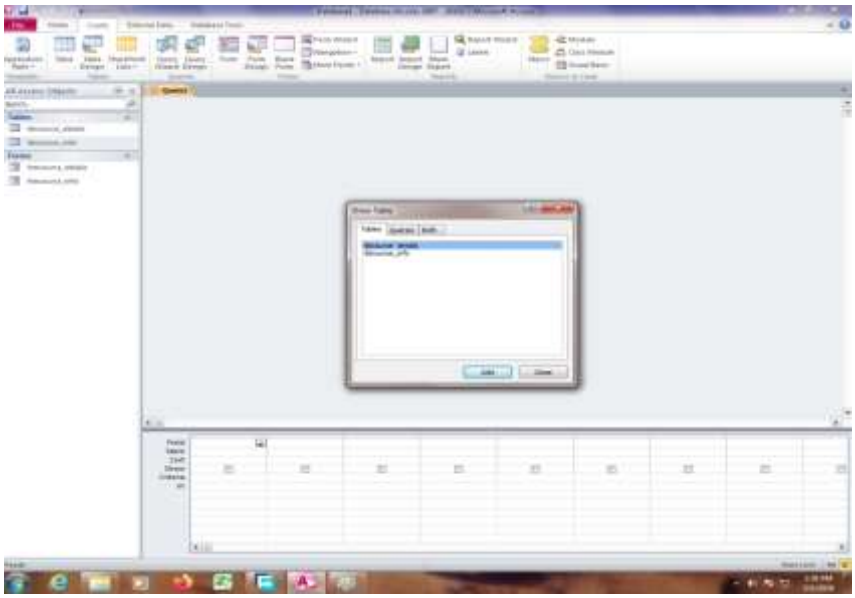


Figure 13: query design view

A *query* defines which database information you want to view. Think of a query as a question you ask the database, such as "Who are the customers that reside in New York?"

For example, if your database contains the names of all your customers who have purchased a particular product, your query might request a list of only those customers that made over a certain income. Another query might request only those customers with children. Essentially, a query limits or *filters* the information in a database. When you use a query as a filter, Access displays only information that satisfies the query.

Why would you use queries? Specifically because you want to work with only a part of your database. Queries make it easy for you to work with only those records that fit particular criteria. Access lets you get as broad or as specific or as complex as you want in the queries you develop.

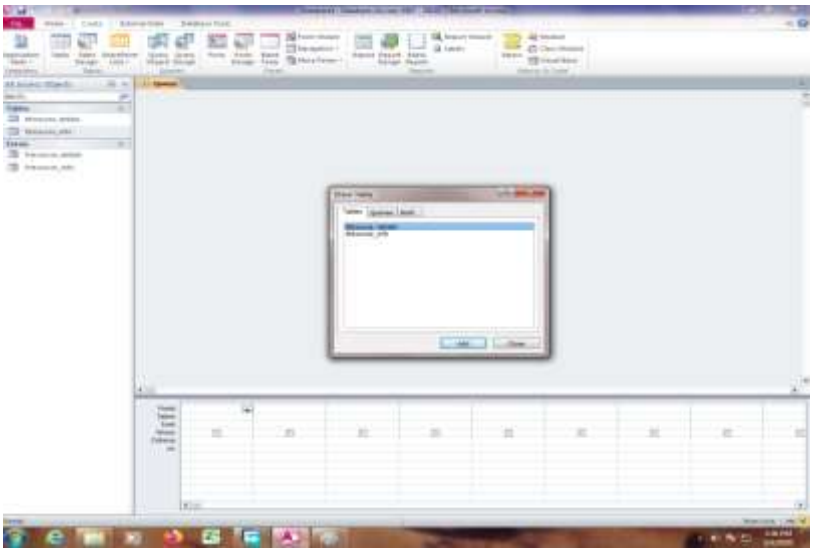


Figure 14: the Query grid view

How to create a Query?

To create a query, select ***Create*** on the ribbon and click on ***query design***. This will take you to a query grid. The grid has the following captions:

The ***show table*** dialog box appears at the centre displaying by default tables existing in the database. The show table dialog box also has **tabs** to show queries existing in the database as well as another tab to show both queries and tables of the database. The ***add button*** on the show table dialog box allows you to add the table one at a time by selecting and clicking the add button. When you are done adding the tables needed to work on, there is a ***close button*** that removes the show table dialog box. You can get the show table dialog box by right clicking anywhere on the grid and selecting the ***show table option***.

Below the show table dialog box there is a the QBE (query by example) grid with the following captions:

- ***Field*** that will show the fieldname
- ***Table*** to indicate the name of the table that the field name belongs
- ***Sort*** where records in the field can be sorted either ascending or descending order, ***Show*** where your specify the fields to be displayed when the query is executed
- ***Criteria/Or*** where you provide the questions or condition you want the query to meet.

Depending on the type of query selected, the QBE (query by example) grid fields will change so as to accommodate the requirements of the functionalities of the query.



Activity 3 - Create Select Query

- Step 1) from the show table dialog box add the tables ***tblcourse_details*** and then ***tblcourses_info*.** Click close to remove the show table dialog.
- Step 2) Drag the field ***courseID*** from table ***tblcourse_details*** and drop it above the ***courseID*** field of ***the tblcourses_info*** table. (Note: these are unique fields in their tables and they all have similar data types which makes creating the relationship between the tables easy). A line connecting the two tables will appear.
- Step 3) double click each field in ***tblcourse_details*** then ***in tblcourses_info***. Each field is to appear once meaning only one ***courseID*** field is to be double clicked. (Note the double clicking is the same as dragging the field one by one and drop into the datasheet grid of the query.
- Step 4) Click RUN on the ***Ribbon*** of the query tools.
- Step 5) Click the ***View*** icon on the ribbon to switch back to the query grid.
- Step 6) Click on ***Course_Dept*** field on ***Sort row*** and select ascending. Run the query to see if the records of the field are in ascending order.
- Step 5) Click the ***View*** icon on the ribbon to switch back to the query grid.
- Step 7) Remove the tick in the ***Show row*** of the *Course_Lecturer.* Run the query to see if the field *Course_Lecturer* will appear.
- Step 8) Click the ***View*** icon on the ribbon to switch back to the query grid.
- Step 9) on ***Criteria row*** under the Course_Lec_Hrs type >64. Run the query to display only those courses that have more than 64 lecture hours.
- Step 10) Click the ***View*** icon on the ribbon to switch back to the query grid.
- Step 13) Press ***'Ctrl + S'*** and Enter the new query name to save the query. queries are saved with a name beginning letters ***qry*** save as ***qrycourse_details***

Note: To see the results of a query you click the exclamation symbol on the query tool marked **RUN**.

There several types of queries in Access and by default the first query that you begin with getting the fields from tables/queries to the grid is the ***Select query***. You can also create a new table by using the ***Make-table query,*** you can also add records to an existing table by using

the ***Append query,*** and you can also update records in an existing table by using the ***Update query.***

Activity 4 - To create a Make table query:

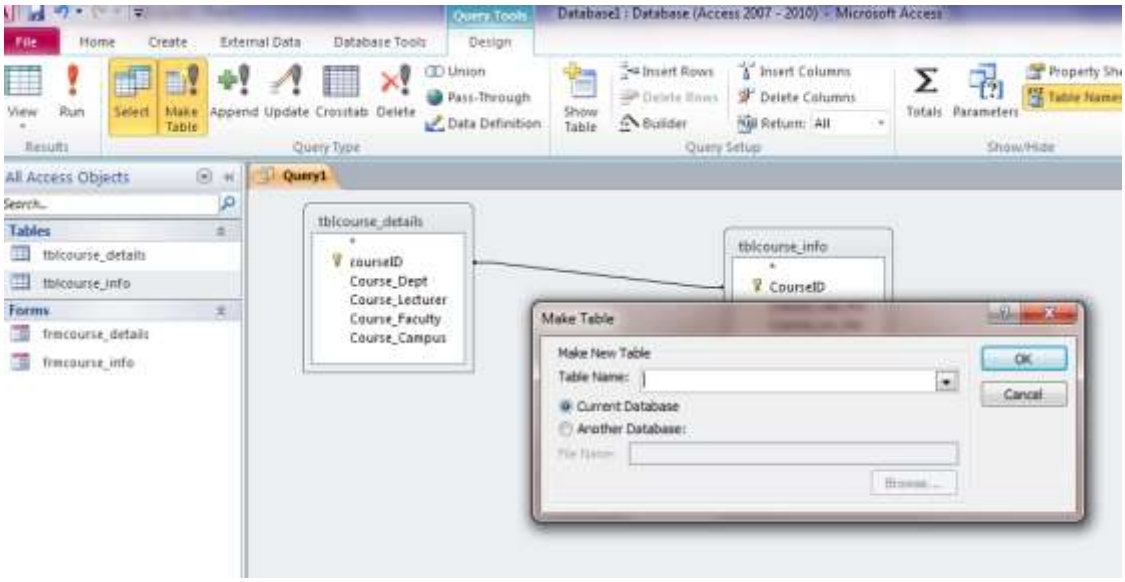


Figure 15: the make table query

- Step 1) from the show table dialog box add the tables ***tblcourse_details*** and then ***tblcourses_info.*** Click close to remove the show table dialog.
- Step 2) Drag the field ***courseID*** from table ***tblcourse_details*** and drop it above the ***courseID*** field of ***the tblcourses_info*** table. (Note: these are unique fields in their tables and they all have similar data types which makes creating the relationship between the tables easy). A line connecting the two tables will appear.
- Step 3) double click each field in ***tblcourse_details*** then ***in tblcourses_info***. Each field is to appear once meaning only one ***courseID*** field is to be double clicked. (Note the double clicking is the same as dragging the field one by one and drop into the datasheet grid of the query.
- Step 4) Click the Make Table icon on the ***Ribbon*** of the query tools. The dialog box will appear asking you to provide the new tables name: Name the table ***tblcourses.*** Click okay.
- Step 5) Click RUN on the ***Ribbon*** of the query tools. A message will appear telling of the number of records to be pasted into a new table. Click ***Yes*** and a new table will appear in the navigation pane under the tables.
- Step 6) Press ***'Ctrl + S'*** and save the query as ***qrymaketable***



Activity 5- To create an Append query

- Step 1) from the show table dialog box add the tables ***tblcourse_details*** and then ***tblcourses_info*.** Click close to remove the show table dialog.
- Step 2) Drag the field ***courseID*** from table ***tblcourse_details*** and drop it above the ***courseID*** field of ***the tblcourses_info*** table. (Note: these are unique fields in their tables and they all have similar data types which makes creating the relationship between the tables easy). A line connecting the two tables will appear.
- Step 3) double click each field in ***tblcourse_details*** then ***in tblcourses_info***. Each field is to appear once meaning only one ***courseID*** field is to be double clicked. (Note the double clicking is the same as dragging the field one by one and drop into the datasheet grid of the query.
- Step 4) Click the ***Append*** query icon on the ***Ribbon*** of the query tools. The dialog box will appear asking you to select the table you want to append (add) the records to: Select the table ***tblcourses.*** Click okay. You will notice on the datasheet grid an addition of a new row APPEND to showing how records will be appended by fields. Note: Access only appends to fields that are same only.
- Step 5) Click RUN on the ***Ribbon*** of the query tools. A message will appear telling of the number of records to be appended into the table. Click ***Yes*** and if you open the table you will see that there are now more records.
- Step 6) Press ***'Ctrl + S'*** and save the query as ***qryappend***



Activity 6 - To create an Update query

- Step 1) from the show table dialog box add the tables ***tblcourse_details*** and then ***tblcourses_info*.** Click close to remove the show table dialog.
- Step 2) Drag the field ***courseID*** from table ***tblcourse_details*** and drop it above the ***courseID*** field of ***the tblcourses_info*** table. (Note: these are unique fields in their tables and they all have similar data types which makes creating the relationship between the tables easy). A line connecting the two tables will appear.
- Step 3) double click each field in ***tblcourse_details*** then ***in tblcourses_info***. Each field is to appear once meaning only one ***courseID*** field is to be double clicked. (Note the double clicking is the same as dragging the field one by one and drop into the datasheet grid of the query.
- Step 4) Click the ***Update*** query icon on the ***Ribbon*** of the query tools. You will notice on the datasheet grid an addition of a new row UPDATE to asking to indicate what has to be update. Indicate **90** on the field ***Course_Grading*** so that the records in the selected field change from 45 to 90.
- Step 5) Click RUN on the ***Ribbon*** of the query tools. A message will appear telling of the number of records to be updated in the table. Click ***Yes*** and if you open the table you will see that the update has occured.
- Step 6) Press ***'Ctrl + S'*** and save the query as ***qryupdate***

6.1.6 Summarising on Queries

Queries are powerful tools that you can use to select, change, and manipulate information in your database. From a data source (a table or another query), a query selects records according to criteria you specify. What Access does with those records depends on the type of query you design. A query is a way for you to pose questions to Access about the information in your tables.

Access lets you create several types of queries: Select, Make Table , Update, Append, Delete and many advanced ones. Queries are another part of your database, just like tables, reports, and forms. You can use the Queries button in the Database window to view which queries are in your database.

You can use the query Design window to design your own queries from scratch. This window, along with the query Design toolbar, lets you specify and create any type of query you can imagine.

The QBE (query by example) grid appears at the bottom of the query Design window. Using the grid, you can specify which fields Access will use to determine if records should be displayed, as well as additional records which Access will display in the query results.

When you design a query, you can specify properties for the fields in the query result. These field properties are similar to those you set when you design a table.

6.1.7 Form

A database exists to store information. After you determine the information your database will contain, you will need to enter the data. Later, you may want to examine, add to, or change that data. Although you can use the Datasheet view to accomplish each of these tasks, you may want instead to create a custom display screen that lets you enter, view, and edit information.

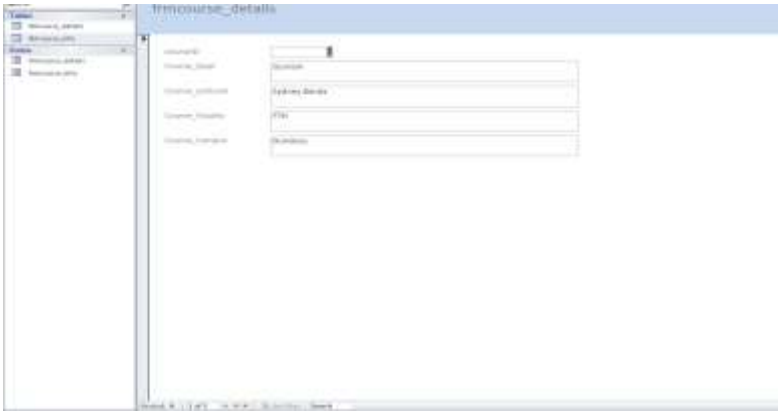


Figure 16: the form view

Within Access, custom display screens are called *forms*. Using forms, you can both display the information in a table and add buttons, text boxes, labels, and other objects that make data entry much easier.

Forms provide a convenient and attractive way to both present information and solicit input from the user. If you are developing an application using Access, it is not unusual to have several different forms, each with a different purpose. Through effectively using forms, you increase the usability of your database tremendously.

Forms are a tool you can use to change or add information in a table or query. From a user's perspective, forms are much more usable than the Datasheet.

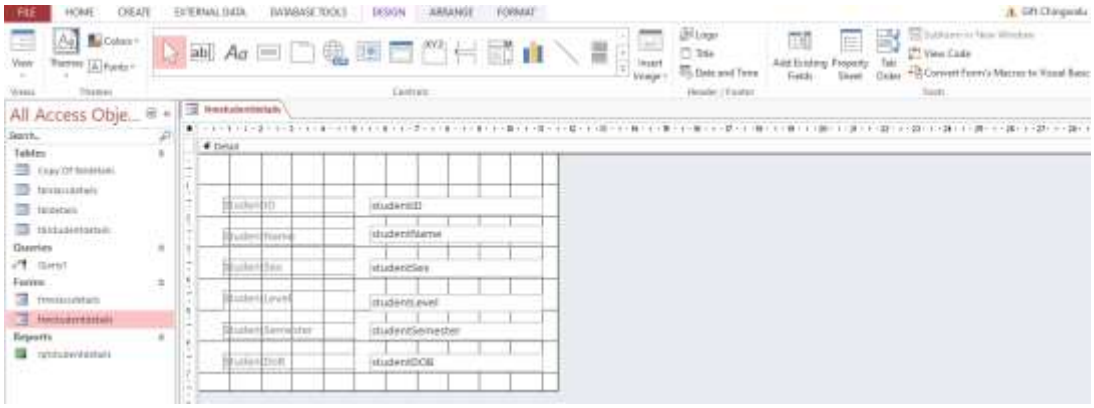


Figure 17: Form design view

In summary a form is a database object that you can use to create a user interface for a database application.

- It mainly used to ease the process of data entry or editing.
- Data in a form can be selected from one or more tables.
- Forms can also be used to control access to data, like which fields or rows of data are visible to which users.
- Forms have a Form View.
- Help you to display live data with easy creation of new data.



Activity 7

To understand the Form lets first create two new Record in ***tblcourses_details***

- Step 1) Select the table from Left Navigation pane.
- Step 2) Create two rows by entering some relevant data in two rows.



Figure 15: form design ribbon

You can create forms using the Form Wizards, or you can create them from scratch. You can create only a limited number of forms with the Form Wizards. You greatly increase your options if you create your forms from scratch. You use the form design toolbox to select objects that can be placed in your form. As with most other objects in Access, you can set properties for objects in your forms. These properties define the attributes which control the object.

As you can see there are several ways to create the form. Some of these methods are explored in this module as follows:

- Form Wizard
- Form
- Multiple Item
- blank Form

Let's have a look at each option to create the form, one by one:



Activity 8- Create using Form Wizard

This option allows the user to create the form with the *wizard* and select the desired fields to be included on the form. Click create on the ribbon

- Step 1) Click on 'Form Wizard.
- Step 2) System will display the screen. You can indicate the table/query name you want to the form for.
- Step 3) Select the columns which you want to be there in final form.
- Step 4) Keep selecting all required column as explained in Step 3 above and then click 'Next.'
- Step 5) Layout selection box will appear which allows the user to select the different type of form layout. Click 'NEXT'

- Step 6) Enter the name of the form. Forms are saved with a name beginning letters ***frm*** save as ***frmcourse_details*** and click 'Finish.'

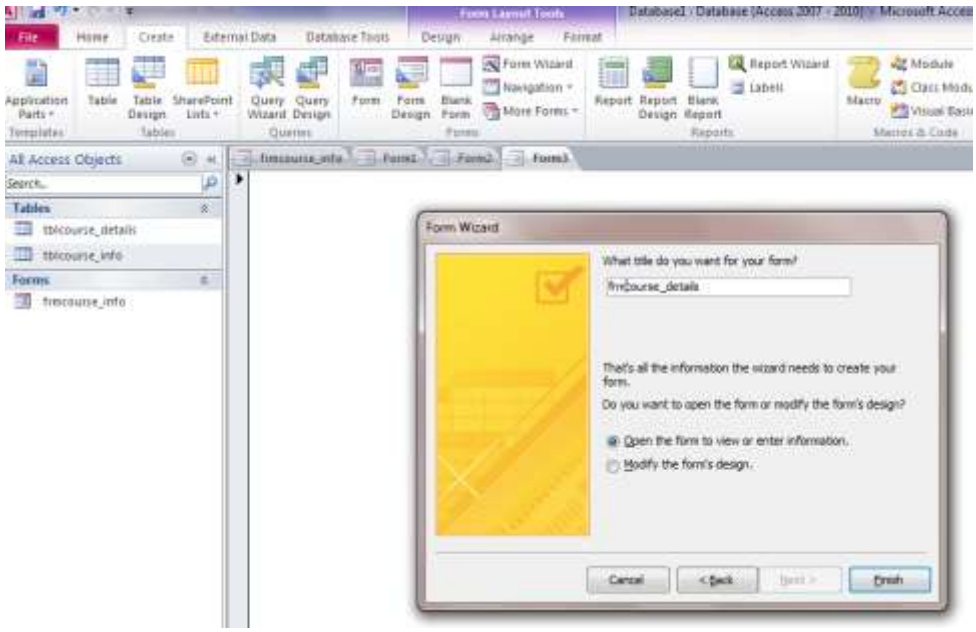


Figure 16: saving a form created using a wizard



Activity 9 - Create using Form

It is the simplest way to create the form which will:

- By default, populate all the column from the selected table in 'form view,'
- The user can delete non-required column manually

Step 1) Select the table for which we want to create the form and click on 'Form.'

Step 2) Below window will appear.

Step 3) Right-click on any cell which we don't want to be part of final forms and click on 'Delete.'

Step 4) Press 'Ctrl+S' and enter new Form Name as ***frmcourses***. Click 'OK'.



Activity 10 - Create form by 'Multiple Item.'

It is another type wherein All the records already created will be displayed in Form with an option to Add new record.

- Step 1) From the 'Create' tab. Click on 'More forms' and select 'Multiple Items.'
- Step 2) Press 'Ctrl+S'. Enter the new form name and click 'OK.'



Activity 11 - Create form by 'Split from'.

It is a mix of simple form and split form in a way that this form provides the view of Form and datasheet in a split window.

Whatever the user enters in Form is visible directly in Datasheet view immediately and vice versa.

- Step 1) From the 'Create' tab, click on 'More forms' and select 'Split Form.'
- Step 2) Press 'Ctrl+S' and enter the new form name. Click 'OK.'

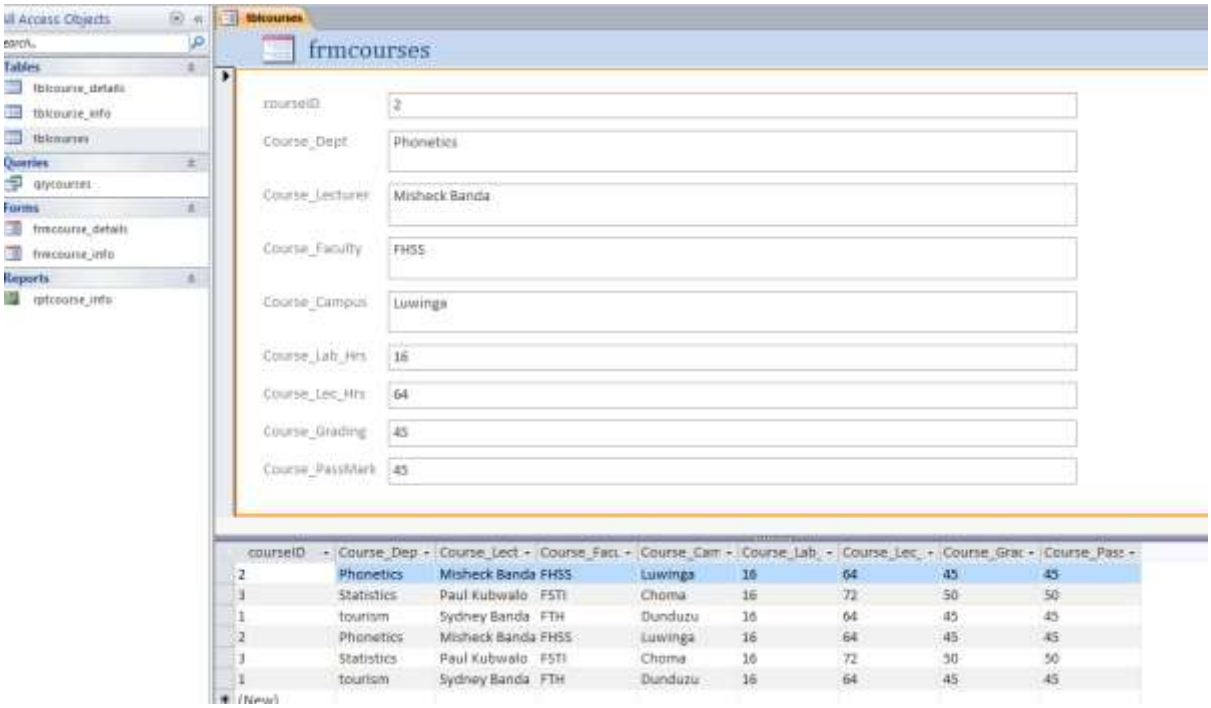


Figure 17: form created using a split form method

Edit the existing record via Form.

One of the features of forms is that we can edit the values and data directly from the form. It is more user-friendly because as compared to row format in the table, forms have better visibility of selected fields and the user can do the direct updates. These values updated from forms will also be reflected in original tables immediately. Forms also give the flexibility to Add records.

Again, this is a user-friendly and appealing way of adding records as compared to adding records in a row form.

6.1.8 Reports

Access has some impressive report capabilities. In fact, about the only limit on what you can do with reports is your imagination. The process of developing these reports has taught you skills you can use to develop other reports.

In Access, a report is much more than a simple printout. Reports give you complete control over what Access prints and how it appears on the page. You can create reports using either the Report Wizards or the report Design window.

You can start a Report Wizard either from the Reports section of the Database window or by choosing Reports from the Insert menu. Report Wizards can be used to create many types of sample reports. These reports are simple in nature, and not nearly as customized as what you can create by using the report Design window.



Activity 12

This option allows the user to create the form with the *wizard* and select the desired fields to be included on the form. Click ***create*** on the ribbon

- Step 1) Click on 'report Wizard.
- Step 2) System will display the screen. You can indicate the table/query name you want to generate report for.

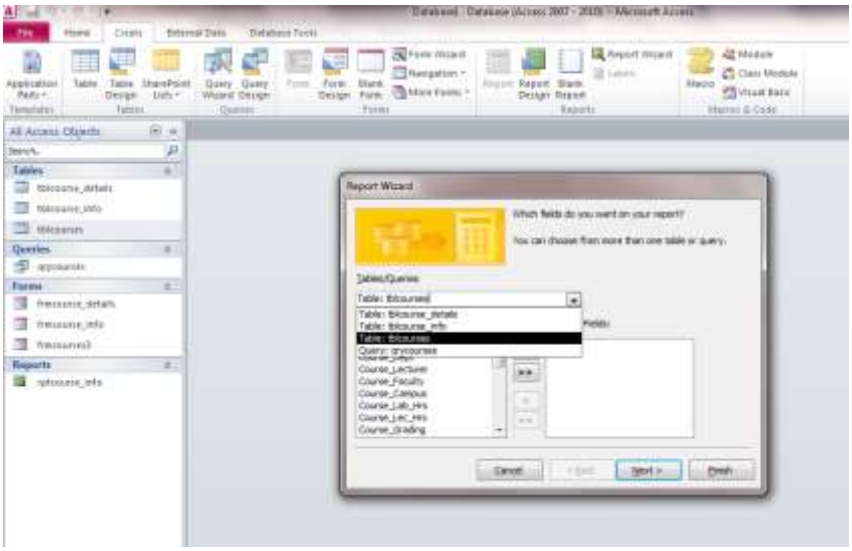


Figure 18: creating a report using a wizard

- Step 3) Select the columns which you want to be there in final report.
- Step 4) Keep selecting all required column as explained in Step 3 above and then click 'Next.'
- Step 5) Layout selection box will appear which allows the user to select the different type of report layout. Click 'NEXT'
- Step 6) Enter the name of the report. Reports are saved with a name beginning letters ***rpt*** save as ***rptcourse_details*** and click 'Finish.'

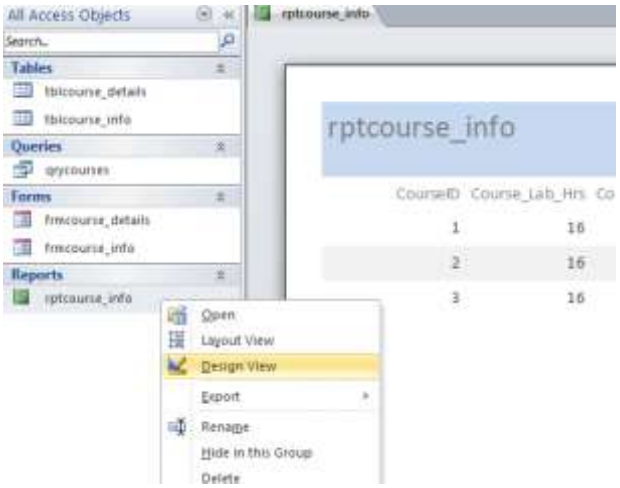


Figure 19: report preview

You can therefore edit the report in design view. This can include adding a report header and tittle as well as some desirable details. Access reports has the same formatting options as covered in Word making customization much easier.

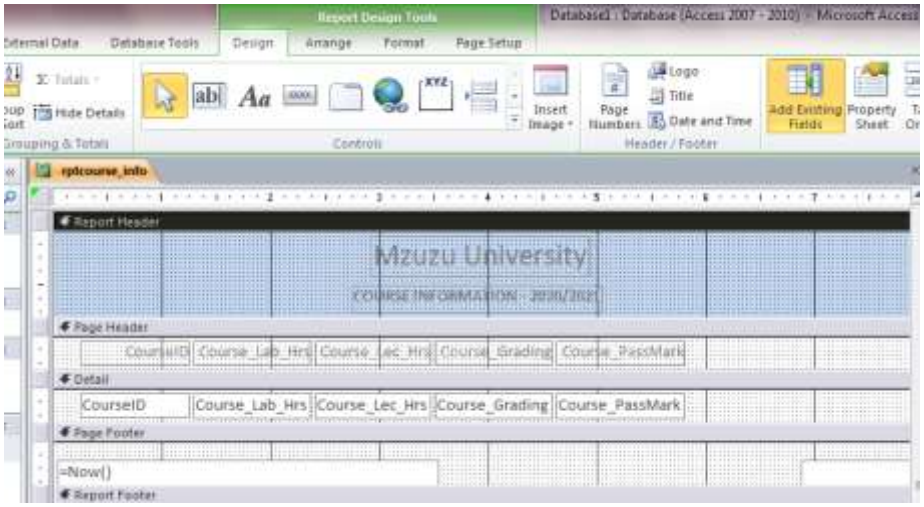


Figure 20: report design view

Reports in access can also be exported analysed further in several formats as indicated on the data tab of the report design ribbon

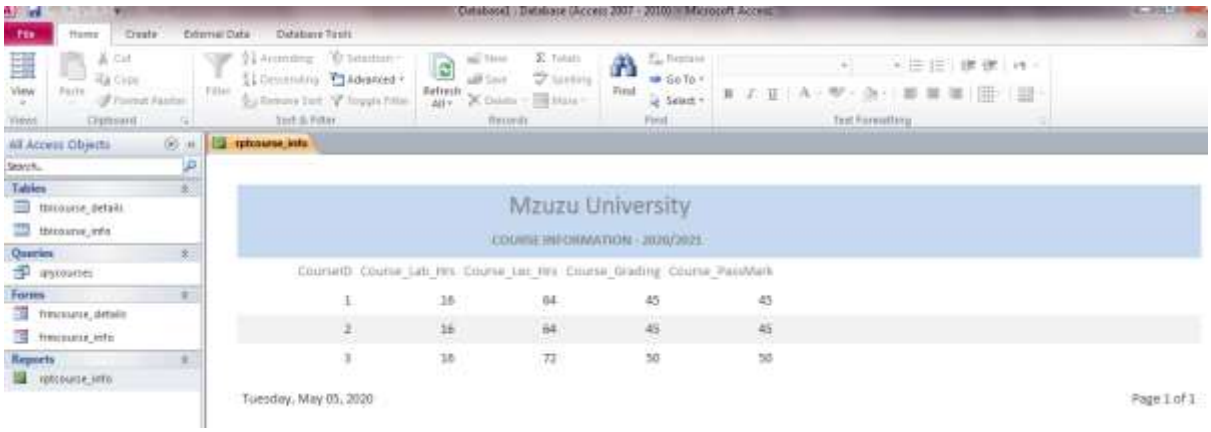


Figure 21 report view

6.1.9 Integration with other applications

Even though Access does a great job of managing data, it is not good at some other tasks. For those purposes, you use programs such as Word and Excel, which join Access as members of the Microsoft Office family of productivity software.

Word can use the information in an Access table in a mail merge document. You can use mail merge documents for a variety of purposes including form letters, mailing labels, envelopes, catalogs, and order forms. To use the Microsoft Word Mail Merge Wizard, you first select the table or query you want to merge with the document, choose Office Links from the Tools menu, and then choose Merge It with MS Word from the submenu. You can merge Access data with either an existing Word document or a new document.

Once the Word document is displayed on your screen, the Access data is already linked. All you need to do is indicate where the merge fields (Access table or query fields) should be inserted in the document. After you position the merge fields, there are several ways you can process the Word document. You can print to the printer, create a new document, or save the file to disk.

To output the contents of an entire table or query in Excel format, first select the table or query you want output, choose Office Links from the Ribbon, Tools menu, and then choose Analyze It with MS Excel from the submenu.

To output a portion of a table or query, you select the portion you want output (rows, columns, or selection), choose Office Links from the Tools menu, and then choose Analyze It with MS Excel from the submenu.

6.1.10 Advantages and Disadvantages of MS-ACCESS

Here, are the pros/benefits for using MS Access application:

- Access offers a fully functional, relational database management system in minutes.
- Easy to import data from multiple sources into Access
- You can easily customize Access according to personal and company needs
- Access works well with many of the development languages that work on Windows OS
- It is robust and flexible, and it can perform any challenging office or industrial database tasks.
- MS-Access allows you to link to data in its existing location and use it for viewing, updating, querying, and reporting.
- Allows you to create tables, queries, forms, and reports, and connect with the help of Macros
- Macros in Access is a simple programming construct with which you can use to add functionality to your database.
- It can perform heterogeneous joins between various data sets stored across different platforms

Here, are the cons for using MS Access

- Microsoft Access is useful for small-to-medium business sectors. However, it is not useful for large-sized organizations
- Lacks robustness compared to DBMS systems like MS SQL Server or Oracle
- All the information from your database is saved into one file. This can slow down reports, queries, and forms
- Technical limit is 255 concurrent users. However, the real-world limit is only 10 to 80 (depending on the type of application which you are using)
- It requires a lot more learning and training compares with other Microsoft programs



Mzuzu University

Access Self-Assessment Test

Instructions

- Answer all questions

1. Create a database called Alendo in the folder you have created. **1 mark**
2. Create a table in the Alendo database with the following fields:
   Motel ID, Weeks Booked, Income, Weeks Not Booked, Location
   10 marks

3. Select appropriate Primary Key for the table. Save the table as **tblMotelDetails**. **2 marks**

4. Enter data in the table as shown below.

Motel Details : Table

Motel ID	Weeks Booked	Income	Weeks Not Booked	Location
R401	22	€3,350.00	2	River
C66	23	€4,660.00	5	City
F642	21	€4,775.00	7	Forest
R103	28	€8,450.00	0	River
R21	26	€5,900.00	2	River
C35	26	€5,250.00	2	City

2 marks

6. Create a Query that will list all motels whose location is River and have an income of more than 5,000.00. Show the following fields only: Motel Code, Income and Location. Save the Query as qry**Rivermorethan5000.**
   5 marks

7. Create another Query that will list all motels that are NOT located at a Forest. Sort the motels in ascending order of **Location**. Show all fields. Save the query as qry**NotForest ascending.**
   3 marks

8. Design a form based on the **Motel Details** table using the Form Wizard. Select only the following fields: Motel ID, Weeks Booked and Location. Choose columnar layout and Ricepaper style. Save the form as frm**Motels.**
   5 marks

9. Create another **table** in the **Alendo** database with the following fields:
   Reference, Last Name, First Name, Date, Event, Motel ID

10 Marks

Select an appropriate primary key and save the table as tbl**Customer Details.**

2 Marks

11. Enter data in the table as shown below.

Customer Details : Table

Customer ID	Last Name	First Name	Date	Event	Motel ID
1	Kawonga	Jane	14-Apr-11	10th Anniversary	R401
2	Kabichi	Yohane	06-Jun-11	Wedding	F642
3	Refalendamu	Andrew	05-May-11	Birthday Party	C35
4	Polanski	Nokia	07-Jun-11	Retirement	R103
5	Kaunda	Benjamin	08-Dec-11	Engagement	R401
6	Naphiri	Maria	07-Jul-11	Christening	R103
7	Mwale	Jonathan	29-Jun-11	Birthday Party	C35

2 Marks

12. Use the split Form method to create a form based on **Customer Details** table that shows only the following fields: Last Name, First Name, Date, and Event. Use columnar layout and International style. Save the form as frm**Customer Details.**

5 Marks

Unit Summary

In this unit we have discussed about databases, the Microsoft access database which is a collection of related objects. The objects include Table, Form, Query, Report, Module and Macro. We have also learnt about data types that exist in access as well as datasheet and design view. Primary and secondary keys are also important when creating tables and they help in creating relationship among tables.

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vii. http://holowczak.com/microsoft-access-2007-and-2010-tutorial/
viii. https://www.techonthenet.com/access/index.php

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END OF MODULE EXAMINATIONS

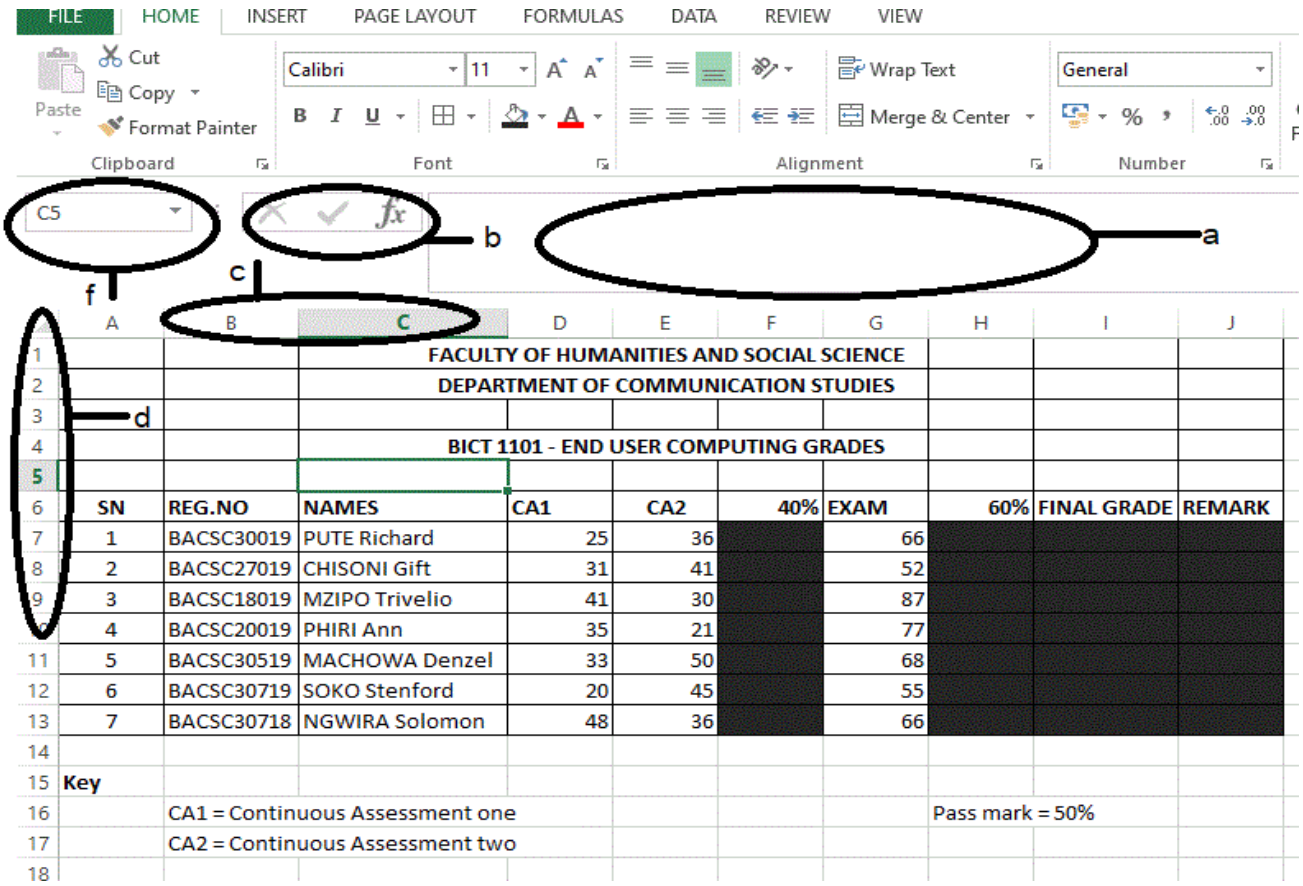
BICT 1101 – END USER COMPUTING

DATE: **TIME: 2-5PM**

INSTRUCTIONS

I. This paper contains **FIVE** pages. Please check.
II. There are **FIVE** questions on this paper. Please check
III. Answer **ALL** questions.
IV. Marks for each question are indicated

QUESTION 1



	A	B	C	D	E	F	G	H	I	J
1			FACULTY OF HUMANITIES AND SOCIAL SCIENCE							
2			DEPARTMENT OF COMMUNICATION STUDIES							
3										
4			BICT 1101 - END USER COMPUTING GRADES							
5										
6	SN	REG.NO	NAMES	CA1	CA2	40%	EXAM	60%	FINAL GRADE	REMARK
7	1	BACSC30019	PUTE Richard	25	36		66			
8	2	BACSC27019	CHISONI Gift	31	41		52			
9	3	BACSC18019	MZIPO Trivelio	41	30		87			
10	4	BACSC20019	PHIRI Ann	35	21		77			
11	5	BACSC30519	MACHOWA Denzel	33	50		68			
12	6	BACSC30719	SOKO Stenford	20	45		55			
13	7	BACSC30718	NGWIRA Solomon	48	36		66			
14										
15	Key									
16		CA1 = Continuous Assessment one						Pass mark = 50%		
17		CA2 = Continuous Assessment two								
18										

Figure 1

A. Explain the use of parts labelled ***a, b, c, d, f*** shown in figure 1 above? **[5 Marks]**

B. In *cell F7* insert a formula to calculate the 40% mark of the total of the continuous grades as shown in *cell D7* and cell *E7*. **[3 Marks]**

C. In *cell H7*, insert a formula to calculate the 60% of the Exam grade as shown in *cell G7*. **[3 Marks]**

D. In *cell I7*, insert a formula for calculating a final grade of the student. (hint: a final grade is composed of 40% continuous and 60% examination grades) **[3 Marks]**

E. In *cell J7*, insert a formula for the remark to indicate "PASS" for those who have attained the passing mark and "FAIL" for those who have not attained the passing mark. **[3 Marks]**

F. What is the difference between Absolute and Relative cell referencing? **[3 Marks]**

QUESTION 2

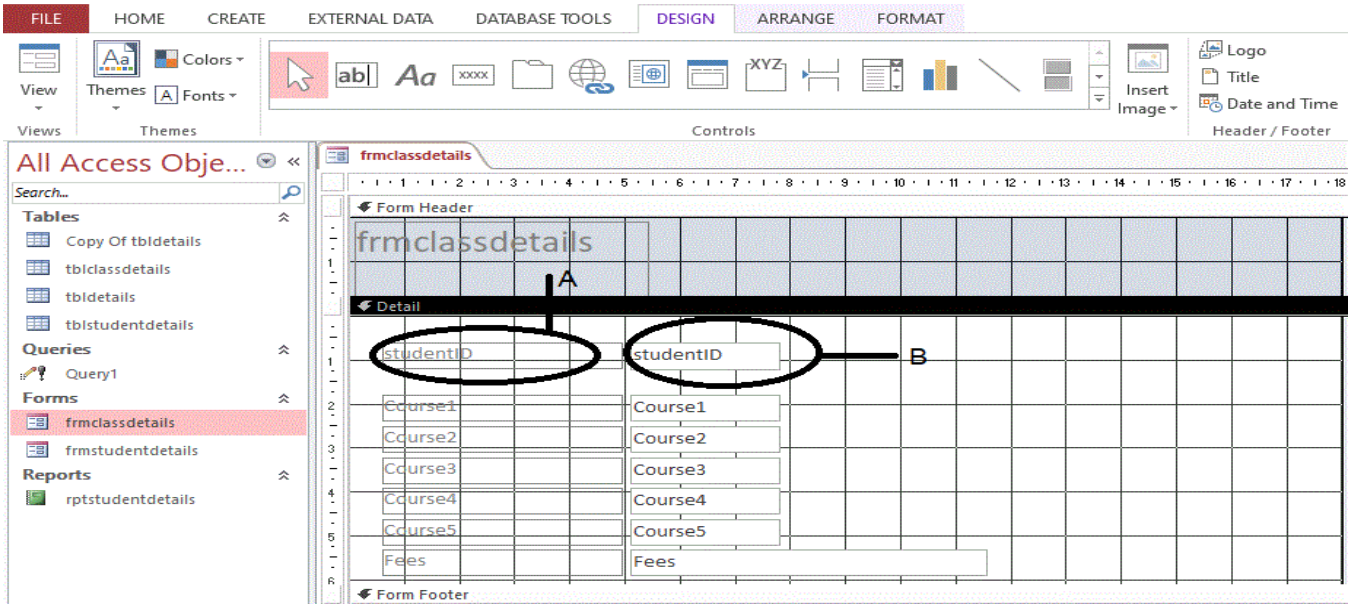


Figure 2

A.

I. In figure 2 above, Name and explain the use of tools marked ***A*** and ***B.*** **[3 Marks]**

II. Define an Access Database? **[3 Marks]**

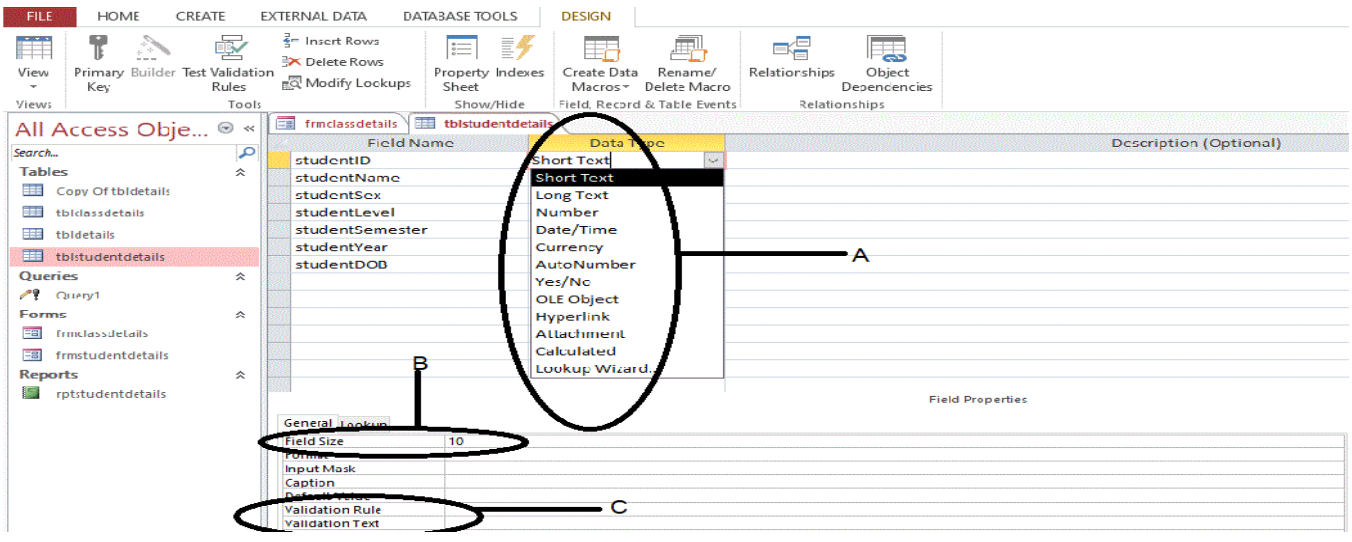


Figure 3

B.

i. For the data types marked ***A*** on the diagram, choose any FIVE from the list and explain when they are used? **[5 Marks]**

ii. Explain the use of the part marked **B** on the diagram. **[3 Marks]**

iii. Explain the difference of the items shown in the figure 3 and marked ***C*** . [**3 Marks]**

iv. Name any THREE types of queries in Microsoft Access database.**[3 Marks]**

QUESTION THREE

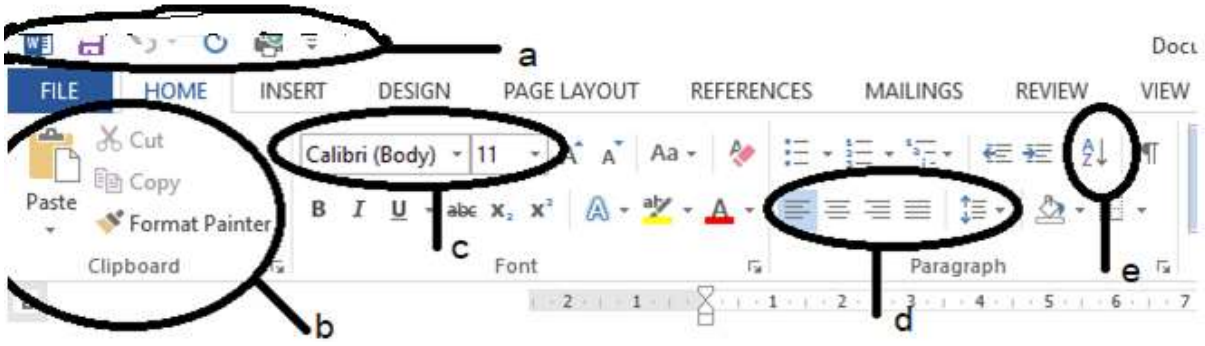


Figure 4

A. What is the use of the part of Microsoft word ribbon marked **A** in the figure 4 above? **[3 Marks]**

B. The part marked **B** on the ribbon has cut, copy and paste menus. Provide the keyboard shortcut for each in achieving their functions? **[3 Marks**]

C. Explain what is contained in the part of the ribbon marked **C**. **[3 Marks]**

D. The part marked **D** contains different types of paragraphs. Apart from *align left* paragraph, List the names of the remaining THREE. **[3 Marks]**

E. What is the use of the part of the ribbon marked **E ?** **[3 Marks]**

F. There are several versions of Microsoft Office, differentiate MS office 2003 word from Ms Office 2010 word. **[5 Marks]**

QUESTION 4

a. Differentiate:

i. Internet and Web **[2 Marks]**

ii. Email and website **[2 Marks]**

iii. In an email, CC and BCC **[2 Marks]**

iv. Browser and Search engine **[2 Marks]**

v. Universal Resource Locator (URL) and Hypertext Transfer Protocol (HTTP) [**2 Marks]**

b. Explain the following as used in Powerpoint:

i. Slide master **[2 Marks]**

ii. Animations **[2 Marks]**

iii. Slide sorter **[2 Marks]**

iv. Slide transitions [**2 Marks]**

v. Themes **[2 Marks]**

QUESTION 5

Discuss any **FOUR** characteristics of a good computer. **[20 Marks]**

END OF QUESTION PAPER

SUGGESTED ANSWERS FOR UNIT 1 AND UNIT 3

Unit 1 Self-Assessment Test Answers

1.Five functions that each digital computer carriers are : -

a. Takes data as input;

b. Stores the data/instructions in its memory and use when required;

c. Processes the data and converts it into useful information;

d. Generates output;

e. Controls all the above steps.

2. RAM and ROM?

RAM: Random Access Memory (RAM) is a memory scheme within the computer system responsible for storing data on a temporary basis, so that it can be promptly accessed by the processor as and when needed

ROM : (Read Only Memory) ROM is a permanent form of storage. ROM stays active regardless of whether power supply to it is turned on or off.

3. The role of Application software in a computer.

Application software is used to accomplish specific tasks other than just running the computer system. Users always interact with the application software when doing different activities. The application software cannot run without the presence of the System software.

List Five ways of avoiding computer viruses?

a. Install an updated anti-virus software from a reputable vendor;

b. Scan for viruses on a regular basis,

c. Use a virus scan before you open any new programs or files that may contain executable code.

d. If you are a member of an online community or chat room, be very careful about accepting files or clicking links that you find or that people send you within the community.

e. Make sure you back up your data regularly

UNIT 3 SELF-ASSESSMENT TEST ANSWERS

1.Define the following terms:

(i) Attachment – it is a file that is included or attached and sent along with your email messages

(ii) CC – carbon copy of the original email message sent to another recipient

(iii) HTML – Hyper Text markup Language used for creating web pages.

(iv) HTTP – Hyper Text Transfer Protocol used to open html files (web pages)

2. What is the purpose of MODEM in dialup networking?

The purpose of a modem is to convert a computer's data stream to analog format so that it can be transmitted over the analog telephone line

3. The Internet and the World Wide Web are two terms used to describe the same thing. Do you agree or disagree? Explain your answer.

The Internet is a global network of networks while the Web, also referred formally as World Wide Web (www) is collection of information which is accessed via the Internet. Another way to look at this difference is; the Internet is infrastructure while the Web is service on top of that infrastructure.

4. What is the main function of a router on the Internet?

A router is a hardware device that allows you to connect several computers and other devices to a single Internet connection

5. Describe what you understand by the term Case Sensitivity with regards to search engines.

Major Search Engines are not case sensitive by default since case sensitivity is not considered a ranking factor by most search engines even if it can improve results in some situation. Moreover, the use of capitalization is not even consistent across the web.

6. What kind of data can you send by e-mail?

An e-mail system allows computer users on a network to send text, graphics, and sometimes sounds and animated images to other users

7. What are the two main differences of web browsers?

The layout engines they use to do the search and the convenience features they offer for navigating and managing the Web and URLs.

8. Given this email address Ekari@yahoo.co.uk

(i) What is a domain name? – the domain name is yahoo.co.uk

(ii) What does .co.uk stand for? – stands for a commercial website in United Kingdom

(iii) Using your answer in (i) what is the Domain type? – the domain type is a commercial organisation

9. Given this address www.mzuni.ac.mw/programs/sciences.html

(i) What is the directory? – the directory is programs

(ii) What is the file in the directory? – the file in the directory which is a resource is sciences.html

10. What are the two things to keep in mind about Uniform resources Locators (URL)

URLs almost never use back slashes (\). All slashes are forward slashes

URLs may contain spaces between characters, they usually do not.

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None

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